

Werk

Titel: Inventiones Mathematicae

Verlag: Springer

Jahr: 1982/83

Kollektion: Mathematica

Digitalisiert: Niedersächsische Staats- und Universitätsbibliothek Göttingen

Werk Id: PPN356556735_0070

PURL: http://resolver.sub.uni-goettingen.de/purl?PPN356556735_0070

LOG Id: LOG_0019

LOG Titel: Introduction.

LOG Typ: article

Übergeordnetes Werk

Werk Id: PPN356556735

PURL: <http://resolver.sub.uni-goettingen.de/purl?PPN356556735>

OPAC: <http://opac.sub.uni-goettingen.de/DB=1/PPN?PPN=356556735>

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Contact

Niedersächsische Staats- und Universitätsbibliothek Göttingen
Georg-August-Universität Göttingen
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37073 Göttingen
Germany
Email: gdz@sub.uni-goettingen.de

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Introduction

Both topics mentioned in the title are interconnected in a very important way in the theory of modular forms. The first explicit connection was given as early as 1932 by H. Petersson [Pet] who expressed the Fourier coefficients of holomorphic Poincaré series as a sum of Kloosterman sums weighted by values of the Bessel functions of integral order. The theory of non-holomorphic automorphic functions developped by H. Maass [Maa 1] and A. Selberg [Sel1] brought more relations into question, the Fourier coefficients of cusp forms of weight zero appearing through the spectral decomposition of the non-euclidean Laplacian.

Special mention should be made of [Sel2] in which Selberg described a relation between cancellation in summation of Kloosterman sums and eigenvalues of the Laplacian asserting that, for $\sigma_0 > 1/2$, the Dirichlet series

$$\sum c^{-2s} S(m, n; c)$$

associated with Kloosterman sums admits an analytic continuation to $\text{Re } s > \sigma_0$ if and only if, the least positive eigenvalue λ_1 of the Laplacian is not less than $\sigma_0(1 - \sigma_0)$. In the case of the full modular group (and also in the case of some congruence subgroups of small index) it has been proved by Roelcke [Roe1] (cf. also [Maa2]) that λ_1 is at least $1/4$. Selberg in [Sel2] expressed the opinion that “possibly [...] this may hold for all congruence subgroups”; we would like to give this opinion the status of a conjecture, which we shall refer to later as Selberg’s eigenvalue conjecture; however, as Selberg noticed, the relation $\lambda_1 \geq \sigma_0(1 - \sigma_0)$ together with Weil’s celebrated estimate for Kloosterman sum implies $\lambda_1 \geq 3/16$.

More recently R.W. Bruggeman [Bru] and N.V. Kuznetsov [Kuz1-2] independently introduced – in the case of the full modular group – trace formulas of a new kind, that of Kuznetsov being stronger and more practical; roughly speaking, Kuznetsov’s formula has the shape

(0.1)

$$\sum_c \frac{1}{c} S(m, n; c) \varphi \left(\frac{\sqrt{mn}}{c} \right)$$

$$= \sum_{\substack{\lambda_j\text{-eigenvalue} \\ \text{of the Laplacian}}} \overline{\rho_j(m)} \rho_j(n) \hat{\varphi}(\lambda_j)$$

directly
manageable
terms

where φ is a kind of Bessel transform of the function φ with compact support and where $\rho_j(n)$ is the n -th Fourier coefficient of the eigen-cusp form associated with λ_j .

Formula (0.1) was then used twice by Kuznetsov. In the first place he deduced mean square estimates for the Fourier coefficients ρ_j from Weil's upper bound of Kloosterman sums in cases where this last upper bound is the best possible, i.e. when the summation in c is short. In the second and more important aspect he used formula (0.1) in cases where the arguments of φ are small, i.e. when the summation in c is long, deriving cancellations between Kloosterman sums from the knowledge of the ρ_j 's. In this way he obtained the upper bound

$$(0.2) \quad \sum_{c < C} \frac{1}{c} S(m, n; c) \ll_{\varepsilon, m, n} C^{1/6 + \varepsilon}$$

whereas a weaker form of the Linnik-Selberg conjecture (cf. [Lin] and [Sel2]) asserts that the above result holds without the exponent $1/6$ and what is more, it should be noticed that the upper bound

$$(0.3) \quad \sum_c \frac{g(c)}{c} S(m, n; c) \ll_{\varepsilon, m, n} C^{\varepsilon}$$

for a non-oscillating smooth function $g \ll 1$ with support in $[C, 2C]$, is an easy consequence of Kuznetsov's work.

The second named author took the matter one step further in [Iwa1] and [Iwa2] by summing the relation (0.1) over m and n with $M < m \leq 2M$, $N < n \leq 2N$. In the first direction (going from Kloosterman sums to the ρ_j 's) it is possible to treat the bilinear form in Kloosterman sums that occur in non trivial ways, so as to obtain a large sieve type result for the ρ_j 's which shows that the Fourier coefficients of cusp forms behave like characters. Going back to the Kloosterman sums this argument leads to a saving effect of order $(MN)^{\frac{1}{2}}$; as we shall see in Theorem 10 it is then possible to get, on average, as good a cancellation as the Linnik-Selberg conjecture predicts.

With arithmetical applications in mind we became interested in summing Kloosterman sums over some arithmetical progressions and we were led to extend the theory to Hecke congruence subgroup $\Gamma_0(q)$. In the meantime, N.V. Proskurin [Pro1] generalized Kuznetsov's formulas for a class of Fuchsian groups including Hecke subgroups; however, for various reasons, his results are not satisfactory for our purpose.

Without entering here into the relevant game that can be played with the different cusps, we shall concentrate our attention on the sums $S_{\infty}^{(q)}(m, n; c) = S(m, n; c)$ for $c \equiv 0 \pmod{q}$ which occur in a natural way (cf. Lemmas 3.3 and 3.4) in the Fourier expansions of holomorphic and non-holomorphic Poincaré series. The extension of Kuznetsov's trace formula (0.1) is carried out in the fourth section. In order to understand how this formula is subsequently used we are going to assume that φ is a very regular non-oscillating function,

bounded by 1 and with support $[X, 2X]$ for some $X < 1$; in this case the Bessel transform $\hat{\varphi}(\lambda)$ is essentially Mellin's transform

$$\int \varphi(t) t^{-\sqrt{1-4\lambda}} \frac{dt}{t}.$$

Besides the problem of keeping the dependence on q in our estimates explicit, the important new problem that arises concerns the λ_j 's which are less than $\frac{1}{4}$, if any! Assuming $\lambda_1 < \frac{1}{4}$, the order of magnitude of $\hat{\varphi}(\lambda_1)$ is $X^{-\sqrt{1-4\lambda_1}}$ and, as we shall see at the end of the fourth section, the sum

$$\sum_c \frac{1}{c} \varphi\left(\frac{1}{c}\right) S_{\infty\infty}^{(q)}(1, 1; c)$$

is indeed equivalent to $X^{-\sqrt{1-4\lambda_1}}$ up to a positive constant; and it is now clear why Weil's estimate implies Selberg's bound $\lambda_1 \geq 3/16$. At this stage we should insist on the fact that Kuznetsov's formula leads to

$$\sum_c \frac{g(c)}{c} S_{\infty\infty}^{(q)}(m, n; c) \ll_{\varepsilon m n q} C^{\sqrt{\max(0, 1-4\lambda_1)+\varepsilon}}$$

which is better than that obtained from Weil's upper bound only if one knows that $\lambda_1 > 3/16 \dots$ (which is the case for the full modular group). In general, averaging over m and n , one retains the $(MN)^{\frac{1}{2}}$ saving effect due to the large sieve inequality, but loses a coefficient $(C/\sqrt{MN})^{\frac{1}{2}}$; i.e. the obtained upper bounds is $(MN)^{3/4}$ -times better than that given by a trivial treatment; Theorem 12 provides a precise statement of this type. As for the dependence of that result on q it is worth noticing that the quality of the upper bounds decreases as q increases, even to the extent that for large q a trivial use of Weil's estimate leads to better results.

The last sections are concerned with the last (!) averaging we perform, that over the level q . Being unfortunately unable to take advantage of this averaging in all cases, we shall at least turn to our profit the contribution of the exceptional λ_j 's, i.e. $\lambda_j < 1/4$, and improve the large sieve inequalities by the introduction of weights; this weighted large sieve will in turn be used to derive upper bounds for quintilinear forms in Kloosterman sums.

The length of the paper is due our wish of presenting a text as self-contained as possible; we tried to give precise references for quoted results (the most important of which concern the spectral decomposition of $L^2(\Gamma \backslash H)$ and Kuznetsov-Bessel transforms), and to avoid invocation of well-known results in the oral tradition.

Acknowledgment. We would like to thank A. Good, N.V. Proskurin and J-P. Serre for informing us about their work, C. Hooley, M.N. Huxley, G. Lachaud and H. Maass for interesting conversations, H. Cohen and D. Zagier for drawing our attention on the eigenvalue conjecture we tacitly assumed in a previous writing, M.-F. Vigneras for sending us a proof of the eigenvalue conjecture for full modular group and R. Rankin for corrections and comments.

This work was initiated when the second named author stayed at the Université de Bordeaux I for the academic year 1979-1980 - he would like to take this occasion to express his thanks for the financial support and the pleasant atmosphere to work-, and pursued in the academic year

1980–81, when the first named author had two opportunities to visit Warsaw – he is very glad to express his gratitude to the Institute of Mathematics of the Polish Academy of Sciences for the invitation and the good working conditions. This work was at last completed when the second named author visited again the University of Bordeaux I in June and September 1981 – he wishes to express additional thanks for the invitation.

1. Definitions and Statement of the Results

1.1 Trace Formulas

Let H denote the upper half plane and q be a positive integer. Consider the group $\overline{\Gamma_0(q)}$ of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with integral entries, $c \equiv 0 \pmod{q}$ and determinant 1; this group acts on H through $z \mapsto \frac{az+b}{cz+d}$; notice that $-1 = \begin{pmatrix} -1 & \\ & -1 \end{pmatrix}$ also belongs to $\overline{\Gamma_0(q)}$ and has the same action as 1; the group $\overline{\Gamma_0(q)}/\{\pm 1\}$ is called Hecke congruence (sub) group of level q and is denoted by $\Gamma_0(q)$; it is a subgroup of the full modular group $\Gamma_0(1)$. We shall abusively confuse an element of $\Gamma_0(q)$ with one of the associated matrices and, when the context is clear, we shall write Γ in place of $\Gamma_0(q)$.

Let σ be an element in $\Gamma_0(q)$; it is called *parabolic* if and only if, it has exactly one fixed point in $\mathbb{C} \cup \{\infty\}$; the fixed point of a parabolic element in Γ is called a *cusp*; if $\mathfrak{a}$ is a cusp of Γ , its *stabilizer* $\Gamma_{\mathfrak{a}}$ is defined by $\Gamma_{\mathfrak{a}} = \{\sigma \in \Gamma; \sigma \mathfrak{a} = \mathfrak{a}\}$. Two cusps $\mathfrak{a}$ and $\mathfrak{b}$ are said to be *equivalent* if and only if, there exists τ in Γ such that $\mathfrak{b} = \tau \mathfrak{a}$ and then $\Gamma_{\mathfrak{b}}$ and $\Gamma_{\mathfrak{a}}$ are conjugate. Notice that the element $\begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$ is parabolic and in $\Gamma_0(q)$ for any q , that the associated cusp is ∞ and that Γ_{∞} is $\left\{ \begin{pmatrix} 1 & b \\ & 1 \end{pmatrix}, b \in \mathbb{Z} \right\}$, which is independent of q . Every cusp of $\Gamma_0(q)$ is equivalent to some $\frac{u}{w}$ with positive coprime u and w such that $w|q$, two such cusps $\frac{u}{w}$ and $\frac{u'}{w'}$ being equivalent if and only if, $w = w'$ and $u \equiv u' \pmod{\left(w, \frac{q}{w}\right)}$. For later use, we introduce the following notation: when $\mathfrak{a}$ is equivalent to some $\frac{u}{w}$ with positive coprime u and w such that $w|q$, we shall write

$$(1.1) \quad \mu(\mathfrak{a}) = \left(w, \frac{q}{w}\right) q^{-1}.$$

Let now $\mathfrak{a}$ and $\mathfrak{b}$ be two cusps of $\Gamma_0(q)$ and let us choose $\sigma_{\mathfrak{a}}$ in $PSL(2, \mathbb{R})$ such that

$$(1.2) \quad \sigma_{\mathfrak{a}} \infty = \mathfrak{a} \quad \text{and} \quad \sigma_{\mathfrak{a}}^{-1} \Gamma_{\mathfrak{a}} \sigma_{\mathfrak{a}} = \Gamma_{\infty}$$

and similarly $\sigma_{\mathfrak{b}}$. For any integers m, n and positive real number γ for which

there exists $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ in $\sigma_a^{-1} \Gamma \sigma_b$, we define the *Kloosterman sum*

$$(1.3) \quad S_{ab}(m, n; \gamma) = \sum'_{\delta \pmod{\gamma \mathbb{Z}}} e \left(m \frac{\alpha}{\gamma} + n \frac{\delta}{\gamma} \right)$$

where the sum is extended to the δ 's, taken modulo $\gamma \mathbb{Z}$, for which there exist α and β such that $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b$. This definition (which is incorrect in that it depends in fact on σ_a and σ_b) is justified because it depends only loosely on σ_a and σ_b , and in fact even on the equivalence class of a and b . More precisely, for given m, n and $\tilde{a}, \tilde{b}$ equivalent to a and b respectively, and for a given choice of $\sigma_a, \sigma_b, \sigma_{\tilde{a}}, \sigma_{\tilde{b}}$, the sets of γ 's for which $S_{ab}(m, n; \gamma)$ and $S_{\tilde{a}\tilde{b}}(m, n; \gamma)$ are defined coincide; moreover, there exist two complex numbers ξ and η with modulus 1, depending on $\sigma_a, \sigma_b, \sigma_{\tilde{a}}, \sigma_{\tilde{b}}$ such that for all γ 's, one has

$$(1.4) \quad S_{ab}(m, n; \gamma) = \xi^m \eta^n S_{\tilde{a}\tilde{b}}(m, n; \gamma).$$

As a first example, consider the case $q=1$ with the only possibility $a=b=\infty$ and choose $\sigma_\infty=1$; the Kloosterman sum $S_{\infty\infty}(m, n; c)$ reduces to the usual one

$$(1.5) \quad S(m, n; c) = \sum_{d \pmod{c}}^* e \left(\frac{m \bar{d} + n d}{c} \right)$$

where the sum is extended over a reduced set of residues modulo c and where $\bar{d}$ denotes the inverse of d modulo c . As a second example, assume $q=rs$ with coprime integers r and s , and consider the two cusps ∞ and $\frac{1}{s}$; choosing $\sigma_{1/s}$

$$= \begin{pmatrix} \sqrt{r} & 1 \\ s\sqrt{r} & \sqrt{r} \end{pmatrix}, \text{ one readily sees that } S_{\infty 1/s}(m, n; \gamma) \text{ is defined if and only if, } \gamma$$

may be written as $\gamma = s\sqrt{r} C$ with integral C coprime with r ; in this case one has

$$(1.6) \quad S_{\infty 1/s}(m, n; \gamma) = e \left(n \frac{\bar{s}}{r} \right) S(m \bar{r}, n; s C)$$

where $s \bar{s} \equiv 1 \pmod{r}$ and $r \bar{r} \equiv 1 \pmod{s C}$.

For a matrix $\sigma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and a complex z , let us write $j(\sigma, z)$ for $cz+d$. A *holomorphic modular form of weight k* with respect to Γ is a holomorphic $\mathbb{C}$ -valued function on H satisfying $f(z) = j(\sigma, z)^{-k} f(\sigma z)$ for any σ in Γ and which is holomorphic at every cusp a ; this means that the holomorphic function φ in $0 < |\xi| < 1$ such that

$$\varphi(e(z)) = j(\sigma_a, z)^{-k} f(\sigma_a z)$$

can be holomorphically prolonged at the origin; this implies that one may write

$$(1.7) \quad j(\sigma_a, z)^{-k} f(\sigma_a z) = \sum_{n=0}^{\infty} c_{an} e(nz)$$

this expression is called the *Fourier expansion* (of f around the cusp a) and the complex numbers $c_{an} = c_{an}(f)$ the *Fourier coefficients* (of f around the cusp a); we say that f is a *cuspidal form* if $c_{a0}(f)$ vanishes for every a .

Of peculiar importance for detecting Fourier coefficients are the *Poincaré series* defined for $m \geq 1$ and $k > 2$ by

$$(1.8) \quad P_{am}(z; k) = \sum_{\tau \in \Gamma_a \backslash \Gamma} j(\sigma_a^{-1} \tau, z)^{-k} e(m \sigma_a^{-1} \tau z)$$

(for $k=2$, the definition is slightly more tricky (cf. [Ran], §5.7)).

Computing the integral $\int_0^1 P_{am}(\sigma_b z; k) e(-nx) dx$ leads (cf. Lemma 3.3) to the expression of the n -th Fourier coefficient of $P_{am}(\cdot; k)$ expanded around the cusp b as a combination of Kloosterman sums weighted by values of Bessel functions of odd order.

On the other hand, the finite dimensional space $\mathfrak{M}_k(\Gamma)$ of holomorphic cusp forms of weight k is conferred a Hilbert structure through the *Petersson inner product*

$$\langle f, g \rangle_k = \iint_{\mathcal{D}} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2}$$

where $\mathcal{D}$ is a fundamental domain for $\Gamma_0(q)$; in case $k=0$, we shall simply write $\langle f, g \rangle$. A simple computation shows that the product $\langle f, P_{am}(\cdot, k) \rangle_k$ is $(k-2)!$ $(4\pi m)^{1-k}$ times the m -th Fourier coefficient of the expansion of f (in $\mathfrak{M}_k$) around the cusp a . Let f_{jk} , for $1 \leq j \leq \theta_k(q)$, be an orthonormal basis of the space $\mathfrak{M}_k(\Gamma)$ of cusp forms of weight k and $\psi_{jk}(a, m)$ denote the m -th Fourier coefficient of f_{jk} around the cusp a ; the Parseval identity then permits one to compute the Fourier coefficients of $P_{am}(\cdot; k)$ in term of the ψ_{jk} 's. Summing over k both expressions, one obtains the first trace formula

Proposition 1. *Let m, n be two positive integers and φ a $\mathcal{C}^3$ -class function with compact support in $]0, \infty)$; let a and b be two cusps of $\Gamma = \Gamma_0(q)$; there exists a real number $b = b_{ab}$ equal to naught when $a = b$ such that one has*

$$(1.9) \quad \sum_{\gamma} \frac{1}{\gamma} S_{ab}(m, n; \gamma) \varphi_B \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) \\ = \frac{1}{2\pi} \delta_{mn} \delta_{ab} e(bn) \int_0^{\infty} J_0(u) \varphi(u) du \\ + \frac{1}{2\pi} \sum_{\substack{k \geq 2 \\ k \text{ even}}} \sum_{1 \leq j \leq \theta_k(q)} \frac{i^k (k-1)!}{(4\pi \sqrt{mn})^{k-1}} \tilde{\varphi}(k-1) \psi_{jk}(b, n) \overline{\psi_{jk}(a, m)},$$

where

$$(1.10) \quad \tilde{\varphi}(l) = \int_0^\infty J_l(y) \varphi(y) \frac{dy}{y} \quad \text{and} \quad \varphi_B(x) = \sum_{\substack{l > 0 \\ l \text{ odd}}} 2l \tilde{\varphi}(l) J_l(x),$$

(1.11) δ_{ab} is 1 if a is equivalent to b and otherwise zero, and the symbol $\sum^r$ means that the summation is performed over the positive real numbers γ for which $S_{ab}(m, n; \gamma)$ is defined.

Here, $\varphi_B(x)$ can be interpreted as the component of $\varphi(x)$ in $L^2(\mathbb{R}^+, dx/x)$ which is orthogonal to all the Bessel functions $J_l(x)$ of odd order; let us define φ_H by

$$(1.12) \quad \varphi(x) = \varphi_B(x) + \varphi_H(x).$$

In the case of the full modular group, Kuznetsov [Kuz1-2] recently obtained trace formulas of the same type with weight function $\varphi_H(x)$ by appealing to the Maass-Selberg theory of non-holomorphic modular forms of weight 0; we are now presenting these formulas in the case of Hecke subgroups.

Let $\Delta = -y^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right)$ denote the non-Euclidean Laplacian. A *non-holomorphic modular form of weight zero* with respect to Γ is a $\mathbb{C}$ -valued function on H which is Γ -invariant, and which is an eigenvector of Δ . Very important examples are *Eisenstein series*, defined for $\text{Re } s > 1$ and $z \in H$ by

$$(1.13) \quad E_a(z; s) = \sum_{\tau \in \Gamma_a \backslash \Gamma} \text{Im}^s(\sigma_a^{-1} \tau z)$$

which have meromorphic continuation to $\mathbb{C}$ and satisfy certain matrix functional equations (cf. [Kub], Theorem 4.4.2). Let f be a Γ -invariant function; for any cusp a , the function $z \mapsto f(\sigma_a z)$ is Γ_∞ -invariant and so $f(\sigma_a z)$ admits the (formal) *Fourier expansion*

$$(1.14) \quad f(\sigma_a z) = \sum_{n \in \mathbb{Z}} c_{an}(y) e(n x).$$

A *cuspidal form* u is a square-integrable (i.e. $\langle u, u \rangle < \infty$) modular form such that the Fourier coefficient $c_{a0}(y)$ vanishes identically for every cusp a . Let u be a cuspidal form associated with the eigenvalue λ , then one readily sees from the equation $(\Delta - \lambda)u = 0$ and the growth condition at infinity that there exist complex numbers $\rho_a(m)$, called the *Fourier coefficients of u around the cusp a* , such that

$$(1.15) \quad u(\sigma_a z) = y^{\frac{1}{2}} \sum_{m \neq 0} \rho_a(m) K_{ik}(2\pi |m| y) e(mx)$$

where

$$(1.16) \quad \kappa^2 = \lambda - \frac{1}{4};$$

the choice of the sign of κ is irrelevant because $K_{ik}(y)$ is even.

Though they are not square-integrable, the Eisenstein series, being eigenvectors of Δ , share many properties with cusp forms; as we shall see their Fourier expansions are quite similar. For integral n and cusps $\mathfrak{a}$ and $\mathfrak{b}$, let

$$\varphi_{\mathfrak{a}\mathfrak{b}n}(s) = \sum \gamma^{-2s} \sum e\left(n \frac{\delta}{\gamma}\right)$$

where the inner sum is extended to the δ 's, modulo $\gamma\mathbb{Z}$, for which there exist α and β with $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ in $\sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{b}}$; then we have

$$(1.17) \quad \begin{aligned} E_{\mathfrak{a}}(\sigma_{\mathfrak{b}} z; s) &= \delta_{\mathfrak{a}\mathfrak{b}} y^s + \pi^{\frac{1}{2}} \Gamma(s - \tfrac{1}{2}) \Gamma^{-1}(s) \varphi_{\mathfrak{a}\mathfrak{b}0}(s) y^{1-s} \\ &+ y^{\frac{1}{2}} \sum_{n \neq 0} 2 \pi^s |n|^{s-\frac{1}{2}} \Gamma^{-1}(s) \varphi_{\mathfrak{a}\mathfrak{b}n}(s) \cdot K_{s-\frac{1}{2}}(2\pi|n|y) e(nx). \end{aligned}$$

Of peculiar importance for detecting Fourier coefficients are the *Poincaré series*. Different objects may play this rôle. Following Selberg (cf. [Sel2]) we consider

$$(1.18) \quad U_{\mathfrak{a}m}(z; s) = \sum_{\tau \in \Gamma_{\mathfrak{a}} \setminus \Gamma} \text{Im}^s(\sigma_{\mathfrak{a}}^{-1} \tau z) e(m \sigma_{\mathfrak{a}}^{-1} \tau z)$$

(notice that the *RHS* depends on $\sigma_{\mathfrak{a}}$, but for notational simplicity we write $U_{\mathfrak{a}m}$ instead of $U_{\sigma_{\mathfrak{a}}m}$).

Computing, as in the holomorphic case, the integral $\int_0^1 U_{\mathfrak{a}m}(\sigma_{\mathfrak{a}} z; s) e(-nx) dx$ leads (cf. Lemma 3.4) to the expression of the n -th Fourier coefficient of $U_{\mathfrak{a}m}(\cdot; s)$ expanded around the cusp $\mathfrak{b}$ as a weighted combination of Kloosterman sums.

The second ingredient in the proof of the trace formulas is more involved than in the holomorphic case, and it depends on the spectral decomposition of $L^2(\Gamma \setminus H)$. In the case of (Hecke) congruence subgroups, one may write $L^2(\Gamma \setminus H) = \mathbb{C} \oplus L_{\text{cusp}}^2(\Gamma \setminus H) \oplus L_{\text{Eis}}^2(\Gamma \setminus H)$, where $L_{\text{Eis}}^2(\Gamma \setminus H)$ is a continuous direct sum spanned by $E_{\mathfrak{a}}(z, \frac{1}{2} + ir) (r \in \mathbb{R})$ and $L_{\text{cusp}}^2(\Gamma \setminus H)$ the space spanned by the cusp forms; we shall denote $\lambda_0 = 0 < \lambda_1 \leq \lambda_2 \leq \dots$ the eigenvalues of Δ , noticing that each eigenvalue has finite order and $\lambda_j \rightarrow \infty$ as $j \rightarrow \infty$. The Parseval identity related to this decomposition (cf. Lemma 3.8) permits us to compute the Fourier coefficients of $U_{\mathfrak{a}m}(\cdot; s)$ in terms of the Fourier coefficients of Eisenstein series and cusp forms.

Combining both expressions, we are first led to rough forms of trace formulas involving real valued parameter t and then using Kuznetsov-Bessel transforms in t we are led to the following

Theorem 1 (Kuznetsov). *Let m, n be two positive integers and φ a $\mathcal{C}^3$ -class function with compact support in $]0, \infty)$; let a and b be two cusps of $\Gamma = \Gamma_0(q)$; denoting by $\sum_{\gamma}^r$ a summation performed over the positive real numbers γ for which $S_{ab}(m, n; \gamma)$ is defined, one has*

$$\begin{aligned}
 (1.19) \quad & \sum_{\gamma}^r \frac{1}{\gamma} S_{ab}(m, n; \gamma) \varphi \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) \\
 &= \frac{1}{2\pi} \sum_{\substack{k \equiv 0 \pmod{2} \\ 1 \leq j \leq \theta_k(q)}} \sum_{k \equiv 0 \pmod{2}} \frac{i^k (k-1)!}{(4\pi \sqrt{mn})^{k-1}} \\
 &\quad \cdot \overline{\psi_{jk}(a, m)} \psi_{jk}(b, n) \check{\varphi}(k-1) \\
 &\quad + \sum_{j \geq 1} \frac{\overline{\rho_{ja}(m)} \rho_{jb}(n)}{\text{ch } \pi \kappa_j} \hat{\varphi}(\kappa_j) \\
 &\quad + \frac{1}{\pi} \sum_{\epsilon} \int_{-\infty}^{\infty} \left(\frac{m}{n} \right)^{-ir} \overline{\varphi_{cam}(\frac{1}{2} + ir)} \varphi_{cbn}(\frac{1}{2} + ir) \hat{\varphi}(r) dr.
 \end{aligned}$$

and

$$\begin{aligned}
 (1.20) \quad & \sum_{\gamma}^r \frac{1}{\gamma} S_{ab}(m, -n; \gamma) \varphi \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) \\
 &= \sum_{j \geq 1} \rho_{ja}(m) \rho_{jb}(n) \frac{\check{\varphi}(\kappa_j)}{\text{ch } \pi \kappa_j} \\
 &\quad + \frac{1}{\pi} \sum_{\epsilon} \int_{-\infty}^{\infty} (mn)^{ir} \varphi_{cam}(\frac{1}{2} + ir) \varphi_{cbn}(\frac{1}{2} + ir) \check{\varphi}(r) dr.
 \end{aligned}$$

where the Bessel transforms are defined by

$$(1.21) \quad \check{\varphi}(l) = \int_0^{\infty} J_l(y) \varphi(y) \frac{dy}{y}$$

$$(1.22) \quad \hat{\varphi}(r) = \frac{\pi}{\text{sh } \pi r} \int_0^{\infty} \frac{J_{2ir}(x) - J_{-2ir}(x)}{2i} \varphi(x) \frac{dx}{x}$$

$$(1.23) \quad \check{\varphi}(r) = \frac{4}{\pi} \text{ch } \pi r \int_0^{\infty} K_{2ir}(x) \varphi(x) \frac{dx}{x}$$

and we recall that κ_j is defined by $\lambda_j = \frac{1}{4} + \kappa_j^2$.

A somewhat weaker form of Kuznetsov's formulas was proved independently by a different method by R.W. Bruggeman. D. Zagier (unpublished) derived them from the well-known spectral decomposition of the Selberg kernel $K(z, z')$ (cf. [Kub]).

1.2 Large Sieve Inequalities for Fourier Coefficients

The trace formulas of Kuznetsov may be used on the one hand to estimate sums of Kloosterman sums, and on the other hand also to investigate the

distribution of Fourier coefficients of modular forms; in this section we speak of the latter matter.

In order to give an idea about our results let us mention briefly some relevant informations from the literature. The Petersson-Ramanujan conjecture, proved by P. Deligne [Del], states that

$$|\psi_{jk}(\mathfrak{a},n)|\ll n^{\frac{k-1}{2}+\varepsilon},$$

while it is only conjectured that the analogous estimate

(1.24)

$$|\rho_{ja}(n)|\ll n^{\varepsilon}$$

is true for the Fourier coefficients of non-holomorphic cusp forms of weight zero. In the case of the full modular group, N.V. Proskurin [Pro3] derived from Kuznietsov’s trace formula, and Weil’s estimate for the magnitude of Kloosterman sums, that

$$|\rho_j(n)|\ll n^{\frac{1}{4}}\tau(n).$$

In a recent letter, J-P. Serre has pointed out to us how it should be possible to replace the exponent 1/4 by 1/5 by appealing to the representation theory of $SL(3,\mathbb{Z})$. On using Rankin’s method which does not depend on Kloosterman sums, one can deduce that

$$\sum_{n\leq X}|\rho_{ja}(n)|^2\sim CX,\text{ }C\text{-constant}$$

whence the Petersson-Ramanujan conjecture (1.24) holds for almost all n .

A question naturally arises here, whether one can improve on the Petersson-Ramanujan conjecture on average, by avoiding to take the absolute values of $\rho_{ja}(n)$, but considering bilinear forms in $\rho_{ja}(n)$ and utilizing a cancellation of terms due to the variations in the argument of $\rho_{ja}(n)$. Various results, in the case of the full modular group, have been established in [Iwa1] and [Iwa2]. We extend them for the Hecke group $\Gamma_0(q)$. Kuznietsov’s trace formula is utilized to reduce the problem to that of bounding bilinear forms in Kloosterman sums. Let us first state such results

Proposition 3. *Let $\mathbf{b}$ be a sequence of complex numbers, N and θ be positive real numbers and $\mathfrak{a}$ be a cusp of $\Gamma_0(q)$. Let $c>0$ be an entry of the left-lower corner of a matrix from $\sigma_{\mathfrak{a}}^{-1}\Gamma_0(q)\sigma_{\mathfrak{a}}$. Write*

$$B_{\mathfrak{a}}(\theta,c,N)=\sum_{N< m,n\leq 2N}b_m\bar{b}_nS_{\mathfrak{a}\mathfrak{a}}(m,n;c)e\left(\frac{2\sqrt{mn}}{c}\theta\right).$$

For any positive ε one has

(1.25)

$$|B_{\mathfrak{a}}(\theta,c,N)|\leq \tau^2(c)c^{\frac{1}{2}}N\|\mathbf{b}_N\|_2^2$$

(1.26)

$$|B_{\mathfrak{a}}(\theta,c,N)|\ll (c+N+\sqrt{\theta cN})\|\mathbf{b}_N\|_2^2$$

(1.27)

$$|B_{\mathfrak{a}}(\theta,c,N)|\ll_{\varepsilon}\theta^{-\frac{1}{2}}c^{\frac{1}{2}}N^{\frac{1}{2}+\varepsilon}\|\mathbf{b}_N\|_2^2$$

the last inequality holding provided $\theta<2$ and $c<N$.

Here, and in what follows, we use the notation

$$\|\mathbf{b}_N\|_2 = \left(\sum_{N < n \leq 2N} |b_n|^2 \right)^{\frac{1}{2}}.$$

As already mentioned, combining the above estimates with Kuznetsov's formula one obtains the following inequalities we shall speak of as *large sieve inequalities for Fourier coefficients of cusp forms* (both holomorphic and non-holomorphic ones) and *Eisenstein series*.

Theorem 2. Let $K \geq 1$, $N \geq \frac{1}{2}$ and $\varepsilon > 0$ be real numbers, $\mathbf{a}$ a sequence of complex numbers and α a cusp of $\Gamma_0(q)$; each of the three expressions

$$(1.28) \quad \sum_{\substack{2 \leq k \leq K \\ k \text{ even}}} \frac{(k-1)!}{(4\pi)^{k-1}} \sum_{1 \leq j \leq \theta_k(q)} \left| \sum_{N < n \leq 2N} a_n n^{-\frac{k-1}{2}} \psi_{jk}(\alpha, n) \right|^2$$

$$(1.29) \quad \sum_{|\kappa_j| \leq K} \frac{1}{\operatorname{ch} \pi \kappa_j} \left| \sum_{N < n \leq 2N} a_n \rho_{j\alpha}(n) \right|^2$$

$$(1.30) \quad \sum_c \int_{-K}^K \left| \sum_{N < n \leq 2N} a_n n^{ir} \varphi_{c\alpha n} \left(\frac{1}{2} + ir \right) \right|^2 dr$$

is majorized up to a constant depending on ε at most by

$$(K^2 + \mu(\alpha) N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2$$

(we recall that $\mu(\alpha)$ is defined in (1.1)).

Remarks. Perhaps, our interpretation of the above inequalities requires some comments. Let us look at (1.29) for $\alpha = \infty$. Since κ_j is either real or purely imaginary with $-\frac{1}{2} < i\kappa_j < \frac{1}{2}$, so $\operatorname{ch} \pi \kappa_j > 0$. We expect, but have not so far been able to prove everything rigorously, that most of $\rho_{j\infty}(n)$ are of order $(\operatorname{ch} \pi \kappa_j / v(q))^{\frac{1}{2}}$ and that the number of points κ_j with $|\kappa_j| \leq K$ is of order $v(q) K^2$ where

$$(1.31) \quad v(q) = [\Gamma_0(q) : \Gamma_0(1)] = q \prod_{p|q} \left(1 + \frac{1}{p} \right).$$

In this light Theorem 2 would mean that

$$\left| \sum_{N < n \leq 2N} a_n \rho_{j\infty}(n) \right| \ll \left(\frac{\operatorname{ch} \pi \kappa_j}{v(q)} \right)^{\frac{1}{2}} \|\mathbf{a}_N\|_2$$

for 'almost all' j with $|\kappa_j| \leq K$, $K^2 \gg 1 + q^{-1} N^{1+\varepsilon}$.

The coefficients $\rho_{j\infty}(n)$ can be expressed by eigenvalues of the Hecke operators T_n defined by

$$T_n f(z) = \frac{1}{\sqrt{n}} \sum_{ad=n} \sum_{b \pmod{d}} f\left(\frac{az+b}{d}\right).$$

This definition is correct for $(n, q) = 1$ in that T_n acts onto the space of cusp forms of $\Gamma_0(q)$. The T_n 's are bounded, hermitian, and satisfy

(1.32)
$$T_m T_n = \sum_{d|(m,n)} T_{mn/d^2}, \quad (m,n,q)=1,$$

therefore they mutually commute. Moreover T_n commutes with the Laplacian Δ , whence the orthonormal basis $u_j(z)$ of cusp forms can be chosen consisting of eigenvectors of all T_n with $(n,q)=1$, i.e.

(1.33)
$$T_n u_j(z) = \tau_j(n) u_j(z),$$

say. Furthermore one observes that T_n acts on Fourier expansions

(1.34)
$$u_j(z) = \sqrt{y} \sum_{m \neq 0} \rho_{j\infty}(m) K_{i\kappa_j}(2\pi|m|y) e(mx)$$

by

(1.35)
$$\rho_{j\infty}(m) = \rho_{j\infty}(d) \tau_j(n)$$

if $m=dn$ with $n \geq 1$, $(n,dq)=1$. The eigenvalues $\tau_j(n)$, $(n,q)=1$ are real because T_n is hermitian (not necessarily positive) and Theorem 2 says essentially that $\tau_j(n)$ changes sign at random, and assumes positive and negative values equally often. It follows from (1.32) that $\tau_j(n)$ is multiplicative, thus suggesting to consider $\tau_j(n)$ as a kind of character. The two first assertions of Theorem 2 remind the large sieve inequality for Dirichlet's characters $\chi(\bmod q)$, namely

$$\sum_{\chi(\bmod q)} \left| \sum_{N < n \leq 2N} a_n \chi(n) \right|^2 = (\varphi(q) + O(N)) \| \mathbf{a}_N \|^2_2$$

(cf. [Mon]) and the third one reminds the mean value theorem for Dirichlet's polynomials, namely

(1.36)
$$\int_0^T \left| \sum_{N < n \leq 2N} a_n n^{it} \right|^2 dt = (T + O(N)) \| \mathbf{a}_N \|^2_2$$

(cf. [Mon]). As a matter of fact, in the case of the full modular group, the Fourier coefficients of Eisenstein series $E(z, \frac{1}{2} + ir)$ are proportional to

$$n^{-ir} \sigma_{2ir}(n) = \sum_{d_1 d_2 = n} \left(\frac{d_1}{d_2} \right)^{ir},$$

thus, the continuous part of Theorem 2 (formula (1.30)) is essentially a consequence of the mean value formula (1.36).

Let us conclude the above remarks by stating an open question: are the three expressions from Theorem 2 majorized up to an absolute constant by

$$(K^2 + q^{-1} N) \| \mathbf{a}_N \|^2_2?$$

1.3 Weighted Large Sieve Inequalities for Exceptional Eigenvalues

For later use we shall call *exceptional* the eigenvalues λ_j such that $0 < \lambda_j < \frac{1}{4}$ and we write $\lambda_j = \frac{1}{4} + \kappa_j^2$ with $i\kappa_j > 0$ for exceptional λ_j 's and with $\kappa_j \geq 0$ for $\lambda_j \geq \frac{1}{4}$.

Theorem 3. *There are no exceptional eigenvalues for the modular group.*

This result goes back to Roelcke [Roe1]. Here we shall give an elegant and short proof kindly shown to us by M.-F. Vignéras.

A. Selberg [Sel2] conjectured that there are no exceptional eigenvalues for congruence subgroups but was however able to show the following (unpublished)

Theorem 4 (A. Selberg). *The smallest positive eigenvalue of the Laplacian acting on $L^2(\Gamma \backslash H)$ is at least $3/16$.*

This result is well-known; the proof we give, though not the shortest *ab initio* is a fairly easy deduction from Kuznetsov formulas, well in the spirit of what will be done in the subsequent chapters. In fact, we shall infer the Selberg estimate (in the case $\Gamma = \Gamma_0(q)$) from the Kuznetsov formula by taking a test function φ with support in $[Y, 2Y]$, letting Y tend to zero, and using Weil's estimate for Kloosterman sums. The conjecture (Selberg's eigenvalue conjecture)

$$(1.37) \quad \lambda_1 \geq \frac{1}{4}$$

would follow along these lines if one assumed an extended Linnik-Selberg hypothesis about the variation of the argument of the Kloosterman sums $S(m, n; q, c)$, but a proof seems more likely to result from geometrical investigations (isoperimetric inequalities on the hyperbolic plane, in the spirit of [Bus] and [Che]?).

In the applications of Kuznetsov's formula for bounding sums of Kloosterman sums in the next section it will turn out that the contribution of the exceptional eigenvalues λ_j 's is dominating, and more over, the smaller λ_j is (i.e. the larger $i\kappa_j$ is) the more important its contribution is; the following weighted large sieve inequality permits to limitate this effect

Theorem 5. *With the notation of Theorem 2 for any $X \geq 1$ and $\varepsilon > 0$ one has*

$$(1.38) \quad \sum_{\lambda_j \text{-except}} X^{2i\kappa_j} \left| \sum_{N < n \leq 2N} a_n \rho_{ja}(n) \right|^2 \ll (1 + \sqrt{\mu(\mathbf{a})NX}) (1 + \sqrt{\mu(\mathbf{a})N^{1+\varepsilon}}) \|\mathbf{a}_N\|_2^2$$

the constant implied in $\ll$ depending on ε alone.

This result is sharper than (1.29) if $\mu(\mathbf{a})N < 1$. Under that assumption we apply Kuznetsov's formula for a smooth test function φ with support $[Y, 2Y]$ where $Y = 8\pi\mu(\mathbf{a})N$ is chosen in such a way that Kloosterman sums $S(m, n; c)$ trivially disappear.

When an averaging over the level q is performed we can show the following

Theorem 6. *Let X, Q, N, ε be positive numbers and $\mathbf{a}$ a sequence of complex numbers. Then one has*

$$(1.39) \quad \sum_{q \leq Q} \sum_{\lambda_j \text{-except}}^{(q)} X^{4i\kappa_j} \left| \sum_{N < n \leq 2N} a_n \rho_{j\infty}(n) \right|^2 \ll (QN)^\varepsilon (Q + N + NX) \|\mathbf{a}_N\|_2^2,$$

the constant implied in $\ll$ depending on ε alone. The sum $\sum^{(q)}$ is extended to the λ_j 's belonging to the exceptional eigenvalues of the Hecke group $\Gamma_0(q)$.

We recall that $\rho_{j\infty}(n)$ stands for n -th Fourier coefficient of Maass form $u_j(z)$ expanded around the cusp $a = \infty$, thus $\mu(a) = q^{-1}$.

This result is sharper than (1.29) if $N < q$ in which case it is sharper than (1.38) as well. It shows a certain rareness of large κ_j 's; this can be expressed by a kind of density theorem (note that by Theorem 4 we know that $i\kappa_j \leq \frac{1}{4}$).

Corollary. For N, Q, ε -positive numbers and $0 < \sigma < 1$ one has

(1.40)
$$\sum_{q \leq Q} \sum_{\sigma < 4i\kappa_j \leq 1} \sum_{N < n \leq 2N}^{(q)} |a_n \rho_{j\infty}(n)|^2 \ll (QN)^\varepsilon (N + N^\sigma Q^{1-\sigma}) \|\mathbf{a}_N\|_2^2,$$

the constant implied in $\ll$ depending on ε alone.

The proof of Theorem 6 begins in much the same way as that of Theorem 5 with only a difference in the support of the test function φ which we choose much smaller. Consequently, the sums of Kloosterman sums $S(m, n; q, c)$ do occur, c being the variable of the summation, and integral valued. The “symmetry” between q and c allows us to reverse their meanings. Considering q as the variable of the summation we apply the Kuznietsov formula for the group $\Gamma_0(c)$, this leads us to an expression looking much like the original one but having different range of the summation over the levels c . In other words we obtain a sort of recurrence relation, and an application of induction on Q concludes the proof.

We should mention that there is an error term in the recurrence relation that imports some weakness to the final estimate. However if this error term is ignored one is led to the following

Conjecture. Theorem 6 holds with the factor $Q + N + \sqrt{N}X$ in place of $Q + N + NX$.

The conjecture is of great significance, because in all applications we know, it is as powerful as the Selberg eigenvalue conjecture. We succeeded to prove our conjecture for $a_n = 1$, more precisely we have

Theorem 7. Let $Q, N, X \geq 1$ and ε be any positive constant. We then have

(1.41)
$$\sum_{q \leq Q} \sum_{\lambda_j\text{-except}}^{(q)} X^{4i\kappa_j} \left| \sum_{n \leq N} \rho_{j\infty}(n) \right|^2 \ll (QN)^\varepsilon (Q + N + \sqrt{N}X) N,$$

the constant implied in $\ll$ depending on ε alone.

Yet, this particular form of the conjecture is very useful especially in estimating sums of incomplete Kloosterman sums. In that context the variable n corresponds to the frequencies of Fourier series arising in the usual method of completing incomplete exponential sums, the limit N is the ratio of the length of the complete sum to that of the incomplete one (see the proof of Theorem 12).

1.4 Upper Bounds for Sums of Kloosterman Sums

We now turn to another use of Kuznietsov's formula, that of estimating sums of Kloosterman sums. Motivated by a number of applications we are led to study sums of the type

$$S_{ab}(\mathbf{a}_M, \mathbf{b}_N; g) = \sum_{\Gamma} \frac{1}{\gamma} \sum_{M < m \leq 2M} a_m \sum_{N < n \leq 2N} b_n g(m, n, \gamma) S_{ab}(m, \pm n; \gamma)$$

where $\Gamma = \Gamma_0(q)$ and $g(m, n, \gamma)$ is a weight function to be suitably specified in each result. We recall that $\sum_{\Gamma}$ means that the summation is taken over positive γ 's that are left-lower corner entries of matrices from $\sigma_a^{-1} \Gamma \sigma_b$, $\mathbf{a}$ and $\mathbf{b}$ being cusps of Γ and $S_{ab}(m, \pm n; \gamma)$, the associated Kloosterman sum.

The simplest weights to which we reduce all the others are

$$(1.42) \quad g(m, n, \gamma) = f(4\pi \sqrt{mn}/\gamma)$$

where $f(x)$ is a function of $\mathcal{C}^2$ class such that

$$(1.43) \quad \text{Supp } f \subset [X, 8X],$$

$$(1.44) \quad \|f\|_{\infty} \leq 1, \quad \|f'\|_1 \leq 2, \quad \|f''\|_1 \leq 6YX^{-1}$$

with some positive X and Y . Some results will depend on the smallest positive eigenvalue $\lambda_1 = \lambda_1(q)$ for $\Gamma = \Gamma_0(q)$, thus it is convenient to introduce the number

$$(1.45) \quad \theta_q = \sqrt{\max(0, 1 - 4\lambda_1)}.$$

First we shall prove

Theorem 8. *Let $\mathbf{a}$ and $\mathbf{b}$ be two cusps of $\Gamma_0(q)$ and let M, N, X, Y be positive real numbers. Let f, g be real valued functions satisfying (1.42), (1.43) and (1.44). For sequences $\mathbf{a}$ and $\mathbf{b}$ of complex numbers and any positive ε , one has*

$$(1.46) \quad S_{ab}(\mathbf{a}_M, \mathbf{b}_N; g) \ll \left(\sqrt{Y} + \frac{1 + |\log X| + X^{-\theta_q}}{1 + X} \right) \cdot (1 + X + \sqrt{\mu(\mathbf{a}) M^{1+\varepsilon}}) (1 + X + \sqrt{\mu(\mathbf{b}) N^{1+\varepsilon}}) \cdot \|\mathbf{a}_M\|_2 \cdot \|\mathbf{b}_N\|_2$$

the constant implied in $\ll$ depending on ε alone.

The introduction of Y permits the following

Corollary. *For positive real numbers T, M, N, ε and complex sequences $\mathbf{a}, \mathbf{b}$ one has*

$$(1.47) \quad \sum_{M < m \leq 2M} a_m \sum_{N < n \leq 2N} b_n \sum_{\substack{c \leq \sqrt{\frac{mn}{MN}} \\ T}} \frac{1}{c} S(m, \pm n, c) \ll T^{\varepsilon} ((MN)^{\frac{1}{2}} + (TMN)^{1/6}) \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2,$$

the constant implied in $\ll$ depending on ε alone.

This result, on one hand, contains the estimate of N.V. Kuznetsov [Kuz 2], namely

$$(1.48) \quad \sum_{c < T} \frac{1}{c} S(m, n; c) \ll_{\varepsilon, m, n} T^{1/6 + \varepsilon}$$

and on the other hand, it means that the following conjecture holds on average.

Conjecture (Linnik-Selberg). For $m, n \geq 1$, $T \geq (m, n)^{\frac{1}{2}}$ and any $\varepsilon > 0$ it holds

$$(1.49) \quad \sum_{c \leq T} \frac{1}{c} S(m, n; c) \ll_{\varepsilon} T^{\varepsilon}.$$

Indeed, if the averaging over m and n in (1.47) is long enough, precisely if $MN \geq T^{\frac{1}{2}}$, we can ignore the term $(TMN)^{1/6}$ on the RHS of (1.47) permitting to interpret the resulting inequality as (1.49) on average. Probably it is interesting to note that the large sieve inequality for Farey points (see H.L. Montgomery [Mon]) yields the bound

$$T^{\varepsilon}((MN)^{\frac{1}{2}} + T) \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2$$

which is as sharp as (1.49) if $MN \geq T^2$. Actually Kuznetsov proved (1.48) with T^{ε} replaced by $(\log T)^{1/3}$.

Another corollary deduced from Theorem 8 is

Theorem 9. Let r and s be positive coprime integers, C, M, N be positive real numbers and g be a real-valued function of $\mathcal{C}^6$ class with support in $[M, 2M] \times [N, 2N] \times [C, 2C]$ such that

$$(1.50) \quad \left| \frac{\partial^{j+k+l}}{\partial m^j \partial n^k \partial c^l} g(m, n, c) \right| \leq M^{-j} N^{-k} C^{-l} \quad \text{for } 0 \leq j, k, l \leq 2.$$

For any $\varepsilon > 0$ and complex sequences $\mathbf{a}, \mathbf{b}$ one has

$$(1.51) \quad \sum_{(c, r)=1} \sum_m a_m \sum_n b_n g(m, n, c) S(m\bar{r}, \pm n; sc) \\ \ll C^{\varepsilon} \left(1 + \frac{s\sqrt{rC}}{\sqrt{MN}} \right)^{\theta_{rs}} \\ \cdot \frac{(s\sqrt{rC} + \sqrt{MN} + \sqrt{sMC})(s\sqrt{rC} + \sqrt{MN} + \sqrt{sNC})}{s\sqrt{rC} + \sqrt{MN}} \\ \cdot \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2$$

the constant implied in $\ll$ depending on ε alone.

If we are concerned with $r=s=1$, the case of modular group in which $\theta_1 = 0$ such as in (1.47), then the above bound becomes

$$(1.52) \quad \ll C^{1+\varepsilon} \sqrt{MN} \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2$$

and this is in C -aspect as strong as the bound from the conjecture of Linnik and Selberg. The substantial difference is that we count Kloosterman sums $S(m, \pm n; c)$ with a smooth weight $g(m, n, c)$ in contrast with the step function from the statement of the conjecture. We should stress, however, that it is often the case one meets in practical situations, thus having something like the Linnik-Selberg conjecture to utilize.

What we know about θ_{rs} in general is that $\theta_{rs} \leq \frac{1}{2}$ which follows from Theorem 4. But another feature of many practical situations is that once we have to deal with Kloosterman sums for congruence subgroups, an averaging over the levels may be taken into account. In such circumstances we may considerably reduce the effect from the factor $(1+s\sqrt{r}C/\sqrt{MN})^{\theta_{rs}}$ by appealing to the weighted large sieve inequalities. For example by Theorem 6 we can prove

Theorem 10. *Let C, M, N, R, S be positive numbers and let $g(c, m, n, r, s)$ be a function of $\mathcal{C}^\infty$ class with compact support in $[C, 2C] \times (0, \infty)^4$ such that*

$$(1.53) \quad \left| \frac{\partial^{v_1+v_2+v_3+v_4+v_5}}{\partial c^{v_1} \partial m^{v_2} \partial n^{v_3} \partial r^{v_4} \partial s^{v_5}} g(c, m, n, r, s) \right| \ll c^{-v_1} m^{-v_2} n^{-v_3} r^{-v_4} s^{-v_5}$$

for any $v_i \geq 0$, $1 \leq i \leq 5$, the constant implied in $\ll$ depending on v_i at most. For $\mathbf{a}_m, \mathbf{b}_{n,r,s}$ complex sequences denote

$$(1.54) \quad \mathcal{L}^\pm(C, M, N, R, S) = \sum_{\substack{R < r \leq 2R \\ S < s \leq 2S \\ (r,s)=1}} \sum_{\substack{0 < n \leq N \\ 0 < m \leq M}} a_m b_{n,r,s} \\ \cdot \sum_{(c,s)=1} g(c, m, n, r, s) S(m\bar{r}, \pm n, s, c).$$

Then for any $\varepsilon > 0$ one has

$$(1.55) \quad \mathcal{L}^\pm(C, M, N, R, S) \ll_\varepsilon (CMNRS)^\varepsilon L(C, M, N, R, S) \|\mathbf{a}_M\|_2 \|\mathbf{b}_{N,R,S}\|_2$$

with

$$(1.56) \quad L^2(C, M, N, R, S) = SR \frac{(S^2 RC^2 + MN + SM C^2)(S^2 RC^2 + MN + SN C^2)}{S^2 RC^2 + MN} \\ + S^2 C^3 \sqrt{(SR + N)RM}.$$

The first term in $L(C, M, N, R, S)$ sets the limit for present methods. The second term can be reduced if Theorem 7 is applied in place of Theorem 6, and this is the case in the following

Theorem 11. *Let the notation and the hypothesis of Theorem 10 be adopted. If $\mathbf{a}_m$ is the characteristic sequence of an interval then (1.55) holds with $L(C, M, N, R, S)$ reduced by the factor $M^{\frac{1}{2}}$ in the second term.*

By standard device from the Fourier series technique, Theorem 11 readily leads to the following estimate for sums of incomplete Kloosterman sums.

Theorem 12. *Let C, D, N, R, S be positive numbers and $\mathbf{b}_{n,r,s}$ be a complex sequence. For $g(c, d, n, r, s)$ having compact support in $[C, 2C] \times [D, 2D] \times (0, \infty)^3$ and satisfying (1.53) denote*

$$\mathcal{K}(C, D, N, R, S) = \sum_{\substack{R < r \leq 2R \\ S < s \leq 2S \\ (r,s)=1}} \sum_{0 < n \leq N} b_{n,r,s} \sum_{\substack{c \leq d \\ (rd,sc)=1}} g(c, d, n, r, s) e\left(n \frac{\overline{rd}}{sc}\right)$$

where $\overline{rd}$ stands for a solution of $\overline{rd}rd \equiv 1 \pmod{sc}$. Then for any $\varepsilon > 0$ one has

(1.57)
$$\mathcal{K}(C, D, N, R, S) \ll (CDNRS)^\varepsilon K(C, D, N, R, S) \|\mathbf{b}_{N,R,S}\|_2$$

with

$$K^2(C, D, N, R, S) = CS(RS + N)(C + DR) + C^2DS\sqrt{(RS + N)R} + D^2NRS^{-1},$$

the constant implied in $\ll$ depending on ε alone. Moreover if the conjecture (1.37) is true then (1.57) holds with

$$K^2(C, D, N, R, S) = CS(RS + N)(C + DR) + C^2DS\sqrt{(RS + N)R} + D^2NRS^{-1},$$

In this paper we do not investigate sums of Kloosterman sums with highly oscillating weight functions because the diversity in assumptions and results prevents us from finding homogenous statements. For this reason we chose to write the large sieve inequalities separately from the estimates for Kloosterman sums.

In all applications of the large sieve inequalities we gave here, we only used their power in the q -aspect; the K -aspect would be important as well if Kloosterman sums with oscillating weights were concerned, such cases being treated in [Iwa 1] and [DI 2].

1.5 Announcement of Applications

There is a lot of interesting problems in analytic theory of numbers that can be related to Kloosterman sums. In several forthcoming papers we applied our estimates for sums of Kloosterman sums; here we state those of major interest for the authors.

An additive divisor problem (cf. [DI 2]): for any $\varepsilon > 0$ the following formula holds

$$\sum_{n \leq x} d(n) d(n+1) = x P(\log x) + O_\varepsilon(x^{\frac{3}{2}+\varepsilon})$$

where $P(t)$ is a quadratic polynomial. With $\frac{2}{3}$ replaced by $\frac{5}{6}$, this result has been obtained independently by E. Wirsing (unpublished) and D.R. Heath-Brown (cf. [H-B]).

The greatest prime factor of $n^2 + 1$ (cf. [DI 1]): there are infinitely many n such that $n^2 + 1$ has a prime factor greater than $n^{6/5}$. Such a result with exponent $11/10$ is due to C. Hooley (cf. [Hoo]).

Mean-values for the Riemann zeta-function (cf. [DI 3]). For $1 \leq N \leq T^{1/5}$ and any complex numbers a_n we have

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 \left| \sum_{n \leq N} a_n n^{it} \right|^2 dt \ll_\varepsilon T^{1+\varepsilon} \sum_{n \leq N} |a_n|^2.$$

When $1 \leq N \leq T^{1/10}$, this is due to H. Iwaniec (cf. [Iwa 3]).

The Brun-Titchmarsh theorem (cf. [DI 4]). Let $x^{\frac{1}{2}} \leq Q \leq x^{1-\varepsilon}$ and $1 \leq |a| \leq x^\varepsilon$. For almost all q in $[Q, 2Q]$ we have

$$\pi(x; q, a) \leq \frac{(\frac{4}{3} + \varepsilon c_1) x}{\varphi(q) \log \frac{x}{q}}$$

with the number of exceptions less than $Qx^{-\varepsilon c_2}$, provided $x > x_0(\varepsilon)$, c_1, c_2 are positive absolute constants; weaker upper bounds of this type have been proved by C. Hooley and H. Iwaniec (cf. [Iwa]).

The greatest prime factor of $p+a$ (cf. [DI 4]). There are infinitely many p such that $p+a$ has a prime factor greater than $p^{21/32}$ ($21/32 = 0.65625$). This result with $\frac{21}{32}$ replaced by $0.6381 \dots$ is due to H. Iwaniec (cf. [Iwa 4]).

2. Congruence Groups and Kloosterman Sums

In this chapter, as well as in the next one, we shall omit the proofs of some elementary arithmetic lemmas.

2.1 Cusps and their Stabilizers, Kloosterman Sums

In this paragraph we shall consider a discrete subgroup Γ of $PSL(2, \mathbb{R})$. The definitions we gave at the beginning of § 1.1 can be extended in a straightforward way; the only precaution to take is to notice that Γ_∞ need not be the group $\left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}; b \in \mathbb{Z} \right\}$ which we shall denote by G in this paragraph and that in (1.2) one should replace Γ_∞ by G . Without any loss for the subsequent parts, the reader may of course restrict himself to Hecke congruence subgroups.

Let σ be an element of $PSL(2, \mathbb{R})$ different from identity; it is parabolic if and only if its trace has absolute value 2 (cf. Proposition 1.12 and 1.13 of [Shi]). If a is a cusp of Γ then its stabilizer is infinite and cyclic; moreover each element of Γ_a different from 1 is parabolic (cf. Proposition 1.17 of [Shi]).

Lemma 2.1. *Let σ in $SL(2, \mathbb{R})$ and z in $\mathbb{C} \cup \{\infty\}$ be such that $\sigma z = z$. Then $\text{tr } \sigma = \pm 2$ if and only if one has*

$$\sigma = \begin{pmatrix} \pm 1 & b \\ & \pm 1 \end{pmatrix} \quad \text{for } z = \infty$$

or

$$\sigma = \begin{pmatrix} \pm 1 + cz & -cz^2 \\ c & \pm 1 - cz \end{pmatrix} \quad \text{for } z \neq \infty.$$

From Lemma 2.1, we easily deduce

Lemma 2.2. *Let Γ be a discrete subgroup of $PSL(2, \mathbb{R})$ and let $\mathfrak{a}$ be a cusp of Γ . Then*

$$\Gamma_{\mathfrak{a}} = \left\{ \begin{pmatrix} 1 & b \\ & 1 \end{pmatrix}, b \in \mathbb{R} \right\} \cap \Gamma \quad \text{if } \mathfrak{a} = \infty$$

and

$$\Gamma_{\mathfrak{a}} = \left\{ \begin{pmatrix} 1 + c\mathfrak{a} & -c\mathfrak{a}^2 \\ c & 1 - c\mathfrak{a} \end{pmatrix}, c \in \mathbb{R} \right\} \cap \Gamma \quad \text{if } \mathfrak{a} \neq \infty.$$

We now prove the correctness of the definition of Kloosterman sums. We must show that $S_{\mathfrak{a}\mathfrak{b}}(m, n; \gamma)$ does not depend on the choice of $\delta \pmod{\gamma}$ neither on the upper row (α, β) . To this end let us consider two matrices

$$\omega = \begin{pmatrix} \alpha & * \\ \gamma & \delta \end{pmatrix} = \sigma_{\mathfrak{a}}^{-1} \tau \sigma_{\mathfrak{b}} \in \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{b}}$$

and

$$\omega' = \begin{pmatrix} \alpha' & * \\ \gamma & \delta' \end{pmatrix} = \sigma_{\mathfrak{a}}^{-1} \tau' \sigma_{\mathfrak{b}} \in \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{b}}$$

with $\delta = \delta' + \gamma l$, $l \in \mathbb{Z}$. Letting $\lambda = \begin{pmatrix} 1 & l \\ & 1 \end{pmatrix}$ and $\varphi = \tau' \sigma_{\mathfrak{b}} \lambda \sigma_{\mathfrak{b}}^{-1} \tau^{-1}$, we obtain $\varphi \in \tau' \sigma_{\mathfrak{b}} G \sigma_{\mathfrak{b}}^{-1} \tau^{-1} = \tau' \Gamma_{\mathfrak{b}} \tau^{-1} \subset \tau' \Gamma \tau^{-1} = \Gamma$ and $\varphi \sigma_{\mathfrak{a}} \omega' \lambda \omega^{-1} = \sigma_{\mathfrak{a}} \begin{pmatrix} * & * \\ & * \end{pmatrix}$. Evaluating both sides at $z = \infty$ we deduce that $\varphi \mathfrak{a} = \mathfrak{a}$, so $\varphi \in \Gamma_{\mathfrak{a}}$; therefore $\omega' \lambda \omega^{-1} \in G$, whence for some $k \in \mathbb{Z}$ one has

$$\omega' = \begin{pmatrix} 1 & k \\ & 1 \end{pmatrix} \omega \lambda^{-1} = \begin{pmatrix} \alpha + \gamma k & * \\ \gamma & \delta' \end{pmatrix}$$

which gives $\alpha' \equiv \alpha \pmod{\gamma}$.

We are now going to prove assertion (1.4): let $\tilde{\mathfrak{a}} = \tau_1 \mathfrak{a}$ and $\tilde{\mathfrak{b}} = \tau_2 \mathfrak{b}$, with τ_1 and τ_2 in Γ ; then, for some

$$\rho_{\mathfrak{a}} = \begin{pmatrix} 1 & b_{\mathfrak{a}} \\ & 1 \end{pmatrix} \quad \text{and} \quad \rho_{\mathfrak{b}} = \begin{pmatrix} 1 & b_{\mathfrak{b}} \\ & 1 \end{pmatrix}$$

one has

$$\sigma_{\tilde{\mathfrak{a}}} = \tau_1 \sigma_{\mathfrak{a}} \rho_{\mathfrak{a}} \quad \text{and} \quad \sigma_{\tilde{\mathfrak{b}}} = \tau_2 \sigma_{\mathfrak{b}} \rho_{\mathfrak{b}}.$$

Hence, one has

$$\sigma_{\tilde{\mathfrak{a}}}^{-1} \Gamma \sigma_{\tilde{\mathfrak{b}}} = \rho_{\mathfrak{a}}^{-1} \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{b}} \rho_{\mathfrak{b}}$$

and so

$$S_{\tilde{\mathfrak{a}}\tilde{\mathfrak{b}}}(m, n; \gamma) = e(n b_{\tilde{\mathfrak{b}}} - m b_{\tilde{\mathfrak{a}}}) S_{\mathfrak{a}\mathfrak{b}}(m, n; \gamma).$$

2.2 Hecke Congruence Subgroups

We first give a complete set of inequivalent cusps for $\Gamma_0(q)$.

Lemma 2.3. *Every cusp of $\Gamma_0(q)$ is equivalent to one among the following*

$$(2.1) \quad \frac{u}{w} \text{ with } u, w > 0, \quad w|q, \quad (u, w) = 1.$$

Two cusps u/w and u_1/w_1 of the above type are $\Gamma_0(q)$ -equivalent if and only if

$$w_1 = w \quad \text{and} \quad u_1 \equiv u \pmod{\left(w, \frac{q}{w}\right)}.$$

Therefore the number of inequivalent cusps is equal to

$$v(q) = \sum_{w|q} \varphi \left(\left(w, \frac{q}{w} \right) \right).$$

Let us now prove (1.6). For the cusp $\alpha = \frac{u}{w}$ with $w|q$, $(u, w) = 1$, $u \neq 0$, it follows from Lemma 2.2 that one has

$$(2.2) \quad \Gamma_\alpha = \left\{ \begin{pmatrix} 1 + c u/w & -c u^2/w^2 \\ c & 1 - c u/w \end{pmatrix}; \quad c \equiv 0 \pmod{[w^2, q]} \right\}.$$

An easy calculation shows that the matrix

$$(2.3) \quad \sigma_\alpha = \begin{pmatrix} \alpha \sqrt{[w^2, q]} & 0 \\ \sqrt{[w^2, q]} & 1/\alpha \sqrt{[w^2, q]} \end{pmatrix}$$

satisfies (1.2).

Assume that $q = rs$ where r and s are coprime positive integers. Consider the two cusps ∞ and $1/s$ of $\Gamma = \Gamma_0(q)$. We may take

$$(2.4) \quad \sigma_\infty = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \quad \text{and} \quad \sigma_{1/s} = \begin{pmatrix} \sqrt{r} & \\ s\sqrt{r} & 1/\sqrt{r} \end{pmatrix}.$$

For $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(q)$ we obtain

$$(2.5) \quad \sigma_\infty^{-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \sigma_{1/s} = \begin{pmatrix} (a+bs)\sqrt{r} & b/\sqrt{r} \\ (c+ds)\sqrt{r} & d/\sqrt{r} \end{pmatrix} = \begin{pmatrix} A\sqrt{r} & B/\sqrt{r} \\ C\sqrt{r} & D/\sqrt{r} \end{pmatrix}$$

say, where A, B, C, D are integers such that

$$(C, r) = 1, \quad C \equiv 0 \pmod{s}, \quad D \equiv s^{-1} C \pmod{r}, \quad (C, D) = 1$$

and

$$AD - BC = 1,$$

so

$$A \equiv \bar{D} \pmod{C}.$$

Conversely, if the integers A, B, C, D satisfy the above conditions then (2.5) is soluble in $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(q)$. Therefore

$$S_{\infty 1/s}(m, n; c) = \sum_{\substack{D \pmod{rC} \\ sD \equiv C \pmod{r} \\ (D, rC) = 1}} e\left(m \frac{\bar{D}}{C} + n \frac{D}{rC}\right).$$

Writing $D = \bar{s}C + rE$ where $s\bar{s} \equiv 1 \pmod{r}$ we see that E runs over the residue classes $\pmod{C}$ coprime with C , thus proving (1.6). ■

For the purpose of Sect. 5 it is important to know $S_{aa}(m, n; c)$ for every cusp a . To find an explicit formula for $S_{aa}(m, n; c)$ we first prove

Lemma 2.4. *Let $a = u/w$, $w|q$, $(u, w) = 1$, $u \neq 0$. Then*

$$\sigma_a^{-1} \Gamma \sigma_a = \left\{ \begin{pmatrix} \frac{\alpha}{u} & \frac{w(w, q/w)}{u^2 q} \beta \\ \frac{q}{(w, q/w)} \gamma & \frac{\delta}{u} \end{pmatrix}; \alpha, \beta, \gamma, \delta \in \mathbb{Z}, \alpha \delta - \beta \gamma w = u^2 \right\}$$

where, besides the above conditions γ, δ satisfy

$$(2.6) \quad (\delta, w) = 1$$

$$(2.7) \quad (\delta - \gamma, \gamma u q/w) = u$$

$$(2.8) \quad (\delta - \gamma) \delta \equiv u^2 \pmod{(w, q/w)}$$

and α is determined $\pmod{\gamma u q/(w, q/w)}$ by

$$(2.9) \quad \alpha + \gamma \equiv 0 \pmod{u}$$

$$(2.10) \quad (\alpha + \gamma)(\delta - \gamma) \equiv u^2 \pmod{\gamma u^2 q/w}$$

$$(2.11) \quad \alpha \delta \equiv u^2 \pmod{(\gamma, u) \gamma w}.$$

Proof. For $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(q)$ we find that

$$\sigma_a^{-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \sigma_a = \begin{pmatrix} a + \alpha^{-1} b & b/\alpha^2 [w^2, q] \\ ((\alpha c - a) + \alpha d - b) [w^2, q] & d - \alpha^{-1} b \end{pmatrix}$$

which assumes the required form on putting

$$(2.12) \quad \alpha = a u + b w, \quad \beta = b, \quad \gamma = \left(\frac{c}{w} u - a\right) u + d u - b w, \quad \delta = d u - b w.$$

We first show that α satisfies the congruences (2.9)–(2.11). Indeed (2.9) and (2.10) follow from

$$\alpha + \gamma = d u + \frac{c}{w} u^2 \quad \text{and} \quad (\alpha + \gamma)(\delta - \gamma) = u^2 - \frac{c}{w} \gamma u^2$$

respectively. For (2.11) notice that $(\beta, u) = (b, u) = (\gamma, u)$ so (2.11) follows from

$$(2.13) \quad \alpha \delta - \beta \gamma w = u^2.$$

Now we show that for integers α, γ, δ satisfying (2.9)–(2.11) and for β given by (2.13) the solutions of (2.12) yield

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(q).$$

Obviously $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = 1$ thus we have to show that a, b, c, d are integers and $c \equiv 0 \pmod{q}$.

By (2.11) and (2.13) it follows that $b = \beta$ is an integer $\equiv 0 \pmod{(\gamma, u)}$. From (2.10) we obtain

$$a u \gamma = (\alpha - b w) \gamma \equiv \alpha \gamma - \alpha \delta + u^2 = \alpha(\gamma - \delta) + u^2 \equiv \gamma(\delta - \gamma) \pmod{\gamma u^2 q/w}$$

which shows that a is an integer $\equiv \frac{\delta - \gamma}{u} \pmod{u q/w}$. Analogously

$$d u \gamma = (\delta + b w) \gamma = \delta \gamma + \alpha \delta - u^2 = \delta(\alpha + \gamma) - u^2 \equiv \gamma(\alpha + \gamma) \pmod{\gamma u^2 q/w}$$

which shows that d is an integer $\equiv \frac{\alpha + \gamma}{u} \pmod{u q/w}$. Finally we obtain

$$\frac{c}{w} u^2 = \gamma - \delta + a u \equiv \gamma - \delta + \delta - \gamma = 0 \pmod{u^2 q/w}$$

which shows that c is an integer $\equiv 0 \pmod{q}$.

To complete the proof of Lemma 2.4 it remains to show that γ and δ satisfy (2.6)–(2.8). Those conditions are equivalent to the existence of solutions α of (2.9)–(2.11) as one can see by a routine calculation. ■

From Lemma 2.4 one deduces the following

Lemma 2.5. *For $\alpha = u/w$, $w|q$, $(u, w) = 1$, $u \neq 0$ we have*

$$(2.14) \quad S_{\alpha\alpha}(m, n; c) = e\left(\frac{m-n}{uq} \left(w, \frac{q}{w}\right)\right) \sum_{\delta_1 \pmod{c}} e\left(m \frac{\alpha_1}{c} + n \frac{\delta_1}{c}\right)$$

where $c = \gamma q \left(w, \frac{q}{w}\right)$, $\delta_1 \pmod{c}$ runs over the solutions of

$$(2.15) \quad (\delta_1, \gamma q/w) = 1, \quad (\gamma + u \delta_1, w) = 1, \quad \delta_1(\gamma + u \delta_1) \equiv u \pmod{\left(w, \frac{q}{w}\right)}$$

and α_1 is determined $\pmod{c}$ by

$$(2.16) \quad \alpha_1 \equiv \bar{\delta}_1 \pmod{\gamma q/w}, \quad \alpha_1 \equiv \gamma \bar{u} + u \overline{(\gamma + u \delta_1)} \pmod{\gamma w/(\gamma, u)}.$$

For the classical Kloosterman sums (1.5) the celebrated result of A. Weil states

$$(2.17) \quad |S(m, n; c)| \leq (m, n, c)^{\frac{1}{2}} c^{\frac{1}{2}} \tau(c).$$

On using (2.14), it can be shown by elementary arguments that the general Kloosterman sum $S_{aa}(m, n; c)$ is a product of Weil type sums with modulus $p^{v_p} \|c$, each of which is majorized by $(m, n, p^{v_p})^{\frac{1}{2}} p^{\frac{1}{2} v_p} \tau(p^{v_p})$, thus giving

Lemma 2.6. *For $a = u/w$, $w|q$, $u \neq 0$, $(u, w) = 1$ we have*

$$(2.18) \quad |S_{aa}(m, n; c)| \leq (m, n, c)^{\frac{1}{2}} c^{\frac{1}{2}} \tau(c).$$

2.3 Bruhat Decomposition

Let Γ be a discrete subgroup of $PSL(2, \mathbb{R})$, a, b -two cusps of Γ and σ_a, σ_b be chosen as in (1.2). For the purpose of expanding the Eisenstein and the Poincaré series into Fourier series (see Sect. 3.2) we need a double coset decomposition of $\sigma_a^{-1} \Gamma \sigma_b$ à la Bruhat. Put

$$\delta_{ab} = \begin{cases} 1 & \text{if } a, b \text{ are } \Gamma\text{-equivalent} \\ 0 & \text{otherwise.} \end{cases}$$

In the first case when $a = \tau b$, $\tau \in \Gamma$ define $\rho_{ab} = \sigma_a^{-1} \tau \sigma_b = \begin{pmatrix} 1 & b \\ & 1 \end{pmatrix}$.

Lemma 2.7. *We have a disjoint decomposition*

$$(2.19) \quad \sigma_a^{-1} \Gamma \sigma_b = \delta_{ab} \rho_{ab} G \cup \bigcup_{\gamma > 0} \bigcup_{\delta \pmod{\gamma}} G \sigma G$$

where $\gamma > 0$ and $\delta \pmod{\gamma}$ run over real numbers such that there exist α and β with

$$\sigma = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b.$$

Remarks. The double coset $G \sigma G$ does not depend on the choice of (α, β) . For a given (γ, δ) the entry α is determined uniquely modulo γ .

3. Eisenstein and Poincaré Series

From now on, Γ will denote the Hecke group $\Gamma_0(q)$.

3.1 Poincaré Series and Fourier Coefficients of Cusp Forms

We first extend a well known formula of Petersson (cf. [Gun], Chap. III, Th. 15) in the following

Lemma 3.1. *Let a be a cusp of Γ and f a holomorphic cusp form of weight k , and let*

$$(3.1) \quad j(\sigma_a, z)^{-k} f(\sigma_a z) = \sum_{m=1}^{\infty} c_{am} e(mz)$$

be the Fourier expansion of f around the cusp $\mathfrak{a}$. Then we have

$$c_{am} = \frac{(4\pi m)^{k-1}}{(k-2)!} \langle f, P_{am}(\cdot; k) \rangle_k.$$

Proof. We restrict ourselves to the case $k > 2$. Denoting $dz = \frac{dx dy}{y^2}$, we have

$$\begin{aligned} \langle f, P_{am}(\cdot; k) \rangle_k &= \iint_{\mathcal{D}} f(z) \overline{P_{am}(z; k)} (\operatorname{Im} z)^k dz \\ &= \iint_{\mathcal{D}} f(z) \sum_{\tau \in \Gamma_{\mathfrak{a}} \setminus \Gamma} \overline{j(\sigma_{\mathfrak{a}}^{-1} \tau, z)^{-k} e(m \sigma_{\mathfrak{a}}^{-1} \tau z)} (\operatorname{Im} z)^k dz; \end{aligned}$$

change the order of summation and integration, and substitute $z = \tau^{-1} \sigma_{\mathfrak{a}} \omega$, getting

$$\langle \cdot, \cdot \rangle_k = \sum_{\tau \in \Gamma_{\mathfrak{a}} \setminus \Gamma} \iint_{\sigma_{\mathfrak{a}}^{-1} \tau \mathcal{D}} f(\tau^{-1} \sigma_{\mathfrak{a}} \omega) \overline{j(\sigma_{\mathfrak{a}}^{-1} \tau, \tau^{-1} \sigma_{\mathfrak{a}} \omega)^{-k} e(m \omega)} (\operatorname{Im} \tau^{-1} \sigma_{\mathfrak{a}} \omega)^k d\omega.$$

But $f(\tau^{-1} \sigma_{\mathfrak{a}} \omega) = j(\tau^{-1}, \sigma_{\mathfrak{a}} \omega)^k f(\sigma_{\mathfrak{a}} \omega)$ since f is modular and

$$j(\tau^{-1}, \sigma_{\mathfrak{a}} \omega)^k \overline{j(\sigma_{\mathfrak{a}}^{-1} \tau, \tau^{-1} \sigma_{\mathfrak{a}} \omega)^{-k}} (\operatorname{Im} \tau^{-1} \sigma_{\mathfrak{a}} \omega)^k = (\operatorname{Im} \omega)^k j(\sigma_{\mathfrak{a}}, \omega)^{-k}$$

by a simple computation, so

$$\langle \cdot, \cdot \rangle_k = \iint_E f(\sigma_{\mathfrak{a}} \omega) j(\sigma_{\mathfrak{a}}, \omega)^{-k} e(-m \bar{\omega}) (\operatorname{Im} \omega)^k d\omega$$

where $E = \bigcup_{\tau \in \Gamma_{\mathfrak{a}} \setminus \Gamma} \sigma_{\mathfrak{a}}^{-1} \tau \mathcal{D}$, which can be represented as the sum of a set of measure zero and the strip $\{z; z = x + iy, y > 0, 0 < x < 1\}$. Introducing the expansion (3.1) and integrating termwise we finally obtain

$$\langle f, P_{am}(\cdot; k) \rangle_k = c_{am} \int_0^{\infty} e^{-4\pi m y} y^{k-2} dy = \frac{(k-2)!}{(4\pi m)^{k-1}} c_{am}. \quad \blacksquare$$

The “non holomorphic” counterpart of Lemma 3.1 is the following

Lemma 3.2. *Let $\mathfrak{a}$ be a cusp of Γ and u a non holomorphic cusp form (of weight zero) associated with the eigenvalue λ . Let the Fourier expansion of u around the cusp $\mathfrak{a}$ be given by (1.15) with $\lambda = \frac{1}{4} + \kappa^2$; then for $m \geq 1$ and $\operatorname{Re} s > 1$ we have*

$$(3.2) \quad \rho_{\mathfrak{a}}(m) = (4\pi m)^{s-\frac{1}{2}} \pi^{-\frac{1}{2}} \frac{\Gamma(s)}{\Gamma(s-\frac{1}{2}-i\kappa) \Gamma(s-\frac{1}{2}+i\kappa)} \langle u, U_{am}(\cdot, \bar{s}) \rangle.$$

Proof. The proof is much the same as that of Lemma 3.1. We have

$$\begin{aligned} \langle u, U_{am}(\cdot, \bar{s}) \rangle &= \iint_{\mathcal{D}} u(z) \sum_{\tau \in \Gamma_{\mathfrak{a}} \setminus \Gamma} \operatorname{Im}(\sigma_{\mathfrak{a}}^{-1} \tau z)^s \overline{e(m \sigma_{\mathfrak{a}}^{-1} \tau z)} dz \\ &= \sum_{\tau \in \Gamma_{\mathfrak{a}} \setminus \Gamma} \iint_{\sigma_{\mathfrak{a}}^{-1} \tau \mathcal{D}} u(\tau^{-1} \sigma_{\mathfrak{a}} z) y^s \overline{e(m z)} dz \\ &= \iint_E u(\sigma_{\mathfrak{a}} z) y^s \overline{e(m z)} dz \\ &= \rho_{\mathfrak{a}}(m) \int_0^{\infty} y^{s-3/2} K_{i\kappa}(2\pi m y) e^{-2\pi m y} dy \end{aligned}$$

which yields (3.2) because

$$(3.3) \quad \int_0^{\infty} e^{-y} y^{s-3/2} K_y(y) dy = \pi^{\frac{1}{2}} 2^{\frac{1}{2}-s} \frac{\Gamma(s-\frac{1}{2}-v) \Gamma(s-\frac{1}{2}+v)}{\Gamma(s)}$$

(see [Kub], p. 50). ■

3.2 Fourier Expansions of Eisenstein and Poincaré Series

Our main profit from the Poincaré series will be extracted from their Fourier expansions. Let us first consider the holomorphic Poincaré series $P_{am}(z; k)$

Lemma 3.3. *Let a and b be cusps of Γ , $m \geq 1$, $k \geq 2$, k even. Let*

$$(3.4) \quad j(\sigma_b, z)^{-k} P_{am}(\sigma_b z; k) = \sum_{n=1}^{\infty} p_{abm}(n, k) e(nz)$$

be the Fourier expansion of $P_{am}(\cdot; k)$ around the cusp b . Then

$$(3.5) \quad p_{abm}(n, k) = \left(\frac{n}{m}\right)^{\frac{k-1}{2}} \left\{ \delta_{mn} \delta_{ab} e(b_{ab} n) + 2\pi i^k \sum_{\gamma} \frac{1}{\gamma} S_{ab}(m, n; \gamma) J_{k-1} \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) \right\}$$

where J_{k-1} is a Bessel function of order $k-1$, $S_{ab}(m, n; \gamma)$ is the Kloosterman sum for cusps a, b , and b_{ab} is a real number depending on σ_a and σ_b .

By the Poisson formulas

$$\begin{aligned} J_k(y) &= \frac{1}{\pi} \int_0^{\pi} \cos(y \sin \varphi - k \varphi) d\varphi \\ &= \frac{2 \left(\frac{y}{2}\right)^k}{\Gamma(k + \frac{1}{2})} \int_0^{\frac{\pi}{2}} \cos(y \sin \varphi) (\cos \varphi)^{2k} d\varphi \end{aligned}$$

(see [EMOT], p. 14) it follows that

$$|J_k(y)| \leq \min \left(1, \frac{\pi}{2} \frac{\left(\frac{y}{2}\right)^k}{(k-1)!} \right)$$

so the series in (3.5) converges.

A similar formula holds for the Fourier coefficients of the non-holomorphic Poincaré series $U_{am}(z; s)$, namely

Lemma 3.4. *Let a and b be cusps of Γ , $m \geq 0$, $\operatorname{Re} s > 1$. Let*

$$(3.6) \quad U_{am}(\sigma_b z; s) = \sum_{n=-\infty}^{\infty} a_{abmn}(y, s) e(nx)$$

be the Fourier expansion of $U_{\mathfrak{a}m}(z; s)$ around the cusp $\mathfrak{b}$. Then

$$(3.7) \quad a_{\mathfrak{a}bmn}(\gamma, s) = \delta_{mn} \delta_{\mathfrak{a}b} e(b_{\mathfrak{a}b} n) y^s e^{-2\pi n y} \\ + y^s \sum_{\Gamma} \gamma^{-2s} S_{\mathfrak{a}b}(m, n; \gamma) B(\gamma, m, n, y, s)$$

where

$$B(\gamma, m, n, y, s) = \int_{-\infty}^{\infty} (x^2 + y^2)^{-s} e\left(-n x - \frac{m}{\gamma^2(x + i y)}\right) dx,$$

and $b_{\mathfrak{a}}$ is a real number depending on $\mathfrak{a}$.

Notice that for $s = \sigma + i t$, $\sigma > 1/2$ we have

$$(3.8) \quad |B(\gamma, m, n, y, s)| \leq \frac{4\sigma}{2\sigma - 1} y^{1-2\sigma} e^{-\pi|n|y}.$$

Indeed, if $n \geq 0$, move the line of the integration to $\operatorname{Im} x = -y/2$ and if $n < 0$, move the line of the integration to $\operatorname{Im} x = y/2$ in either case obtaining

$$|B(\gamma, m, n, y, s)| \leq e^{-\pi|n|y} \int_{-\infty}^{\infty} \frac{dx}{((x + i y/2)^2 + y^2)^{\sigma}} \\ = 2y^{1-2\sigma} e^{-\pi|n|y} \int_0^{\infty} (x^2 + \frac{3}{4})^{-\sigma} dx \\ = \frac{4\sigma}{2\sigma - 1} y^{1-2\sigma} e^{-\pi|n|y}.$$

Therefore, by Lemma 3.2 and (3.8), the Fourier expansion (3.6) together with (3.7) yield the analytical continuation of $U_{\mathfrak{a}m}(z; s)$ to the half plane $\operatorname{Re} s > 3/4$.

The proof of Lemma 3.3 may be found in § 5.3 of [Ran] and that of Lemma 3.4 may be conducted along the same lines. In the special case $m=0$, the evaluation of $B(\gamma, 0, n, y; s)$ in terms of Γ and Bessel functions (cf. [Kub], p. 15) leads to (1.17).

The Fourier expansion of $E_{\mathfrak{a}}(\sigma_{\mathfrak{b}} z; s)$ is very similar to that of the cusp forms (cf. (1.15)); this is not surprising since Eisenstein series and cusp forms are eigenvectors of the Laplacian. Now, the arguments from the proof of Lemma 3.2 are easily adaptable to give the following Lemma, the notation $\varphi_{\mathfrak{a}bm}$ being introduced in Sect. 1.1

Lemma 3.5. For $m \geq 1$, $\operatorname{Re} s_1 > 1$, $\operatorname{Re} s > 1$ and cusps $\mathfrak{a}$ and $\mathfrak{b}$ of Γ we have

$$m^{s-s_1} \varphi_{\mathfrak{a}bm}(s) = \pi^{s_1-s-1} 2^{2s_1-2} \frac{\Gamma(s)\Gamma(s_1)}{\Gamma(s+s_1-1)\Gamma(s_1-s)} \langle E_{\mathfrak{b}}(\cdot; s), U_{\mathfrak{a}m}(\cdot; \bar{s}_1) \rangle.$$

Remark. It follows by analytic continuation that the above formula is valid for $s = \frac{1}{2} + i r$.

3.3 Poles of Eisenstein Series for $\Gamma_0(q)$

Our aim is to prove that $E_{\mathfrak{a}}(z; s)$ has no pole in $(\frac{1}{2}, 1)$ for any cusp $\mathfrak{a}$ of $\Gamma = \Gamma_0(q)$. To this end we look at the Fourier expansion of $E_{\mathfrak{a}}(z; s)$ around ∞ .

Let $\mathfrak{a} = u/w$, $w|q$, $u \neq 0$, $(u, w) = 1$ and take

$$\sigma_{\mathfrak{a}} = \begin{pmatrix} u\sqrt{q/(w^2, q)} & 0 \\ w\sqrt{q/(w^2, q)} & \frac{1}{u\sqrt{q/(w^2, q)}} \end{pmatrix}$$

which satisfies (1.2). If $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ belongs to Γ then

$$\begin{aligned} \sigma_{\mathfrak{a}}^{-1} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} &= \begin{pmatrix} \frac{\alpha}{u\sqrt{q/(w^2, q)}} & \frac{\beta}{u\sqrt{q/(w^2, q)}} \\ (u\gamma - w\alpha)\sqrt{q/(w^2, q)} & (u\delta - w\beta)\sqrt{q/(w^2, q)} \end{pmatrix} \\ &= \begin{pmatrix} * & * \\ C\sqrt{q/(w^2, q)} & -D\sqrt{q/(w^2, q)} \end{pmatrix} \end{aligned}$$

where $\begin{pmatrix} -\delta & \beta \\ \gamma & -\alpha \end{pmatrix} \frac{u}{w} = \frac{D}{C}$, in other words D/C describes the orbit of u/w in Γ .

The next Lemma is straightforward.

Lemma 3.6. *The orbit of the cusp u/w in $\Gamma_0(q)$ consists of the rational numbers D/C , $C > 0$ satisfying*

$$(3.9) \quad (C, D) = 1$$

$$(3.10) \quad (C, q) = w$$

$$(3.11) \quad CD \equiv uw \pmod{(w^2, q)}.$$

From Lemma 3.6 it follows that

$$\varphi_{\mathfrak{a}\infty n}(s) = \left(\frac{(w, q/w)}{wq} \right)^s \sum_{(\gamma, q/w)=1} \gamma^{-2s} \sum_{\substack{\delta \pmod{\gamma w} \\ (\delta, \gamma w)=1 \\ \delta\gamma \equiv uw \pmod{(w^2, q)}}} e\left(-n \frac{\delta}{\gamma w}\right).$$

For $n \neq 0$, the inner sum is $\ll(n, \gamma w)$, thus $\varphi_{\mathfrak{a}\infty n}(s)$ is holomorphic in $\operatorname{Re} s > 1/2$. For $n = 0$, we have

$$\begin{aligned} \varphi_{\mathfrak{a}\infty 0}(s) &= \left(\frac{(w, q/w)}{wq} \right)^s \sum_{(\gamma, q/w)=1} \gamma^{-2s} \varphi(\gamma w) / \varphi((w, q/w)) \\ &= \frac{\varphi(w)}{\varphi((w, q/w))} \left(\frac{(w, q/w)}{wq} \right)^s \prod_{p|q} \left(1 - \frac{1}{p^{2s}} \right)^{-1} \prod_{p \nmid \frac{q}{w}} \left(1 - \frac{1}{p^{2s-1}} \right) \frac{\zeta(2s-1)}{\zeta(2s)} \end{aligned}$$

therefore $\varphi_{\mathfrak{a}\infty 0}(s)$ is meromorphic in $\operatorname{Re} s > 1/2$ with only a simple pole at $s = 1$ with residuum

$$\operatorname{res}_{s=1} \varphi_{\mathfrak{a}\infty 0}(s) = \frac{1}{2} \zeta^{-1}(2) q^{-1} \prod_{p|q} \left(1 + \frac{1}{p} \right)^{-1}.$$

Notice that the residuum does not depend on the cusp $\mathfrak{a}$ and also that

$$[\Gamma_0(1) : \Gamma_0(q)] = q \prod_{p|q} \left(1 + \frac{1}{p}\right).$$

By the above discussion we conclude that the Fourier expansion of $E_{\mathfrak{a}}(z; s)$ around $z = \infty$ yields the meromorphic continuation for $E_{\mathfrak{a}}(z; s)$ on the half-plane $\operatorname{Re} s > 1/2$. Moreover

Lemma 3.7. *In the half-plane $\operatorname{Re} s > 1/2$ each Eisenstein series has only a simple pole at $s = 1$ with residuum*

$$(3.12) \quad \operatorname{res} E_{\mathfrak{a}}(z; s) = \frac{3}{\pi} q^{-1} \prod_{p|q} \left(1 + \frac{1}{p}\right)^{-1}.$$

3.4 Spectral Decomposition of $L^2(\Gamma \backslash H)$

In this short subsection, we quote a few facts about the spectral decomposition of $L^2(\Gamma \backslash H)$ with respect to Δ in the case when $\Gamma = \Gamma_0(q)$; proofs may be found in [Kub], [Roe2], [Fad]. Let f be a square-integrable bounded function on $\Gamma \backslash H$, one has

$$(3.13) \quad f(z) = \sum_{j \geq 0} \langle f, u_j \rangle u_j(z) + \frac{1}{4\pi} \sum_{\mathfrak{c}} \int_{-\infty}^{\infty} \langle f, E_{\mathfrak{c}}(\cdot; \tfrac{1}{2} + ir) \rangle E_{\mathfrak{c}}(z; \tfrac{1}{2} + ir) dr$$

where u_0 denotes the constant function and u_j ($j \geq 1$) runs over an orthonormal basis of the space spanned by the cusp forms, and where $\mathfrak{c}$ runs over a complete set of Γ -inequivalent cusps. It should be stressed that we implicitly used in (3.13) the fact that the Eisenstein series have no pole in $(\frac{1}{2}, 1)$ (cf. Lemma 3.7) and that the scalar product $\langle f, E_{\mathfrak{c}}(\cdot; \frac{1}{2} + ir) \rangle$ exists for any square-integrable bounded f , though the Eisenstein series are not square-integrable on $\mathscr{D}$. The last important result we quote is the following

Lemma 3.8 (Parseval identity). *For $f, g \in L^2(\Gamma \backslash H)$ it holds*

$$(3.14) \quad \langle f, g \rangle = \sum_{j \geq 0} \langle f, u_j \rangle \overline{\langle g, u_j \rangle} + \frac{1}{4\pi} \sum_{\mathfrak{c}} \int_{-\infty}^{\infty} \mathscr{E}_{\mathfrak{c}}(r, f) \overline{\mathscr{E}_{\mathfrak{c}}(r, g)} dr$$

where

$$(3.15) \quad \mathscr{E}_{\mathfrak{c}}(r, f) = \iint_{\mathscr{D}} f(z) \overline{E_{\mathfrak{c}}(z; \tfrac{1}{2} + ir)} dz.$$

4. Trace Formulas for Kloosterman Sums

4.1 The Holomorphic Case

We first deal with the holomorphic case which is slightly simpler and different. Our aim is to prove Proposition 1.

Let $f_{jk}(z)$, $1 \leq j \leq \theta_k(\Gamma)$, be an orthonormal basis of the space $\mathfrak{M}_k(\Gamma)$ of cusp forms of weight $k \geq 2$, k -even. Let

$$j(\sigma_a, z)^{-k} f_{jk}(\sigma_a z) = \sum_{m=1}^{\infty} \psi_{jk}(a, m) e(mz)$$

be the Fourier expansion of $f_{jk}(z)$ around the cusp a . By Lemma 3.1 we first deduce that

$$(4.1) \quad P_{am}(z; k) = \frac{(k-2)!}{(4\pi m)^{k-1}} \sum_{1 \leq j \leq \theta_k(\Gamma)} \overline{\psi_{jk}(a, m)} f_{jk}(z).$$

Let b be another cusp of Γ . Then by Lemma 3.1 again we obtain

$$(4.2) \quad \langle P_{am}(\cdot; k), P_{bn}(\cdot; k) \rangle_k = \frac{(k-2)!}{(4\pi n)^{k-1}} P_{abm}(n; k).$$

On the other hand by the Parseval identity it follows from (4.1) that

$$(4.3) \quad \langle P_{am}(\cdot; k), P_{bn}(\cdot; k) \rangle_k = \frac{((k-2)!)^2}{(16\pi^2 mn)^{k-1}} \sum_{1 \leq j \leq \theta_k(\Gamma)} \overline{\psi_{jk}(a, m)} \psi_{jk}(b, n).$$

Combining (4.2), (4.3) and (3.5) we obtain

$$(4.4) \quad \begin{aligned} \delta_{mn} \delta_{ab} e(b_{ab} n) + 2\pi i^k \sum_{\gamma} \frac{1}{\gamma} S_{ab}(m, n; \gamma) J_{k-1} \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) \\ = \frac{(k-2)!}{(4\pi \sqrt{mn})^{k-1}} \sum_{1 \leq j \leq \theta_k(\Gamma)} \overline{\psi_{jk}(a, m)} \psi_{jk}(b, n). \end{aligned}$$

This formula can be generalized by forming linear combinations of both sides with certain coefficients depending on k . The Bessel functions J_{k-1} of odd integral order span a subspace of functions we called $\varphi_B(x)$ in (1.10).

In order to prove (1.9), we multiply both sides of (4.4) by $\frac{i^k}{\pi} (k-1) \tilde{\varphi}(k-1)$ and sum over positive even k ; in this way we obtain all the terms of (1.9), with the exception of the $\delta_{mn} \dots$ term; this last term is evaluated by the elementary identity

$$2 \sum_{k=1}^{\infty} (-1)^k (2k-1) J_{2k-1}(y) = -y J_0(y)$$

(this identity easily follows from the recurrence relation $J_{n-1}(y) + J_{n+1}(y) = \frac{2n}{y} J_n(y)$ (cf. [EMOT], p. 12)).

4.2 Auxiliary Formulas in the Non-holomorphic Case

We begin by evaluating the scalar product of two Poincaré series in two ways in order to establish, in the non-holomorphic case, formulas analogous to (4.2), (4.3) and (4.4). Let a and b be two cusps of Γ .

For $m \geq 1$, $n \geq 1$, $\text{Re } s_1 > 1$ and $\text{Re } s_2 > 1$, the two Poincaré series $U_{am}(z; s_1)$ and $U_{bn}(z; s_2)$ are in $L^2(\Gamma \backslash H)$ and therefore the scalar product

$$\langle U_{am}(\cdot; s_1), U_{bn}(\cdot; \bar{s}_2) \rangle$$

is well defined. Modifying the arguments of the proof of Lemma 3.1 we deduce the following

$$\begin{aligned}
 (4.5) \quad \langle U_{am}(\cdot; s_1), U_{bn}(\cdot; \bar{s}_2) \rangle &= \iint_{\mathcal{D}} U_{am}(z; s_1) \sum_{\tau \in \Gamma_b \setminus \Gamma} \operatorname{Im}^{s_2}(\sigma_b^{-1} \tau z) \overline{e(n \sigma_b^{-1} \tau z)} dz \\
 &= \sum_{\tau \in \Gamma_b \setminus \Gamma} \iint_{\sigma_b^{-1} \tau \mathcal{D}} U_{am}(\tau^{-1} \sigma_b z; s_1) y^{s_2} \overline{e(n z)} dz \\
 &= \int_0^1 \int_0^1 U_{am}(\sigma_b z; s_1) y^{s_2} \overline{e(n z)} dz \\
 &= \int_0^\infty a_{abmn}(y, s_1) y^{s_2-2} e^{-2\pi n y} dy.
 \end{aligned}$$

Hence by (3.7) we further obtain

$$\begin{aligned}
 &= \delta_{mn} \delta_{ab} e(b_{ab} n) (4\pi n)^{1-s_1-s_2} \Gamma(s_1+s_2-1) \\
 &\quad + \sum_{\gamma} \gamma^{-2s_1} S_{ab}(m, n; \gamma) \int_0^\infty y^{s_1+s_2-2} e^{-2\pi n y} B(\gamma, m, n, y, s_1) dy.
 \end{aligned}$$

The last integral has been evaluated by Kuznetsov (cf. [Kuz 2], formulas (4.27)–(4.31)).

Lemma 4.1 (Kuznetsov). *For $\operatorname{Re} s_1, \operatorname{Re} s_2 > 1$ and $m, n \geq 1$ we have*

$$\begin{aligned}
 &\int_0^\infty y^{s_1+s_2-2} e^{-2\pi n y} B(\gamma, m, n, y, s_1) dy \\
 &= -i 2^{3-s_1-s_2} \gamma^{s_1-s_2} \left(\frac{m}{n}\right)^{(s_2-s_1)/2} \int_{-i}^i K_{s_1-s_2} \left(\frac{4\pi \sqrt{mn}}{\gamma} v\right) \left(v + \frac{1}{v}\right)^{s_1+s_2-2} \frac{dv}{v}
 \end{aligned}$$

where the integration is performed through the half-circle $|z|=1$, $\operatorname{Re} z > 0$ starting from the point $-i$.

By Lemma (1.6), the formula (4.5) yields the analytic continuation of the scalar product of two Poincaré series on the domain $\operatorname{Re} s_1 > \frac{3}{4}$, $\operatorname{Re} s_2 > \frac{3}{4}$ and, due to the absolute convergence of the Kloosterman sums series we can consider a particularly important case where $s_1 = 1 + it$, $s_2 = 1 - it$, giving

Lemma 4.2. *For $m, n \geq 1$ and cusps a and b of Γ , there exists $b = b_{ab}$ such that*

$$\begin{aligned}
 &\langle U_{am}(\cdot; 1 + it), U_{bn}(\cdot; 1 + it) \rangle \\
 &= \delta_{ab} \delta_{mn} \frac{e(bn)}{4\pi n} - 2i \left(\frac{n}{m}\right)^{it} \sum_{\gamma} \gamma^{-2} S_{ab}(m, n; \gamma) \int_{-i}^i K_{2it} \left(\frac{4\pi \sqrt{mn}}{\gamma} v\right) \frac{dv}{v}.
 \end{aligned}$$

To get a result involving the Kloosterman sums $S_{ab}(m, -n; \gamma)$ we evaluate the scalar product $\langle U_{am}(\cdot; s_1), U_{bn}(\cdot; s_2) \rangle$. It requires a slight modification of the previous arguments to get

$$\begin{aligned}
 (4.6) \quad & \langle U_{am}(\cdot; s_1), \overline{U_{bn}(\cdot; s_2)} \rangle \\
 &= \sum^R \gamma^{-2s_1} S_{ab}(m, -n; \gamma) \int_0^\infty y^{s_1+s_2-2} e^{-2\pi n y} B(\gamma, m, -n, y, s_1) dy.
 \end{aligned}$$

Surprisingly the integral can be computed explicitly (cf. [Kuz1])

Lemma 4.3 (Kuznietsov). *For $\operatorname{Re} s_1, \operatorname{Re} s_2 > 1$ and $m, n \geq 1$ we have*

$$\begin{aligned}
 & \int_0^\infty y^{s_1+s_2-2} e^{-2\pi n y} B(\gamma, m, -n, y, s_1) dy \\
 &= \pi 2^{2-s_1-s_2} \gamma^{s_1-s_2} \frac{\Gamma(s_1+s_2-1)}{\Gamma(s_1)\Gamma(s_2)} \left(\frac{m}{n}\right)^{(s_2-s_1)/2} K_{s_1-s_2} \left(\frac{4\pi\sqrt{mn}}{\gamma}\right).
 \end{aligned}$$

By (4.6) and Lemma 4.3, on putting $s_1 = 1 + it$ and $s_2 = 1 - it$ we get

Lemma 4.4. *For $m, n \geq 1$*

$$\begin{aligned}
 & \langle U_{am}(\cdot; 1 + it), \overline{U_{bn}(\cdot; 1 - it)} \rangle \\
 &= \frac{sh\pi t}{t} \left(\frac{n}{m}\right)^{it} \sum^R \gamma^{-2} S_{ab}(m, -n; \gamma) K_{2it} \left(\frac{4\pi\sqrt{mn}}{\gamma}\right).
 \end{aligned}$$

Proof. (We give here a proof of Lemma 4.3 since [Kuz1] is not easily accessible for all readers.) We have

$$B(\gamma, m, -n, s_1) = \int_{-\infty}^{\infty} (x^2 + y^2)^{-s_1} e\left(nx - \frac{m}{\gamma^2(x + iy)}\right) dx$$

which becomes, by changing x into y

$$y^{-2s_1+1} \int_{-\infty}^{\infty} (1 + \xi^2)^{-s_1} e\left(n\xi y - \frac{m}{\gamma\gamma^2(i + \xi)}\right) d\xi.$$

Hence, changing the order of the integrations with respect to ξ and y we get

$$\begin{aligned}
 & \int_0^\infty y^{s_1+s_2-2} e^{-2\pi n y} B(\gamma, m, -n, y, s_1) dy \\
 &= \int_{-\infty}^{\infty} (1 + \xi^2)^{-s_1} \int_0^\infty y^{s_2-s_1-1} e\left((i + \xi)ny - \frac{m}{\gamma\gamma^2(i + \xi)}\right) dy d\xi.
 \end{aligned}$$

It follows from [EMOT], p. 82, formula (24) that for $\operatorname{Im} a > 0$

$$\int_0^\infty y^{\nu-1} e\left(ay - \frac{b}{y}\right) dy = 2e\left(\frac{\nu}{4}\right) \left(\frac{b}{a}\right)^{\frac{\nu}{2}} K_{-\nu}(4\pi\sqrt{ab}).$$

Letting $\nu = s_2 - s_1$, $a = (i + \xi)n$, $b = \frac{m}{\gamma^2(i + \xi)}$, it gives

$$\begin{aligned} & \int_0^\infty y^{s_2-s_1-1} e\left((i+\xi)ny - \frac{m}{y\gamma^2(i+\xi)}\right) dy \\ &= 2 \left(\frac{m}{n\gamma^2}\right)^{\frac{s_2-s_1}{2}} (1-i\xi)^{s_1-s_2} K_{s_1-s_2}\left(\frac{4\pi\sqrt{mn}}{\gamma}\right). \end{aligned}$$

By formula 6.375 of [G-R], we also have

$$\begin{aligned} \int_{-\infty}^{\infty} (1+\xi^2)^{-s_1} (1-i\xi)^{s_1-s_2} d\xi &= \frac{\pi 2^{2-s_1-s_2}}{(s_1+s_2-1)B(s_1, s_2)} \\ &= \pi \frac{2^{2-s_1-s_2} \Gamma(s_1+s_2-1)}{\Gamma(s_1) \cdot \Gamma(s_2)} \end{aligned}$$

and this completes the proof. ■

An other way of computing the scalar product of Poincaré series is based on the Parseval identity (Lemma 3.8). The problem is reduced to the computation of the scalar product of a Poincaré series with a cusp form u_j or with an Eisenstein series $E_a(z; s)$, which has already been performed in Lemma 3.2 or 3.4 (with $m=0$). Hence, by the Parseval identity we infer

$$\begin{aligned} \langle U_{am}(\cdot; s_1), U_{bn}(\cdot; \bar{s}_2) \rangle &= \frac{\pi m^{\frac{1}{2}-s_1} n^{\frac{1}{2}-s_2}}{(4\pi)^{s_1+s_2-1} \Gamma(s_1) \Gamma(s_2)} \\ &\cdot \left\{ \sum_{j \geq 1} \overline{\rho_{ja}(m)} \rho_{jb}(n) \Lambda(s_1, s_2; \kappa_j) \right. \\ &+ \sum_c \int_{-\infty}^{\infty} \left(\frac{n}{m}\right)^{ir} \overline{\varphi_{cam}(\tfrac{1}{2}+ir)} \\ &\cdot \varphi_{cbn}(\tfrac{1}{2}+ir) \frac{\Lambda(s_1, s_2; r)}{|\Gamma(\tfrac{1}{2}+ir)|^2} dr \Big\} \end{aligned}$$

where

$$\Lambda(s_1, s_2; r) = \Gamma(s_1 - \tfrac{1}{2} - ir) \Gamma(s_1 - \tfrac{1}{2} + ir) \Gamma(s_2 - \tfrac{1}{2} - ir) \Gamma(s_2 - \tfrac{1}{2} + ir).$$

In the case when $s_1 = 1 + it$, $s_2 = 1 - it$, this gives

Lemma 4.5. *For $m, n \geq 1$, a, b cusps of Γ and t in $\mathbb{R}$ we have*

$$\begin{aligned} & \langle U_{am}(\cdot; 1+it), U_{bn}(\cdot; 1+it) \rangle \\ &= \frac{1}{4\sqrt{mn}} \left(\frac{n}{m}\right)^{it} \frac{\operatorname{sh} \pi t}{t} \left\{ \pi \sum_{j \geq 1} \frac{\overline{\rho_{ja}(m)} \rho_{jb}(n)}{\operatorname{ch} \pi(\kappa_j - t) \operatorname{ch} \pi(\kappa_j + t)} \right. \\ &\quad \left. + \sum_c \int_{-\infty}^{\infty} \left(\frac{n}{m}\right)^{ir} \frac{\overline{\varphi_{cam}(\tfrac{1}{2}+ir)} \varphi_{cbn}(\tfrac{1}{2}+ir)}{\operatorname{ch} \pi(r-t) \operatorname{ch} \pi(r+t)} \operatorname{ch} \pi r dr \right\}. \end{aligned}$$

For the next formula we assume, without loss of generality that u_j is real, so

$$\rho_{ja}(-m) = \overline{\rho_{ja}(m)}.$$

and similarly

$$\varphi_{\mathfrak{c}a-m}(\tfrac{1}{2}-ir) = \overline{\varphi_{\mathfrak{c}am}(\tfrac{1}{2}+ir)}$$

because the Eisenstein series take real values for real s .

As before, the Parseval identity leads to

$$\begin{aligned} & \langle U_{am}(\cdot; s_1), \overline{U_{bn}(\cdot; s_2)} \rangle \\ &= \frac{\pi m^{\frac{1}{2}-s_1} n^{\frac{1}{2}-s_2}}{(4\pi)^{s_1+s_2-1} \Gamma(s_1) \Gamma(s_2)} \left\{ \sum_{j \geq 1} \overline{\rho_{ja}(m) \rho_{jb}(n)} A(s_1, s_2; \kappa_j) \right. \\ & \quad \left. + \sum_{\mathfrak{c}} \int_{-\infty}^{\infty} (mn)^{-ir} \overline{\varphi_{\mathfrak{c}am}(\tfrac{1}{2}+ir)} \varphi_{\mathfrak{c}bn}(\tfrac{1}{2}+ir) \frac{A(s_1, s_2; \kappa_j)}{|\Gamma(\tfrac{1}{2}+ir)|^2} dr \right\}. \end{aligned}$$

In particular on taking $s_1 = 1+it$, $s_2 = 1-it$, $t \in \mathbb{R}$ we obtain

Lemma 4.6. *For $m, n \geq 1$, a, b cusps of Γ and real t we have*

$$\begin{aligned} & \langle U_{am}(\cdot; 1+it), \overline{U_{bn}(\cdot; 1-it)} \rangle \\ &= \frac{1}{4\pi \sqrt{mn}} \left(\frac{n}{m}\right)^{it} \frac{\text{sh } \pi t}{t} \left\{ \pi \sum_{j \geq 1} \frac{\overline{\rho_{ja}(m) \rho_{jb}(b)}}{\text{ch } \pi(\kappa_j - t) \text{ch } \pi(\kappa_j + t)} \right. \\ & \quad \left. + \sum_{\mathfrak{c}} \int_{-\infty}^{\infty} (mn)^{-ir} \overline{\varphi_{\mathfrak{c}am}(\tfrac{1}{2}+ir)} \varphi_{\mathfrak{c}bn}(\tfrac{1}{2}+ir) \frac{\text{ch } \pi r dr}{\text{ch } \pi(r-t) \text{ch } \pi(r+t)} \right\}. \end{aligned}$$

For notational simplicity we put $H(r, t) = \text{ch } \pi r / \text{ch } \pi(r-t) \text{ch } \pi(r+t)$ and

$$\begin{aligned} (4.7) \quad D_{2it}(x) &= -\frac{2it}{\text{sh } \pi t} \int_{-i}^i K_{2it}(xv) \frac{dv}{v} \\ &= \frac{t}{\text{sh } 2\pi t} \int_x^\infty (J_{2it}(v) + J_{-2it}(v)) \frac{dv}{v}. \end{aligned}$$

Then, combining Lemmas 4.2 and 4.5 we obtain a rough form of a trace formula

Lemma 4.7. *For $m, n \geq 1$, a, b cusps of Γ and real t we have*

$$\begin{aligned} (4.8) \quad & \sum_{\gamma} \frac{4\pi \sqrt{mn}}{\gamma^2} S_{ab}(m, n; \gamma) D_{2it} \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) + \delta_{ab} \delta_{mn} e(b_{ab}, n) \frac{t}{\pi \text{sh } \pi t} \\ &= \pi \sum_{j \geq 1} \frac{\overline{\rho_{ja}(m) \rho_{jb}(n)}}{\text{ch } \pi \kappa_j} H(\kappa_j, t) \\ & \quad + \sum_{\mathfrak{c}} \int_{-\infty}^{\infty} \left(\frac{m}{n}\right)^{-ir} \overline{\varphi_{\mathfrak{c}am}(\tfrac{1}{2}+ir)} \varphi_{\mathfrak{c}bn}(\tfrac{1}{2}+ir) H(r, t) dr. \end{aligned}$$

Next, combining Lemmas 4.3 and 4.6 we obtain another formula

Lemma 4.8. *For $m, n \geq 1$, a, b cusps of Γ and real t we have*

$$\begin{aligned}
 (4.9) \quad & \sum^r \frac{4\pi \sqrt{mn}}{\gamma^2} S_{ab}(m, -n; \gamma) K_{2it} \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) \\
 &= \pi \sum_{j \geq 1} \frac{\rho_{ja}(m) \rho_{jb}(n)}{\operatorname{ch} \pi \kappa_j} H(\kappa_j, t) \\
 &+ \sum_c \int_{-\infty}^{\infty} (mn)^{ir} \varphi_{cam}(\tfrac{1}{2} + ir) \varphi_{cbn}(\tfrac{1}{2} + ir) H(r, t) dr.
 \end{aligned}$$

4.3 Kuznetsov's Trace Formulas

Here we modify the formulas (4.8) and (4.9) with respect to the weight functions attached to the Kloosterman sums. To this end, we freely borrow from [Kuz1] Kuznetsov's results about integral transforms of Bessel functions.

Let φ be a $\mathcal{C}^3$ -class function with compact support in $(0, \infty)$. For real t put

$$(4.10) \quad f(t) = - \int_0^{\infty} (J_{2it}(x) + J_{-2it}(x)) \left(\frac{\varphi(x)}{x} \right)' dx$$

and recall that $\hat{\varphi}$ has been defined in (1.22) and φ_H in (1.12). We have, for $t \in \mathbb{R}$ and $ir \in [0, \frac{1}{4}]$ or $r \geq 0$

$$(4.11) \quad - \int_{-\infty}^{\infty} t \operatorname{sh} 2\pi t H(r, t) K_{2it}(\theta) dt = \theta K_{2ir}(\theta)$$

$$(4.12) \quad \int_{-\infty}^{\infty} \frac{t f(t)}{\operatorname{sh} \pi t} dt = \int_0^{\infty} J_0(x) \varphi(x) dx$$

$$(4.13) \quad x \int_x^{\infty} \int_{-\infty}^{\infty} \left(\frac{t f(t)}{\operatorname{sh} 2\pi t} (J_{2it}(u) + J_{-2it}(u)) \right) \frac{du}{u} = \varphi_H(x)$$

$$(4.14) \quad \int_{-\infty}^{\infty} H(r, t) f(t) dt = \frac{2}{\pi} \hat{\varphi}(r).$$

For the proofs, see formulas (6.20)–(6.23), (6.19) and (6.27) of [Kuz2], the extension to the imaginary r 's being straightforward.

On multiplying both sides of (4.8) by $f(t)$ and integrating over the real line, we deduce from (4.12) and (4.13) the following

Proposition 2 (Kuznetsov). *If $\varphi(x)$ is a $\mathcal{C}^3$ -class function with compact support $(0, \infty)$, a, b two cusps of Γ , there exists a real $b = b_{ab}$ vanishing when $a = b$ such that*

$$\begin{aligned}
 (4.15) \quad & \sum^r \frac{1}{\gamma} S_{ab}(m, n; \gamma) \varphi_H \left(\frac{4\pi \sqrt{mn}}{\gamma} \right) + \frac{1}{2\pi} \delta_{ab} \delta_{mn} e(bn) \int_0^{\infty} J_0(x) \varphi(x) dx \\
 &= \sum_{j \geq 1} \frac{\overline{\rho_{ja}(m)} \rho_{jb}(n)}{\operatorname{ch} \pi \kappa_j} \hat{\varphi}(\kappa_j) \\
 &+ \sum_c \int_{-\infty}^{\infty} \left(\frac{m}{n} \right)^{-ir} \overline{\varphi_{cam}(\tfrac{1}{2} + ir)} \varphi_{cbn}(\tfrac{1}{2} + ir) \hat{\varphi}(r) dr.
 \end{aligned}$$

Gathering together the results of Propositions 1 and 2 we obtain (1.19) of Theorem 1. In order to get (1.20) in Theorem 1, we multiply both sides of (4.9) by

$$\frac{4}{\pi^2} t \operatorname{sh} 2\pi t \int_0^\infty K_{2it}(\theta) \varphi(\theta) \frac{d\theta}{\theta^2}$$

and then integrate both sides through the real line $(-\infty, \infty)$. In the *LHS* we get (with $x=4\pi\sqrt{mn}/\gamma$) the double integral

$$(4.16) \quad \frac{4}{\pi^2} \int_{-\infty}^\infty t \operatorname{sh} 2\pi t K_{2it}(x) \left(\int_0^\infty K_{2it}(\theta) \varphi(\theta) \frac{d\theta}{\theta^2} \right) dt = x^{-1} \varphi(x)$$

by the Kontorovitch-Lebedev formula (see [EMOT], p. 75), whence the *LHS* of (1.20). In the right-hand side we get the double integral

$$(4.17) \quad \begin{aligned} & \frac{4}{\pi} \int_0^\infty \left(- \int_{-\infty}^\infty t \operatorname{sh} 2\pi t H(r, t) K_{2it}(\theta) \right) dt \cdot \varphi(\theta) \frac{d\theta}{\theta^2} \\ &= \frac{4}{\pi} \int_0^\infty K_{2ir}(\theta) \varphi(\theta) \frac{d\theta}{\theta} = \frac{\check{\varphi}(r)}{\operatorname{ch} \pi r} \end{aligned}$$

by (4.11), whence the *RHS* of (1.20).

In the above arguments we omitted several delicate questions about the convergence of the involved sums and integrals. For relevant details the reader is referred to the original papers of Kuznetsov.

5. Large Sieve Inequalities for Fourier Coefficients

In this section, we apply the trace formulas to derive Theorem 2.

5.1 Quadratic Forms with Kloosterman Sums

Let $\mathfrak{a}$ be a cusp of $\Gamma_0(q)$ and $S_{\mathfrak{a}\mathfrak{a}}(m, n; c)$ a Kloosterman sum associated with $\mathfrak{a}$. For $\theta > 0$, we investigate quadratic forms of the type

$$B_{\mathfrak{a}}(\theta, c, N) = \sum_{N < m, n \leq 2N} b_m \bar{b}_n S_{\mathfrak{a}\mathfrak{a}}(m, n; c) e\left(\frac{2\sqrt{mn}}{c} \theta\right)$$

with the aim of proving Proposition 3. We shall assume, without loss of generality, that $\mathfrak{a} = \frac{u}{w}$, with $w|q$, $(u, w) = 1$ and $u \neq 0$.

The first inequality (1.25) follows easily from Lemma 2.6.

The second inequality (1.26) is a consequence of a hybrid large sieve inequality for Farey points, namely

$$(5.1) \quad \int_{-T}^T \sum_{d \pmod{c}}^* \left| \sum_{N < n \leq 2N} b_n n^{it} e\left(n \frac{d}{c}\right) \right|^2 dt \ll (cT + N) \|\mathbf{b}_N\|_2^2.$$

We leave the proof of (5.1) to the reader as an exercise. A minor difficulty arises when separating the variables m and n in $e(2\theta\sqrt{mn}/c)$. To this end we applied to the Mellin transform inversion formula

$$M(s) = \int_0^\infty m(x) x^{s-1} dx, \quad m(x) = \frac{1}{2\pi i} \int_{(\sigma)} M(s) x^{-s} ds, \quad \sigma > 0.$$

Notice that in the range of the summation in $B_a(\theta, c, N)$ we have $N < \sqrt{mn} \leq 2N$. Therefore $B_a(\theta, c, N)$ remains unchanged if a smooth weight $\eta(\sqrt{mn}/N)$ is attached to each term provided that

$$\eta(x) = \begin{cases} 1 & \text{if } x \in (1, 2], \\ 0 & \text{if } x \notin (\frac{1}{2}, 3). \end{cases}$$

In what follows we demand $\eta(x)$ to be of $\mathcal{C}^\infty$ class. Then

$$\eta(x) e(2\theta c^{-1} x N) = \frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} M(s) x^{-s} ds$$

with

$$M(s) = \int_0^\infty \eta(x) e(2\theta c^{-1} x N) x^{it} dx \ll \begin{cases} (1+|t|)^{-\frac{1}{2}} \\ t^{-2} & \text{for } |t| > 16\pi\theta c^{-1} N. \end{cases}$$

Hence, by (5.1) and (2.14) we infer

$$\begin{aligned} B_a(\theta, c, N) &= \frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} M(s) \sum_{N < m, n \leq 2N} b_m \bar{b}_n \left(\frac{\sqrt{mn}}{N} \right)^{-s} S_{aa}(m, n; c) ds \\ &\ll (c + N + \sqrt{\theta c N}) \|b_N\|_2^2. \end{aligned}$$

Now we proceed to prove (1.27) for $c \leq N^{1-\varepsilon}$, the remaining range $N^{1-\varepsilon} < c \leq N$ being covered by (1.26). By the Cauchy-Schwarz inequality we get

$$\begin{aligned} |B_a(\theta, c, N)|^2 &\leq \|b_N\|_2^2 \sum_n \eta\left(\frac{n}{N}\right) \left| \sum_m b_m S_{aa}(m, n; c) e\left(\frac{2\sqrt{mn}}{c} \theta\right) \right|^2 \\ &= \|b_N\|_2^2 \sum_{\substack{m_1, m_2 \\ \delta_1, \delta_2}} b_{m_1} \bar{b}_{m_2} e\left(\frac{m_1 \delta_1 - m_2 \delta_2}{c}\right) \sum_n f(n) \end{aligned}$$

where

$$f(n) = \eta\left(\frac{n}{N}\right) e\left(\frac{\alpha_1 - \alpha_2}{c} n + \frac{2(\sqrt{m_1} - \sqrt{m_2})\theta}{c} \sqrt{n}\right) = \eta\left(\frac{n}{N}\right) e(A n + B \sqrt{n}),$$

say, and the pairs (α_1, δ_1) , (α_2, δ_2) run over the solutions of (2.15), (2.16). For the inner sum we apply the Poisson summation formula

$$\sum_{n \in \mathbf{Z}} f(n) = \sum_{u \in \mathbf{Z}} \hat{f}(u)$$

where – in this subsection – $\hat{f}(u)$ is the Fourier transform

$$\hat{f}(u) = \int \eta \left(\frac{t}{N} \right) e((A-u)t + B\sqrt{t}) dt.$$

If $u \neq A$ then $|A-u| \geq \frac{1}{c}$ while $|B|/2\sqrt{t} \leq (\sqrt{2}-1)\theta c^{-1} \leq 2(\sqrt{2}-1)c^{-1}$. Therefore by partial integration p -times we obtain

$$\begin{aligned} \hat{f}(u) &= \left(\frac{-1}{2\pi i} \right)^p \int \left(\dots \left(\eta \left(\frac{t}{N} \right) \frac{1}{A-u+B/2\sqrt{t}} \right)' \frac{1}{A-u+B/2\sqrt{t}} \right)' \dots \\ &\quad \cdot e((A-u)t + B\sqrt{t}) dt \\ &\ll N(|A-u|N)^{-p}. \end{aligned}$$

Hence on taking $p = [2\varepsilon^{-1}]$ we deduce that

$$\sum_{u \neq A} \hat{f}(u) \ll N \left(\frac{c}{N} \right)^p \ll c^{-1}.$$

It remains to consider the case $u=A$. Since u is an integer it implies $\alpha_1 \equiv \alpha_2 \pmod{c}$ and consequently $\delta_1 \equiv \delta_2 \pmod{c}$. In this case, $\hat{f}(A)$ is independent of δ_1 , so we can carry out the summation over δ_1 first, giving

$$\left| \sum_{\delta_1} e \left(\frac{m_1 - m_2}{c} \delta_1 \right) \right| = |S_{aa}(0, m_1 - m_2; c)| \leq (m_1 - m_2, c).$$

For $\hat{f}(A)$ we have, besides the trivial estimate $\hat{f}(A) \ll N$, also

$$\hat{f}(A) = \int \eta \left(\frac{t}{N} \right) e(B\sqrt{t}) dt \ll |B|^{-1} N^{\frac{1}{2}} \ll \frac{cN}{\theta|m_1 - m_2|}$$

by partial integration, provided $m_1 \neq m_2$. Gathering all the above results together we conclude that

$$\begin{aligned} |B_a(\theta; c, N)|^2 &\ll cN \|\mathbf{b}_N\|_2^4 + \theta^{-1} cN \sum_{m_1 \neq m_2} |b_{m_1} b_{m_2}| \frac{(m_1 - m_2, c)}{|m_1 - m_2|} \|\mathbf{b}_N\|_2^2 \\ &\ll \theta^{-1} cN^{1+\varepsilon} \|\mathbf{b}_N\|_2^4. \end{aligned}$$

This completes the proof of (1.27) and of Proposition 3. ■

5.2 Large Sieve Inequalities for Fourier Coefficients of Holomorphic Cusp Forms

Our primary aim is to prove (1.28), however it needs no extra effort to prove the following asymptotic formula

Proposition 4. For $K \geq 1$, $N \geq 1$, $\varepsilon > 0$ and any complex numbers a_n we have

$$\begin{aligned}
 (5.2) \quad & \sum_{\substack{k \equiv 0 \pmod{2} \\ k \geq 2}} e^{-(k-1)/K} \frac{(k-1)!}{(4\pi)^{k-1}} \sum_{1 \leq j \leq \theta_k(q)} \left| \sum_{N < n \leq 2N} a_n n^{-\frac{k-1}{2}} \psi_{jk}(\mathbf{a}, n) \right|^2 \\
 & = \left(\frac{1}{2} K^2 + O(\mu(\mathbf{a}) N^{1+\varepsilon}) \right) \|\mathbf{a}_N\|_2^2,
 \end{aligned}$$

the constant implied in the symbol O depending on ε alone.

Proof. We follow the arguments of [Iwa2] where the case of the full modular group is treated. We begin with the formula (4.4) for $\mathbf{a}=\mathbf{b}$ in which case the factor $e(bn)$ in front of the δ -terms is one. On multiplying both sides of (4.4) by

$$(k-1) e^{-(k-1)/K} \bar{a}_m a_n$$

then summing over $m, n \in (N, 2N]$ and $k \geq 2$, $k \equiv 0 \pmod{2}$ on one side we get exactly the expression from the left-hand side of (5.2) while the other side becomes

$$(5.3) \quad D_K \|\mathbf{a}_N\|_2^2 + \pi \sum_c \frac{1}{c} \sum_{m,n} \bar{a}_m a_n S_{\mathbf{a}\mathbf{a}}(n, m; c) E_K \left(\frac{4\pi \sqrt{mn}}{c} \right)$$

where

$$(5.4) \quad D_K = \sum_{l=1}^{\infty} (2l-1) e^{-(2l-1)/K} = \frac{1}{2} \left(\operatorname{ch} \frac{1}{K} \right) \left(\operatorname{sh} \frac{1}{K} \right)^{-2} = \frac{1}{2} K^2 + O(1)$$

and for $x = 4\pi \sqrt{mn}/c$

$$E_K(x) = 2 \sum_{l=1}^{\infty} (2l-1) (-1)^l e^{-(2l-1)/K} J_{2l-1}(x).$$

It remains to estimate the quadratic forms

$$F_{\mathbf{a}}(c, N) = \sum_{N < m, n \leq 2N} \bar{a}_m a_n S_{\mathbf{a}\mathbf{a}}(n, m; c) E_K \left(\frac{4\pi \sqrt{mn}}{c} \right).$$

We shall show that

$$(5.5) \quad F_{\mathbf{a}}(c, N) \ll c^{-\varepsilon} N^{1+5\varepsilon} \|\mathbf{a}_N\|_2^2.$$

To this end we borrow from [Iwa2] (see Lemma 6) the result

$$E_K(x) = \frac{1}{2} \left(\operatorname{sh} \frac{2}{K} \right) \int_0^1 \xi x J_0(\xi x) \left[\left(\operatorname{ch} \frac{1}{K} \right)^2 - \xi^2 \right]^{-3/2} d\xi$$

where J_0 is the Bessel function. Given Δ with $0 < \Delta \leq \frac{\pi}{2}$ we rearrange the Poisson integral as follows

$$\begin{aligned}
 \frac{\pi}{2} J_0(y) &= \int_0^{\frac{\pi}{2}} \cos(y \cos \theta) d\theta \\
 &= \int_0^{\Delta} \cos(y \cos \theta) d\theta + \frac{1}{y} \int_{\Delta}^{\frac{\pi}{2}} \frac{\cos \theta}{\sin^2 \theta} \sin(y \cos \theta) d\theta + \frac{\sin(y \cos \Delta)}{y \sin \Delta}.
 \end{aligned}$$

Accordingly we write $E_K(x) = E_{K_1}(x) - E_{K_2}(x) + E_{K_3}(x)$ where

$$\pi E_{K_1}(x) = \left(\operatorname{sh} \frac{2}{K} \right) \int_0^1 \int_0^\xi \xi x \left[\left(\operatorname{ch} \frac{1}{K} \right)^2 - \xi^2 \right]^{-3/2} \cos(\xi x \cos \theta) d\theta d\xi$$

$$\pi E_{K_2}(x) = \left(\operatorname{sh} \frac{2}{K} \right) \int_0^1 \int_\Delta \frac{\cos \theta}{\sin^2 \theta} \left[\left(\operatorname{ch} \frac{1}{K} \right)^2 - \xi^2 \right]^{-3/2} \sin(\xi x \cos \theta) d\theta d\xi$$

and

$$\pi E_{K_3}(x) = \left(\operatorname{sh} \frac{2}{K} \right) \int_0^1 \frac{1}{\sin \Delta} \left[\left(\operatorname{ch} \frac{1}{K} \right)^2 - \xi^2 \right]^{-3/2} \sin(\xi x \cos \Delta) d\xi.$$

By Proposition 2

$$\sum_{N < m, n \leq 2N} \bar{b}_m b_n S_{aa}(m, n; c) e \left(\frac{4\pi \sqrt{mn}}{c} \xi \cos \theta \right) \\ \ll (\xi \cos \theta)^{-\frac{1}{2}} (cN)^\varepsilon D(c, N) \|\mathbf{b}_N\|_2^2$$

where

$$D(c, N) = \begin{cases} c^{\frac{1}{2}} N & \text{if } c > N^2 \\ c & \text{if } N < c \leq N^2 \\ c^{\frac{1}{2}} N^{\frac{1}{2}} & \text{if } c < N. \end{cases}$$

Hence, decomposing $F_a(c, N) = F_1(\Delta, c, N) + F_2(\Delta, c, N) + F_3(\Delta, c, N)$ as we did it with $E_K(x)$, we deduce that

$$F_1(\Delta, c, N) \ll \Delta(cN)^\varepsilon \frac{N}{c} D(c, N) \|\mathbf{a}_N\|_2^2,$$

$$F_2(\Delta, c, N) \ll \Delta^{-1} (cN)^\varepsilon D(c, N) \|\mathbf{a}_N\|_2^2,$$

and

$$F_3(\Delta, c, N) \ll \Delta^{-1} (cN)^\varepsilon D(c, N) \|\mathbf{a}_N\|_2^2.$$

If $c < N$ we take $\Delta = \sqrt{c/N}$ getting

$$F(c, N) \ll \left(\Delta \frac{N}{c} + \Delta^{-1} \right) (cN)^\varepsilon c^{\frac{1}{2}} N^{\frac{1}{2}} \|\mathbf{a}_N\|_2^2 \\ \ll (cN)^\varepsilon N \|\mathbf{a}_N\|_2^2 \ll c^{-\varepsilon} N^{1+2\varepsilon} \|\mathbf{a}_N\|_2^2.$$

If $c \geq N$ we take $\Delta = \pi/2$ getting

$$F(c, N) = F_1 \left(\frac{\pi}{2}, c, N \right) \ll (cN)^\varepsilon \frac{N}{c} D(c, N) \|\mathbf{a}_N\|_2^2 \ll c^{-\varepsilon} N^{1+5\varepsilon} \|\mathbf{a}_N\|_2^2.$$

This completes the proof of (5.5). Combining (5.3), (5.4) and (5.5) we deduce that the *RHS* of (5.2) is equal to

$$\left\{ \frac{1}{2} K^2 + O \left(\sum_{c \equiv 0 \pmod{q/(w, q/w)}} c^{-1-\varepsilon} N^{1+5\varepsilon} \right) \right\} \|\mathbf{a}_N\|_2^2$$

which yields (5.2) after reinterpreting 5ε by ε . Of course (1.28) follows from (5.2). ■

5.3 Large Sieve Inequalities for Fourier Coefficients of Non-holomorphic Cusp Forms and Eisenstein Series

This section is devoted to the proof of Theorem 2.

We first make use of the summation formula (4.8) with $a=b$: we multiply both sides of (4.8) by

$$t \operatorname{sh} \pi t e^{-(t/K)^2} \bar{a}_m a_n$$

and then sum over $m, n \in (N, 2N]$ and integrate over $t \in (-\infty, \infty)$. Since

$$\int_{-\infty}^{\infty} t (\operatorname{sh} \pi t) H(r, t) e^{-(t/K)^2} dt \geq \int_{|r|+1}^{|r|+2} \gg (|r|+1) e^{-(r/K)^2}$$

we obtain

$$\begin{aligned} (5.5) \quad & \sum_j \frac{\kappa_j + 1}{\operatorname{ch} \pi \kappa_j} e^{-(\kappa_j/K)^2} \left| \sum_n a_n \rho_{ja}(n) \right|^2 \\ & + \sum_{\epsilon} \int_{-\infty}^{\infty} (|r|+1) e^{-(r/K)^2} \left| \sum_n a_n n^{ir} \varphi_{\epsilon a n} \left(\frac{1}{2} + ir \right) \right|^2 dr \\ & \ll K^3 \|\mathbf{a}_N\|_2^2 + \sum \frac{1}{c} \left| \sum_{m,n} \bar{a}_m a_n \frac{4\pi \sqrt{mn}}{c} S_{aa}(m, n; c) \Phi \left(\frac{4\pi \sqrt{mn}}{c} \right) \right| \end{aligned}$$

where for $x = 4\pi \sqrt{mn/c}$ we put

$$\Phi(x) = \int_{-\infty}^{\infty} t^2 e^{-(t/K)^2} \int_{-i}^i K_{2it}(xv) \frac{dv}{v}.$$

For $\operatorname{Re} y > 0$ we have (see [EMOT], p. 82)

$$\begin{aligned} K_{2it}(y) &= \int_0^{\infty} e^{-y \operatorname{ch} \xi} \cos(2t \xi) d\xi \\ &= \frac{y}{2t} \int_0^{\infty} e^{-y \operatorname{ch} \xi} \operatorname{sh} \xi \sin(2t \xi) d\xi \end{aligned}$$

the latter following by partial integration. Moreover we have (see [G-R], p. 214)

$$\int_{-\infty}^{\infty} t e^{-(t/K)^2} \sin(2t \xi) dt = \sqrt{\pi} K^3 \xi e^{-(\xi K)^2}.$$

Hence

$$(5.6) \quad \Phi(x) = \sqrt{\pi} i K^3 \int_0^{\infty} e^{-(\xi K)^2} \xi \operatorname{th} \xi \sin(x \operatorname{ch} \xi) d\xi.$$

On integrating by parts we obtain another formula

$$(5.7) \quad x \Phi(x) = \sqrt{\pi} i K^3 \int_0^{\infty} e^{-(\xi K)^2} (1 - \xi \operatorname{th} \xi - 2\xi^2 K^2) \cos(x \operatorname{ch} \xi) \frac{d\xi}{\operatorname{ch} \xi}.$$

When treated trivially (5.6) yields $\Phi(x) \ll 1$ while (5.7) yields $\Phi(x) \ll x^{-1} K^2$. Therefore it is optimal to use (5.6) if $c > K^{-2} N$ and (5.7) if $c \leq K^{-2} N$. Denote

$$\Phi(c, K, N) = \sum_{m, n} \bar{a}_m a_n \frac{4\pi \sqrt{mn}}{c} S_{aa}(m, n; c) \Phi\left(\frac{4\pi \sqrt{mn}}{c}\right).$$

If $c > N^2$ then by (5.6) and (1.25) we get

$$\Phi(c, K, N) \ll \tau^2(c) c^{-\frac{1}{2}} N^2 \|\mathbf{a}_N\|_2^2.$$

If $N < c \leq N^2$ then by (5.6) and (1.26) we get

$$\Phi(c, K, N) \ll N \|\mathbf{a}_N\|_2^2.$$

If $K^{-2}N < c \leq N$ then by (5.6) and (1.26) if $\xi \leq 1$ and (1.27) if $\xi \leq 1$ we get

$$\Phi(c, K, N) \ll (K e^{-K^2} c^{-1} N^2 + c^{\frac{1}{2}} N^{(3/2)+\varepsilon}) \|\mathbf{a}_N\|_2^2.$$

If $c \leq K^{-2}N$ then by (5.7) and (1.26) if $\xi \geq 1$ and (1.27) if $\xi \leq 1$ we get

$$\Phi(c, K, N) \ll (K e^{-K^2} c^{-1} N^2 + K^2 c^{\frac{1}{2}} N^{\frac{1}{2}+\varepsilon}) \|\mathbf{a}_N\|_2^2.$$

Gathering the above estimates together we obtain

$$\sum_{c \equiv 0 \pmod{q/(w, q/w)}} c^{-1} |\Phi(c, K, N)| \ll \mu(\mathfrak{a}) KN(N^\varepsilon + \mu(\mathfrak{a}) NK e^{-K^2}) \|\mathbf{a}_N\|_2^2.$$

Hence by (5.5)

$$\begin{aligned} & \sum_{|\kappa_j| \leq K} \frac{|\kappa_j| + 1}{\text{ch } \pi \kappa_j} \left| \sum_n a_n \rho_{ja}(n) \right|^2 + \sum_c \int_{-K}^K (|r| + 1) \left| \sum_n a_n n^{ir} \varphi_{ca n} \left(\frac{1}{2} + ir \right) \right|^2 dr \\ & \ll K(K^2 + \mu(\mathfrak{a}) N^{1+\varepsilon} + (\mu(\mathfrak{a}) N)^2 K e^{-K^2}) \|\mathbf{a}_N\|_2^2. \end{aligned}$$

Since the left-hand side is nondecreasing in K we may replace K in the right-hand side by any number $K_1 > K$. On putting $K_1 = K + (\mu(\mathfrak{a}) N)^\varepsilon$ it yields

$$\ll K(K^2 + \mu(\mathfrak{a}) N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2.$$

Hence by partial summation the left-hand sides of (1.29) and (1.30) are both

$$\begin{aligned} & \ll (K^2 + \mu(\mathfrak{a}) N^{1+\varepsilon} \log 2K) \|\mathbf{a}_N\|_2^2 \\ & \ll (K^2 + \mu(\mathfrak{a}) N^{1+2\varepsilon}) \|\mathbf{a}_N\|_2^2. \end{aligned}$$

This completes the proof of Theorem 2.

6. Exceptional Eigenvalues

We shall prove in this section Theorems 3 and 4.

6.1 The Case of Modular Group

Here we deal with $\Gamma = SL(2, \mathbb{Z})$ and we actually prove

$$\lambda_1 \geq \frac{3\pi^2}{2}$$

which is much more than what Theorem 3 asserts. Somewhat surprisingly, this has no advantage in practice. For numerical experiments about λ_1 and other points from the spectrum see D.A. Hejhal [Hej].

For the purpose of the proof we take as a fundamental domain $\mathcal{D}$ in H for the action of Γ the normal polygon

$$\mathcal{D} = \{z = x + iy; -\frac{1}{2} \leq x \leq \frac{1}{2}, |z| \geq 1\}.$$

Let $\tau\mathcal{D}$ be the image of $\mathcal{D}$ under the transform $\tau z = -1/z$ from $PSL(2, \mathbb{Z})$, so $\mathcal{D} \cup \tau\mathcal{D}$ covers completely the rectangle

$$I = \{z = x + iy; -\frac{1}{2} \leq x \leq \frac{1}{2}, y \geq y_0\}$$

where $y_0 = \sqrt{3}/2$. Let $u(z)$ be a cusp form such that

$$\int_{\mathcal{D}} |u|^2 dz = 1, \quad \Delta u = \lambda u, \quad \lambda > 0.$$

By the periodic boundary conditions

$$\begin{aligned} 2\lambda &= \int_{\mathcal{D} \cup \tau\mathcal{D}} \bar{u} \Delta u dz = \int_{\mathcal{D} \cup \tau\mathcal{D}} (u_x^2 + u_y^2) dx dy \\ &\geq \int_I (u_x^2 + u_y^2) dx dy \geq \int_I u_x^2 dx dy. \end{aligned}$$

Further, from the Fourier expansion

$$u(x + iy) = \sum_{n \neq 0} a_n(y) e(n x)$$

it follows that

$$\begin{aligned} \int_I u_x^2 dx dy &= \int_{y_0}^{\infty} \sum_{n \neq 0} (2\pi n a_n(y))^2 dy \\ &\geq (2\pi y_0)^2 \int_{y_0}^{\infty} \sum_{n \neq 0} a_n^2(y) y^{-2} dy \\ &= (2\pi y_0)^2 \int_I |u|^2 dz \geq (2\pi y_0)^2 \int_{\mathcal{D}} |u|^2 dz = 3\pi^2 \end{aligned}$$

whence $\lambda \geq \frac{3}{2}\pi^2$.

6.2 Selberg's Lower Bound

The ideas behind the above proof completely fail for $\Gamma_0(q)$ with large q . Here we use Kuznetsov's formula (1.19) and A. Weil estimate (Lemma 2.6) to show that $\lambda_1 \geq 3/16$.

Let $\varphi(x)$ be a function of $\mathcal{C}^3$ class whose graph is

$$\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \quad , \quad Y > 10,$$

$\frac{1}{Y} \qquad \qquad \qquad 2/Y$

and whose derivatives satisfy $\varphi^{(l)}(x) \ll Y^l$, $l=0, 1, 2, 3$. By Kuznetsov's formula (1.19) for $m=n$, $a=b=\infty$ we get

$$\begin{aligned} & \sum_{c \equiv 0 \pmod{q}} \frac{1}{c} S(n, n; c) \varphi(4\pi n/c) \\ &= \sum_{j \geq 1} \frac{\hat{\varphi}(\kappa_j)}{\operatorname{ch} \pi \kappa_j} |\rho_{j\infty}(n)|^2 + \frac{1}{\pi} \sum_c \int_{-\infty}^{\infty} \hat{\varphi}(t) |\varphi_{c\infty n}(\tfrac{1}{2} + it)|^2 dt \\ & \quad + \frac{1}{2\pi} \sum_{k \equiv 0 \pmod{2}} \frac{(k-1)!}{(4\pi n)^{k-1}} \tilde{\varphi}(k-1) \sum_{1 \leq j \leq \theta_k(q)} |\psi_{jk}(\infty, n)|^2. \end{aligned}$$

By (2.18) the *LHS* is majorized by

$$\sum_{\substack{c < 4\pi n Y \\ c \equiv 0 \pmod{q}}} c^{-\frac{1}{2}} (n, c)^{\frac{1}{2}} \tau(c) \ll (nY)^{\frac{1}{2}} \log nY.$$

In the *RHS* we have the transforms $\hat{\varphi}(r)$ and $\tilde{\varphi}(r)$ which by Lemma 7.1 satisfy

$$\hat{\varphi}(r), \tilde{\varphi}(r) \ll (r^2 + 1)^{-1}$$

for all real r . Therefore all the series and the integrals except the finite sum with $\lambda_j < \frac{1}{4}$ converge to constants depending at most on n and q ; to see this, look at the large sieve inequalities from Theorem 2. In the remaining terms, i.e. in those with exceptional λ_j 's, we have

$$\hat{\varphi}(\kappa_j) = \frac{\pi i}{2 \operatorname{sh} \pi \kappa_j} \int_0^{\infty} (J_{2i\kappa_j}(x) - J_{-2i\kappa_j}(x)) \varphi(x) \frac{dx}{x}$$

with $0 < i\kappa_j < \frac{1}{2}$. From the power series expansion

$$J_\nu(x) = \sum_{l=0}^{\infty} \frac{(-1)^l}{l! \Gamma(l+1+\nu)} \left(\frac{x}{2}\right)^{2l+\nu}, \quad Y^{-1} < x < 2Y^{-1}$$

one can easily infer that $\hat{\varphi}(\kappa_j) > 0$ and even that

$$\hat{\varphi}(\kappa_j) \gg Y^{2i\kappa_j} \quad \text{as } Y \rightarrow \infty.$$

Combining all the above observations we conclude

$$\sum_{\lambda_j \text{ except}} Y^{2i\kappa_j} |\rho_{j\infty}(n)|^2 \ll Y^{\frac{1}{2}} \log Y,$$

the constant implied in $\ll$ depending on n and q at most. Taking n such that $\rho_{j\infty}(n) \neq 0$ and letting Y tend to infinity this implies $i\kappa_j \leq \frac{1}{4}$ which is equivalent to $\lambda_j \geq 3/16$.

7. Upper Bounds for Sums of Kloosterman Sums I

This paragraph is devoted to bounds for sums of Kloosterman sums connected with a single given group $\Gamma_0(q)$.

7.1 A General Estimate

In order to prove Theorem 8, we shall apply Theorem 1 and so we need some estimates for the Bessel-Kuznietsov transforms defined in (1.21)–(1.23)

Lemma 7.1. *If $f(x)$ satisfies (1.43), (1.44) then for real r , one has*

$$(7.1) \quad \check{f}(ir), \hat{f}(ir) \ll \frac{1 + X^{-2r}}{1 + X} \quad \text{for } 0 < r < \frac{1}{2}$$

$$(7.2) \quad \check{f}(r), \hat{f}(r), \tilde{f}(r) \ll \frac{1 + |\log X|}{1 + X} \quad \text{for all } r$$

$$(7.3) \quad \check{f}(r), \hat{f}(r), \tilde{f}(r) \ll |r|^{-3/2} + r^{-2} X \quad \text{for } |r| \geq 1$$

$$(7.4) \quad \check{f}(r), \hat{f}(r), \tilde{f}(r) \ll (|r|^{-3/2} + r^{-2} X) |r|^{-1} Y \quad \text{for } |r| \geq 1$$

the constants implied in $\ll$ being absolute.

Proof. Let us first prove (7.2). In case of $\check{f}(r)$ we appeal to the formula

$$K_{2ir}(x) = \int_0^\infty e^{-x \operatorname{ch} \xi} \cos(2r \xi) d\xi$$

(see [EMOT], p. 82). Integrating by parts we obtain

$$\int_0^\infty e^{-x \operatorname{ch} \xi} \frac{f(x)}{x} dx = \frac{1}{\operatorname{ch} \xi} \int_0^\infty \left(\frac{f(x)}{x} \right)' e^{-x \operatorname{ch} \xi} dx \ll \min(1, X^{-1} e^{-\xi}).$$

Therefore

$$\begin{aligned} \check{f}(r) &= \int_0^\infty \cos(2r \xi) \int_0^\infty e^{-x \operatorname{ch} \xi} \frac{f(x)}{x} dx d\xi \\ &\ll \int_0^\infty \min(1, X^{-1} e^{-\xi}) d\xi \ll \frac{1 + |\log X|}{1 + X}. \end{aligned}$$

In case of $\hat{f}(r)$ we appeal to the Mehler-Sonine formula

$$J_\nu(x) = \frac{2}{\pi} \int_0^\infty \sin \left(x \operatorname{ch} \xi - \frac{\pi}{2} \nu \right) \operatorname{ch}(\nu \xi) d\xi$$

(see [EMOT], p. 82) which yields

$$\frac{J_{2ir}(x) - J_{-2ir}(x)}{\operatorname{sh} \pi r} = \frac{4i}{\pi} \int_0^\infty \cos(x \operatorname{ch} \xi) \cos(2r \xi) d\xi.$$

On the other hand integrating by parts we obtain

$$\int_0^{\infty} \cos(x \operatorname{ch} \xi) \frac{f(x)}{x} dx = \int_0^{\infty} \left(\frac{f(x)}{x} \right)' \frac{\sin(x \operatorname{ch} \xi)}{\operatorname{ch} \xi} dx \ll \min(1, X^{-1} e^{-\xi}).$$

Therefore

$$\begin{aligned} \hat{f}(r) &= 2 \int_0^{\infty} \cos(2r\xi) \int_0^{\infty} \cos(x \operatorname{ch} \xi) \frac{f(x)}{x} dx d\xi \\ &\ll \int_0^{\infty} \min(1, X^{-1} e^{-\xi}) \ll \frac{1 + |\log X|}{1 + X}. \end{aligned}$$

In case of $\tilde{f}(r)$, r positive integer, we appeal to the Bessel representation

$$J_r(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \xi - r\xi)} d\xi$$

(see [EMOT], p. 14). Integrating by parts we obtain

$$\int_0^{\infty} e^{ix \sin \xi} \frac{f(x)}{x} dx = \frac{i}{\sin \xi} \int_0^{\infty} e^{ix \sin \xi} \left(\frac{f(x)}{x} \right)' dx \ll \min(1, X^{-1} |\sin \xi|^{-1}).$$

Therefore

$$\begin{aligned} \tilde{f}(r) &= \frac{1}{2\pi} \int_0^{2\pi} e^{-ir\xi} \int_0^{\infty} e^{ix \sin \xi} \frac{f(x)}{x} dx d\xi \\ &\ll \int_0^{2\pi} \min(1, X^{-1} |\sin \xi|^{-1}) d\xi \ll \frac{1 + |\log X|}{1 + X}. \end{aligned}$$

The proof of (7.1) follows the same line, except that the upper bound $\cos(2r\xi) \leq 1$ is replaced by $\cos(2ir\xi) \ll e^{2ir\xi}$ and so $\tilde{f}(ir)$ – as well as $\hat{f}(ir)$ – is $\ll \int_0^{\infty} e^{2r\xi} \min(1, X^{-1} e^{-\xi}) d\xi \ll \frac{1 + X^{-2r}}{1 + X}$.

Now we proceed to prove (7.3) and (7.4). In case of $\check{f}(r)$ we appeal to

$$\int_0^{\infty} K_\nu(x) x^{s-1} dx = 2^{s-1} \Gamma\left(\frac{s+\nu}{2}\right) \Gamma\left(\frac{s-\nu}{2}\right), \quad \operatorname{Re}(s \pm \nu) > 0$$

(see [EMOT], p. 51) which, by the Mellin's inversion formula, yields

$$K_{2ir}(x) = \frac{1}{2\pi i} \int_{(\sigma)} 2^{s-1} x^{-s} \Gamma\left(\frac{s}{2} + ir\right) \Gamma\left(\frac{s}{2} - ir\right) ds.$$

On moving the line of integration to $\sigma = -1$ we meet poles at $s = \pm 2ir$ with residua

$$2^{-1 \pm 2ir} x^{\mp 2ir} \Gamma(\pm 2ir).$$

Moreover, integrating one time by parts we obtain

$$(7.5) \quad \int f(x) x^{-s-1} dx = -\frac{1}{s} \int f'(x) x^{-s} dx.$$

Therefore by Stirling's formula

$$\begin{aligned} \check{f}(r) &= -2^{-1+2ir} \Gamma(2ir) \frac{\text{ch } \pi r}{2ir} \int f'(x) x^{-2ir} dx \\ &\quad - 2^{-1-2ir} \Gamma(-2ir) \frac{\text{ch } \pi r}{2ir} \int f'(x) x^{2ir} dx \\ &\quad - \frac{\text{ch } \pi r}{2\pi i} \int f'(x) \int_{(-1)} 2^{s-2} x^{-s} \Gamma\left(\frac{s}{2} + ir\right) \Gamma\left(\frac{s}{2} - ir\right) \frac{ds}{s} dx \\ &\ll |r|^{-3/2} + r^{-2} X \end{aligned}$$

which is (7.3). Integrating another time by parts in (7.5) we may then derive (7.4) for $\check{f}(r)$ in a similar way.

The proofs of (7.3) and (7.4) for $\hat{f}(r)$ are much similar. In this case we appeal to the Mellin-Barnes integral

$$J_{2ir}(x) = \frac{1}{4\pi i} \int_{(\sigma)} \left(\frac{x}{2}\right)^{-s} \frac{\Gamma\left(ir + \frac{s}{2}\right)}{\Gamma\left(1 + ir - \frac{s}{2}\right)} ds, \quad 0 < \sigma < 1$$

(see [EMOT], p. 21). On moving the line of integration to $\sigma = -1$ we meet a pole at $s = -2ir$ with residuum

$$\left(\frac{x}{2}\right)^{2ir} \Gamma(1 + 2ir)^{-1}.$$

Therefore, by (7.5) and Stirling's formula

$$\begin{aligned} \hat{f}(r) &= -2^{-2ir} (\text{sh } \pi r)^{-1} \Gamma(1 + 2ir)^{-1} (2ir)^{-1} \int f'(x) x^{2ir} dx \\ &\quad - 2^{2ir} (\text{sh } \pi r)^{-1} \Gamma(1 - 2ir)^{-1} (2ir)^{-1} \int f'(x) x^{-2ir} dx \\ &\quad + \frac{1}{4\pi i} (\text{sh } \pi r)^{-1} \int f'(x) \int_{(-1)} \left(\frac{x}{2}\right)^{-s} \frac{\Gamma\left(ir + \frac{s}{2}\right)}{\Gamma\left(1 + ir - \frac{s}{2}\right)} \frac{ds}{s} dx \\ &\quad - \frac{1}{4\pi i} (\text{sh } \pi r)^{-1} \int f'(x) \int_{(-1)} \left(\frac{x}{2}\right)^{-s} \frac{\Gamma\left(-ir + \frac{s}{2}\right)}{\Gamma\left(1 - ir - \frac{s}{2}\right)} \frac{ds}{s} dx \\ &\ll |r|^{-3/2} + r^{-2} X \end{aligned}$$

which is (7.3) for $\hat{f}(r)$. Integrating another time by parts with respect to x we deduce (7.4).

Finally to prove (7.3) and (7.4) for $\tilde{f}(r)$ we appeal to the Mellin-Barnes formula again giving

$$J_r(x) = \frac{1}{4\pi i} \int_{(-1)} \left(\frac{x}{2}\right)^{-s} \frac{\Gamma\left(\frac{r+s}{2}\right)}{\Gamma\left(1+\frac{r-s}{2}\right)} ds, \quad r \geq 1.$$

Therefore, by (1.10) we obtain

$$\begin{aligned} \tilde{f}(r) &= \frac{1}{\pi} \int f'(x) \int_{-\infty}^{\infty} \left(\frac{x}{2}\right)^{-1+it} \frac{\Gamma(\frac{1}{2}(r+3+it))}{\Gamma(\frac{1}{2}(r+3-it))} \\ &\quad \cdot \frac{dt}{(r-1+it)(r+1+it)(1-it)} \ll r^{-2} X \end{aligned}$$

which is slightly stronger than (7.3). On integrating by parts with respect to x we deduce

$$\tilde{f}(r) \ll r^{-3} XY.$$

This completes the proof of Lemma 7.1. ■

Now, we are ready to prove Theorem 8. Let us first consider the case when both coefficients m, n in the Kloosterman sum have the same sign. By Theorem 1 we obtain

$$\begin{aligned} S_{ab}(\mathbf{a}_M, \mathbf{b}_N; f) &\leq \frac{1}{2\pi} \sum_{k \equiv 0 \pmod{2}} |\tilde{f}(k-1)| \frac{(k-1)!}{(4\pi)^{k-1}} \sum_{1 \leq j \leq \theta_k(q)} \\ &\quad \cdot |\sum_m \bar{a}_m m^{-\frac{k-1}{2}} \psi_{jk}(\mathbf{a}, m)| |\sum_n b_n n^{-\frac{k-1}{2}} \psi_{jk}(\mathbf{b}, n)| \\ &\quad + \sum_{j \geq 1} \frac{|\hat{f}(\kappa_j)|}{\operatorname{ch} \pi \kappa_j} |\sum_m \bar{a}_m \rho_{ja}(m)| |\sum_n b_n \rho_{jb}(n)| \\ &\quad + \frac{1}{\pi} \sum_c \int_{-\infty}^{\infty} |\hat{f}(r)| |\sum_m \bar{a}_m m^{-ir} \varphi_{cam}(\frac{1}{2} + ir)| \\ &\quad \cdot |\sum_n b_n n^{-ir} \varphi_{cbn}(\frac{1}{2} + ir)| dr. \end{aligned}$$

Take two numbers $R_2 \geq R_1 \geq 1$ and replace the transforms $\tilde{f}, \hat{f}$ by the upper bounds (7.2) if $|r| \leq R_1$, (7.3) if $R_1 < |r| \leq R_2$ and (7.4) if $R_2 < |r|$. Of course $R_1 = R_2$ means that (7.3) is not applied in this case. Then by Cauchy-Schwarz and the large sieve inequalities (cf. Theorem 2) we get

$$S_{ab}(\mathbf{a}_M, \mathbf{b}_N; f) \ll (D_1 + D_{12} + D_2) \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2$$

where

$$D_1 = \frac{1 + |\log X| + X^{-\theta_q}}{1 + X} (R_1^2 + \mu(\mathbf{a}) M^{1+\varepsilon})^{\frac{1}{2}} (R_1^2 + \mu(\mathbf{b}) N^{1+\varepsilon})^{\frac{1}{2}}$$

$$D_{12} = \frac{1 + X}{R_1^2} (R_1^2 + \mu(\mathbf{a}) M^{1+\varepsilon})^{\frac{1}{2}} (R_1^2 + \mu(\mathbf{b}) N^{1+\varepsilon})^{\frac{1}{2}} + R_2^{\frac{1}{2}}$$

and

$$D_2 = Y(1 + R_2^{-\frac{1}{2}} X) R_2^{-5/2} (R_2^2 + \mu(a) M^{1+\varepsilon})^{\frac{1}{2}} (R_2^2 + \mu(b) N^{1+\varepsilon})^{\frac{1}{2}}.$$

On taking $R_1 = X + 1$ and $R_2 = \max(X, Y) + 1$ it yields (1.46); when $Y \leq X$ we omitted the middle term D_{12} because $R_1 = R_2$.

The case when the coefficients of Kloosterman sums have different signs is similar and simpler because the Fourier coefficients of holomorphic cusp forms do not occur. With this remark, we conclude the proof of Theorem 8. ■

7.2 Linnik-Selberg Conjecture on Average and other Particular Cases

Now it is a matter of simple calculation to derive Corollary from Theorem 8. First, by a usual device we restrict the range of summation over c in (1.47) into intervals of the type $(\frac{1}{2} C \sqrt{mn/MN}, C \sqrt{mn/MN}]$. Letting $X = 4\pi \sqrt{MN}/C$ we reinterpret the latter range for c by

$$2X \leq \frac{4\pi \sqrt{mn}}{c} < 4X.$$

Next we assume that $C > \sqrt{MN}$ because otherwise the result follows from the large sieve inequality. Let $2 \leq Y \leq C$ and $f(\xi)$ be a function of $\mathcal{C}^2$ class such that

$$\begin{aligned} f(\xi) &= 1 & \text{if } \xi \in [2X, 4X] \\ f(\xi) &= 0 & \text{if } \xi \leq 2X \left(1 - \frac{1}{Y}\right) \quad \text{or } \xi \geq 4X \left(1 + \frac{1}{Y}\right) \\ |f^{(p)}(\xi)| &\leq 8 \left(\frac{Y}{X}\right)^p & \text{if } \xi \in I(X, Y) \quad \text{for } p = 0, 1, 2 \end{aligned}$$

where $I(X, Y) = (2X(1 - Y^{-1}), 2X) \cup (4X, 4X(1 + Y^{-1}))$. For c with $4\pi \sqrt{mn}/c$ in $I(X, Y)$, we apply Weil's result, obtaining

$$\sum_m \sum_n \sum_{4\pi \sqrt{mn}/c \in I(X, Y)} |a_m b_n| \frac{1}{c} S(m, \pm n, c) \ll C^\varepsilon \sqrt{CMN} Y^{-1} \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2.$$

For the complementary sum we have

$$\sum_m \sum_n \sum_c |a_m b_n| \frac{1}{c} S(m, \pm n, c) f\left(\frac{4\pi \sqrt{mn}}{c}\right) \ll (\sqrt{Y} + C^\varepsilon \sqrt{MN}) \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2$$

by applying Theorem 8 with $q = 1$ (check the conditions (1.43) and (1.44)).

The optimal value for Y is $(CMN)^\frac{1}{2}$ which yields the right-hand side of (1.47).

Another consequence of Theorem 8 is the following

Theorem 13. Let $C, M, N \geq 1$ and let $g(m, n, c)$ be a function of $\mathcal{C}^6$ class such that

$$(7.6) \quad \text{Supp } g(m, n, c) \subseteq [M, 2M] \times [N, 2N] \times [C, 2C]$$

$$(7.7) \quad \left| \frac{\partial^{p_1+p_2+p_3}}{(\partial m)^{p_1}(\partial n)^{p_2}(\partial c)^{p_3}} g(m, n, c) \right| \leq M^{-p_1} N^{-p_2} C^{-p_3},$$

$$0 \leq p_1, p_2, p_3 \leq 2.$$

Then one has, for any positive ε

$$\sum_c \frac{1}{c} \sum_{M < m \leq 2M} a_m \sum_{N < n \leq 2N} b_n g(m, n, c) S_{ab}(m, \pm n; c) \ll D(C, M, N) \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2$$

with

$$D(C, M, N) = (CMN)^\varepsilon \frac{(C + \sqrt{MN} + C\sqrt{\mu(a)M})(C + \sqrt{MN} + C\sqrt{\mu(b)N})}{C(C + \sqrt{MN})} \left(1 + \frac{C}{\sqrt{MN}}\right)^{\theta_q}$$

We begin the proof with separating the variables m, n, c in $g(m, n, c)$ by means of Fourier integral

$$g(x_1, x_2, 4\pi\sqrt{x_1 x_2}/x) = \iint G(t_1, t_2; x) e(t_1 x_1 + t_2 x_2) dt_1 dt_2$$

where, by the inversion formula

$$G(t_1, t_2; x) = \iint g(x_1, x_2, 4\pi\sqrt{x_1 x_2}/x) e(-t_1 x_1 - t_2 x_2) dx_1 dx_2.$$

For $t_1 t_2 \neq 0$, integrating by parts p_1 times with respect to x_1 and p_2 times with respect to x_2 and then differentiating p times with respect to x yield

$$\begin{aligned} \frac{\partial^p}{\partial x^p} G(t_1, t_2; x) &= (2\pi i t_1)^{-p_1} (2\pi i t_2)^{-p_2} \iint \frac{\partial^{p_1+p_2+p}}{\partial x_1^{p_1} \partial x_2^{p_2} \partial x^p} \\ &\quad \cdot g\left(x_1, x_2, \frac{4\pi\sqrt{x_1 x_2}}{x}\right) e(-t_1 x_1 - t_2 x_2) dx_1 dx_2 \\ &\ll (t_1 M)^{-p_1} (t_2 N)^{-p_2} (\sqrt{MN}/C)^{-p} MN. \end{aligned}$$

Therefore for ε a sufficiently small positive constant the function

$$f(x) = \varepsilon(t_1 M)^{p_1} (t_2 N)^{p_2} (MN)^{-1} G(t_1, t_2; x)$$

satisfies (1.43), (1.44) with $X = 2\pi\sqrt{MN}/C$ and $Y = 1$. Theorem 8 gives

$$(7.8) \quad \sum_c \frac{1}{c} \sum_m \sum_n a_m b_n e(t_1 m + t_2 n) G\left(t_1, t_2; \frac{4\pi\sqrt{mn}}{c}\right) S_{ab}(m, \pm n; c) \\ \ll MN(t_1 M)^{-p_1} (t_2 N)^{-p_2} D(C; M, N) \|\mathbf{a}_M\|_2 \|\mathbf{b}_N\|_2.$$

Take $p_1 = 0$ if $|t_1|M \leq 1$, $p_2 = 0$ if $|t_2|N \leq 1$ and $p_1 = p_2 = 2$ otherwise with the effect that

$$\iint MN(t_1 M)^{-p_1} (t_2 N)^{-p_2} dt_1 dt_2 = 16.$$

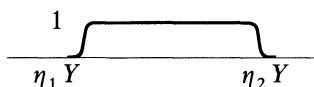
On integrating both sides of (7.8) with respect to t_1 and t_2 we complete the proof of Theorem 13.

By Theorem 13 for the group $\Gamma = \Gamma_0(q)$, $q = rs$, $(r, s) = 1$ and the cusps $\mathfrak{a} = \infty$, $\mathfrak{b} = 1/s$ and by (1.6), we obtain (1.51) after a simple reinterpretation of the variable c and the coefficient (b_n) .

8. Weighted Large Sieve Inequalities

We had already an occasion in the former section to see how the exceptional eigenvalues weaken the estimates for sums of Kloosterman sums. Here, we turn this defect into an extra force for strengthening the large sieve inequalities.

Let us call, for the use in this section, a test function $\varphi(x)$ any function which is of $\mathcal{C}^\infty$ class, whose graph is



and whose derivatives satisfy $\varphi^{(l)}(x) \ll x^{-l}$, the constant implied in $\ll$ depending on l and η_1, η_2 at most. If $Y < 1$, the transform $\hat{\varphi}$ behaves much like the Mellin transform, precisely we have

$$(8.1) \quad \hat{\varphi}(\kappa) = \frac{\pi i}{2 \operatorname{sh} \pi \kappa} \int \left(\frac{(x/2)^{2i\kappa}}{\Gamma(1+2i\pi)} - \frac{(x/2)^{-2i\kappa}}{\Gamma(1-2i\kappa)} \right) \varphi(x) \frac{dx}{x} + O((|\kappa|^2 + 1)^{-1})$$

by the power series expansion for J (cf. [EMOT]) and partial integration. Here the error term is good enough to bring results as sharp as that which follow from the conjecture $\theta_q = 0$. The difference between the cases of real and purely imaginary κ 's is obvious, since in the first one we have

$$(8.2) \quad \hat{\varphi}(\kappa) \ll (\kappa^2 + 1)^{-1}$$

while in the latter ($i\kappa > 0$)

$$(8.3) \quad \hat{\varphi}(\kappa) \gg Y^{-2i\kappa}.$$

8.1 The Case of Fixed Level

We wish to prove Theorem 5. Let $q \geq 1$, $\Gamma = \Gamma_0(q)$ and $\mathfrak{a}$ be a cusp of Γ . We first observe that (1.38) follows from (1.29) and Theorem 4 if $\mu(\mathfrak{a})N$ is bigger than a constant. Thus we assume that $Y := 8\pi\mu(\mathfrak{a})N$ is sufficiently small. Let $\varphi(x)$ be a test function with support in $[Y, 2Y]$. Apply the Kuznetsov formula (1.19) for $\mathfrak{a} = \mathfrak{b}$, multiply its both sides by $\bar{a}_m a_n$ and sum over m, n from $[N, 2N]$. The resulting sum of Kloosterman sums $S_{\mathfrak{a}\mathfrak{a}}(m, n; \gamma)$ with respect to γ is void because for some positive integer d

$$\gamma = d/\mu(\mathfrak{a}) \geq 1/\mu(\mathfrak{a}),$$

so the argument of $\varphi(4\pi\sqrt{mn}/\gamma)$ lies beyond the support of $\varphi(x)$. The other side becomes

$$\sum_{\lambda_j \text{ except.}} \left| \frac{\hat{\varphi}(\kappa_j)}{\text{ch } \pi \kappa_j} \right| \left| \sum_{N < n \leq 2N} a_n \rho_{ja}(n) \right|^2 + O((1 + \mu(a) N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2)$$

the error term coming from estimates for bilinear forms in Fourier coefficients of holomorphic cusp forms, Eisenstein series and non-holomorphic cusp forms with $\lambda_j \geq \frac{1}{4}$, each obtained by means of Theorem 2.

By (8.3) this yields

$$\sum_{\lambda_j \text{ except.}} Y^{-2i\kappa_j} \left| \sum_{N < n \leq 2N} a_n \rho_{ja}(n) \right|^2 \ll (1 + \mu(a) N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2.$$

Letting $L(X)$ be the expression from the LHS of (1.38) we deduce by Theorem 4 that

$$L(X) \leq (1 + \sqrt{XY}) L(Y) \ll (1 + \sqrt{\mu(a)NX}) (1 + \mu(a) N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2$$

which, together with (1.29), completes the proof of (1.38). ■

8.2 Results on Average

Our aim is to prove Theorem 6. The crucial point in what follows is the use of test functions $\varphi(x)$ with smaller support, so the sums of Kloosterman sums $S(m, n; q; c)$ with respect to c will not be void anymore. There is no way to treat such sums by the Kuznetsov's trace formula for the group $\Gamma_0(q)$ since it must lead to the same expression we started from. However, if we sum over q first, there is another possibility to apply the trace formula for the group $\Gamma_0(c)$ that permits to break the vicious circle. What really happens can be expressed by a kind of recurrence relation for

$$(8.4) \quad S(Q, Y, N; s) = \sum_{Q < q \leq 16Q} \sum_{\lambda_j \text{ except.}}^{(q)} Y^{2i\kappa_j} \left| \sum_{N < n \leq 2N} a_n n^s \rho_{j\infty}(n) \right|^2.$$

Lemma 8.1. *Let $Q, N, Y \geq 1$ and $\mathbf{a}$ be a complex sequence. For any $\varepsilon > 0$ there exists a constant $c(\varepsilon)$ such that*

$$(8.5) \quad S(Q, Y, N, 0) \leq c(\varepsilon) \int_{-\infty}^{\infty} S(\pi NYQ^{-1}, Y, N, it) \frac{dt}{t^4 + 1} + c(\varepsilon) (YN)^\varepsilon (Q + N + NYQ^{-1}) \|\mathbf{a}_N\|_2^2.$$

Proof: Let $\varphi(x)$ be a test function whose graph is

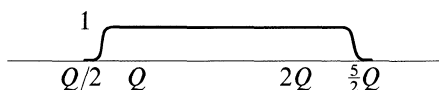
$$(8.6) \quad \begin{array}{c} 1 \\ \text{-----} \\ 1/2Y \quad 1/Y \quad 2/Y \quad 5/2Y \end{array}$$

Apply Kuznetsov's formula (1.19) for the cusps $a=b=\infty$ of the group $\Gamma_0(q)$, multiply its both sides by $\bar{a}_m a_n$ and sum over m, n from $[N, 2N]$. This gives

$$(8.7) \quad \sum_{\lambda_j \text{-except.}}^{(q)} \left| \frac{\hat{\phi}(\kappa_j)}{\operatorname{ch} \pi \kappa_j} \right| \left| \sum_{N < n \leq 2N} a_n \rho_{j\infty}(n) \right|^2 \\ = \sum_{N < m, n \leq 2N} \bar{a}_m a_n \sum_{c=1}^{\infty} \frac{1}{qc} \varphi \left(\frac{4\pi \sqrt{mn}}{qc} \right) S(m, n; qc) \\ + O((1+q^{-1}N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2),$$

the error term coming from the estimates for bilinear forms in Fourier coefficients of holomorphic cusp forms, Eisenstein series and non-holomorphic cusp forms with $\lambda_j \geq \frac{1}{4}$, all by means of Theorem 2.

Let $f(x)$ be another test function whose graph is



Multiply both sides of (8.7) by $f(q)$ and sum over q . Next apply Kuznetsov's formula (1.19) again for cusps $a=b=\infty$ of the group $\Gamma_0(c)$ giving

$$(8.8) \quad \sum_{q=1}^{\infty} \frac{1}{qc} f(q) \varphi \left(\frac{4\pi \sqrt{mn}}{qc} \right) S(m, n; qc) \\ = \sum_{\lambda_j}^{(c)} \frac{\overline{\rho_{j\infty}(m)} \rho_{j\infty}(n)}{\operatorname{ch} \pi \kappa_j} \hat{h}(\kappa_j) \\ + \text{similar terms corresponding} \\ \text{to the holomorphic cusp forms} \\ \text{and the Eisenstein series.}$$

Here, for brevity, we denoted $h(\xi) = f(4\pi \sqrt{mn}/\xi c) \varphi(\xi)$, so $h(\xi)$ depends on m, n and c . In order to separate these variables we write

$$f(\xi) = \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} \chi(s) \xi^{-s} ds$$

with

$$(8.9) \quad \chi(s) = \int_0^{\infty} f(\xi) \xi^{s-1} d\xi \ll (t^4 + 1)^{-1}, \quad s = it.$$

All terms on the *RHS* of (8.8) except those with the exceptional λ_j 's can be estimated (after summing over m, n with coefficients $\bar{a}_m a_n$) by means of Theorem 2 showing that they contribute at most

$$(8.10) \quad O((1+c^{-1}N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2).$$

For the exceptional λ_j 's, we approximate $\hat{h}(\kappa_j)$ by the integral (8.1), of course with φ replaced by h . If $0 < i\kappa_j \leq \varepsilon$ we obtain

$$|\hat{h}(\kappa_j)| \ll Y^{2\varepsilon} \log Y$$

thus these terms, as in the case of $\lambda_j \geq \frac{1}{4}$, contribute not more than (8.10) multiplied by $Y^{2\varepsilon} \log Y$. If $\varepsilon < i\kappa_j \leq \frac{1}{4}$ we obtain

$$\begin{aligned} \hat{h}(\kappa_j) = & \frac{\pi 4^{i\kappa_j}}{2 \sin(\pi i\kappa_j) \Gamma(1-2i\kappa_j)} \int \xi^{-2i\kappa_j} f\left(\frac{4\pi\sqrt{mn}}{\xi c}\right) \varphi(\xi) \frac{d\xi}{\xi} \\ & + O_\varepsilon(|\kappa_j|^2 + 1)^{-1}. \end{aligned}$$

The error terms contribute to the *RHS* of (8.8) not more than (8.10). The integral can be rewritten as follows

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \chi(it) \left(\frac{c}{4\pi\sqrt{mn}} \right)^{it} \int_0^{\infty} \xi^{i(t-2\kappa_j)-1} \varphi(\xi) d\xi$$

where $\chi(it) \ll (t^4 + 1)^{-1}$ by (8.9). Hence, collecting (8.7), (8.10) we deduce that

$$\begin{aligned} & \sum_q f(q) \sum_{\lambda_j \text{-except.}}^{(q)} \left| \frac{\hat{\phi}(\kappa_j)}{\text{ch } \pi \kappa_j} \right| \left| \sum_{N < n \leq 2N} a_n \rho_{j\infty}(n) \right|^2 \\ &= \int_{-\infty}^{\infty} \chi(it) \sum_{C < c \leq 16C} \left(\frac{c}{4\pi} \right)^{it} \sum_{\lambda_j \text{-except.}}^{(c)} \frac{4^{i\kappa_j-1}}{\sin(\pi i\kappa_j) \Gamma(1-2i\kappa_j)} \varphi^*(2i\kappa_j - it) \\ & \cdot \left(\sum_m \overline{a_m \rho_{j\infty}(m)} m^{-it/2} \right) \left(\sum_n a_n \rho_{j\infty}(n) n^{-it/2} \right) dt \\ &+ O((YN)^{3\varepsilon} (Q + C + N) \|\mathbf{a}_N\|_2^2) \end{aligned}$$

where φ^* is the Mellin transform of φ and $C = \pi NYQ^{-1}$. We trivially have

$$\hat{\phi}(\kappa_j) \gg Y^{2i\kappa_j} \quad \text{and} \quad |\varphi^*(2i\kappa_j - it)| \ll Y^{2i\kappa_j}$$

thus by the Cauchy-Schwarz inequality we obtain

$$\begin{aligned} & \sum_{Q < q \leq 2Q} \sum_{\lambda_j \text{-except.}}^{(q)} \left| Y^{2i\kappa_j} \right| \left| \sum_{N < n \leq 2N} a_n \rho_{j\infty}(n) \right|^2 \\ & \ll \int_{-\infty}^{\infty} \sum_{C < c \leq 16C} \sum_{\lambda_j \text{-except.}}^{(c)} \left| Y^{2i\kappa_j} \right| \left| \sum_{N < n \leq 2N} a_n n^{it} \rho_{j\infty}(n) \right|^2 \frac{dt}{t^4 + 1} \\ &+ (YN)^{3\varepsilon} (Q + N + NYQ^{-1}) \|\mathbf{a}_N\|_2^2. \end{aligned}$$

All what is left to do is to extend the range of summation over q from $[Q, 2Q]$ to $[Q, 16Q]$ with no change of the range $[C, 16C]$ for c . To this end apply the above estimate for pairs

$$2^l Q, \quad 2^l Y, \quad l = 0, 1, 2, 3$$

and add the resulting inequalities sides by sides. This completes the proof of Lemma 8.1. ■

Proceeding to the proof of Theorem 6 we remark that it suffices to show that

$$(8.11) \quad S(Q, Y, N, 0) \ll (QN)^\varepsilon (Q + \sqrt{Y}N) \|\mathbf{a}_N\|_2^2$$

for any $Q, Y, N \geq 1$ and $\varepsilon > 0$. First, we prove by induction on Q that for $1 \leq N \leq Q^{1-\varepsilon}$ and $Y = (Q^{1-\varepsilon}/N)^2$ it holds

$$(8.12) \quad S(Q, Y, N, 0) \ll Q^{1+\varepsilon} \|\mathbf{a}_N\|_2^2.$$

If $Q \leq Q_0(\varepsilon)$ or $Q^{1-2\varepsilon} < N \leq Q^{1-\varepsilon}$, we have

$$S(Q, Y, N, 0) \leq Y^{\frac{1}{2}} S(Q, 1, N, 0) \ll N^{-1} Q^{1-\varepsilon} (Q + N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2$$

by the large sieve inequality (Theorem 2) and this yields (8.12). If $Q > Q_0(\varepsilon)$ and $N < Q^{1-2\varepsilon}$, then we apply Lemma 8.1 giving

$$S(Q, Y, N, 0) \ll \sup_t S(Q_1, Y, N, it) + Q^{1+\frac{\varepsilon}{2}} \|\mathbf{a}_N\|_2^2$$

where $Q_1 = \pi N Y Q^{-1} = \pi Q^{1-2\varepsilon} N^{-1} \leq Q-1$, provided the constant $Q_0(\varepsilon)$ is chosen sufficiently large. To apply the induction hypothesis we write $Y_1 = (Q_1^{1-\varepsilon}/N)^2$, so $Y = \pi^{-2} N^2 Q^{4\varepsilon(1-\varepsilon)} Y_1$ and

$$\begin{aligned} S(Q_1, Y, N, it) &\leq \pi^{-1} N Q^{2\varepsilon(1-\varepsilon)} S(Q_1, Y_1, N, it) \\ &\ll N Q^{2\varepsilon(1-\varepsilon)} Q_1^{1+\varepsilon} \|\mathbf{a}_N\|_2^2 \\ &\ll Q^{1+\varepsilon-2\varepsilon^2} \|\mathbf{a}_N\|_2^2. \end{aligned}$$

Here the extra factor $Q^{-2\varepsilon^2}$ annihilates the constants implied in the symbol $\ll$ and a routine manipulation with the other constants completes the proof of (8.12).

Now, the general case (8.11) results from (8.12) as follows

$$\begin{aligned} S(Q, Y, N, 0) &\leq (1 + \sqrt{Y} N Q^{\varepsilon-1}) S(Q, (Q^{1-\varepsilon}/N)^2, N, 0) \\ &\ll (1 + \sqrt{Y} N Q^{\varepsilon-1}) Q^{1+\varepsilon} \|\mathbf{a}_N\|_2^2 \\ &\leq (QN)^{2\varepsilon} (Q + \sqrt{Y} N) \|\mathbf{a}_N\|_2^2 \end{aligned}$$

if $N \leq Q^{1-\varepsilon}$, and

$$\begin{aligned} S(Q, Y, N, 0) &\leq \sqrt{Y} S(Q, 1, N, 0) \ll \sqrt{Y} (Q + N^{1+\varepsilon}) \|\mathbf{a}_N\|_2^2 \\ &\ll (QN)^\varepsilon (Q + \sqrt{Y} N) \|\mathbf{a}_N\|_2^2 \end{aligned}$$

if $N > Q^{1-\varepsilon}$. The proof of Theorem 6 is complete. ■

One might be surprised by the fact we did not use the recurrence relation (8.5) for $Y = Q^{2-\varepsilon} N^{-1}$ that fits to its structure much better and presumably would yield sharper results. This is indeed true as long as the second term on the *RHS* of (8.5) is ignored. This term matters only if $N > Q$. Although the case $N > Q$ is not important for applications it is necessary to deal with it in order to complete the induction arguments. In the next subsection we treat sums of Kloosterman sums by *ad hoc* methods getting, in special cases with $N > Q$, a satisfactory bound to incorporate with the induction process with prescribed Y .

8.3 Auxiliary Estimates for Kloosterman Sums

Some of these results having their own interest will be given a status of theorems; the first one is

Theorem 14. For $C, M, N \geq 1$ we have

$$(8.13) \quad \sum_{c \leq C} \left| \sum_{m \leq M} \sum_{n \leq N} S(m, n; c) \right| \ll_e (CMN)^e C(C + MN).$$

We begin the proof by showing the very simple following

Lemma 8.2. For $M, N \geq 1$ we have

$$(8.14) \quad \left| \sum_{0 < m \leq M} \sum_{0 < n \leq N} S(m, n; c) \right| \leq \sum_{-\infty < m, n < \infty} \hat{f}_M(m) \hat{f}_N(n) S(m, n; c)$$

where $\hat{f}_M(m)$ and $\hat{f}_N(n)$ are some numbers such that

$$(8.15) \quad |\hat{f}_M(m)| \leq 1 + 4 \log M$$

$$(8.16) \quad |\hat{f}_M(m)| \leq (M/m)^2 \quad \text{if } m \neq 0,$$

the same bounds holding for $\hat{f}_N(n)$.

At first glance this inequality seems to be superfluous because the *RHS* contains about the same terms (in number and shape) and it even contains the zero terms (the Ramanujan sums). But what is won at that price is the lack of absolute value in the *RHS* which implies a considerable flexibility when summing over the modulus c .

Proof of Lemma 8.2. For any real ξ we have

$$\left| \sum_{0 < m \leq M} e(m\xi) \right| \leq \min(M, 2 \|\xi\|^{-1}) = f_M(\xi),$$

say

$$= \sum_{-\infty < m < \infty} \hat{f}_M(m) e(m\xi)$$

where $\|\xi\|$ stands for the distance from ξ to the nearest integer and $\hat{f}_M(m)$ denotes – in this subsection – the Fourier transform

$$\begin{aligned} \hat{f}_M(m) &= \int_0^1 e(-m\xi) f_M(\xi) d\xi \\ &= 2 \int_0^{\frac{1}{2}} \min(M, 2\xi^{-1}) \cos(2\pi m\xi) d\xi. \end{aligned}$$

Hence for all m we trivially obtain (8.15) and for $m \neq 0$, by twice iterated partial integration, we deduce (8.16). Finally, by the definition of Kloosterman sum we infer that

$$\begin{aligned}
\left| \sum_{0 < m \leq M} \sum_{0 < n \leq N} S(m, n; c) \right| &\leq \sum_{d \pmod{c}}^* \left| \sum_{0 < m \leq M} e\left(m \frac{\bar{d}}{c}\right) \right| \left| \sum_{0 < n \leq N} e\left(n \frac{d}{c}\right) \right| \\
&\leq \sum_{d \pmod{c}}^* f_M\left(\frac{\bar{d}}{c}\right) f_N\left(\frac{d}{c}\right) \\
&= \sum_{-\infty < m, n < \infty} \hat{f}_M(m) \hat{f}_N(n) S(m, n; c).
\end{aligned}$$

The proof of (8.14) is complete. ■

Proof of Theorem 14. It suffices to prove (8.13) for sums over c restricted to an interval of the type $[C, 2C]$. Let $g(c)$ be a test function majorizing the characteristic function of that interval. By Lemma (8.2) we get

$$\begin{aligned}
\sum_{C \leq c \leq 2C} \left| \sum_{0 < m \leq M} \sum_{0 < n \leq N} S(m, n; c) \right| \\
\leq \sum_m \sum_n \hat{f}_M(m) \hat{f}_N(n) \sum_c g(c) S(m, n; c).
\end{aligned}$$

The zero terms, i.e. those with $mn=0$, by the well-known estimate of Ramanujan sums, contribute less than

$$\begin{aligned}
&\hat{f}_M(0) \hat{f}_N(0) \sum_c g(c) \varphi(c) + 2 \hat{f}_M(0) \sum_{n=1}^{\infty} |f_N(n)| \sum_c g(c) (n, c) \\
&\quad + 2 \hat{f}_N(0) \sum_{m=1}^{\infty} |f_M(m)| \sum_c g(c) (m, c) \\
&\ll (CMN)^{\varepsilon} (C^2 + CM + CN)
\end{aligned}$$

by (8.15) and (8.16). For the non-zero terms one can apply Theorem 9 (see the remark (1.52) after the statement) showing that they contribute

$$\ll C^{1+\varepsilon} MN.$$

All together complete the proof of Theorem 14. ■

Now we are in a position to prove Theorem 7. Of course it is sufficient to show that

$$(8.17) \quad S(Q, Y, N, 0) \ll (QN)^{5\varepsilon} (Q + N + \sqrt{NY}) N$$

for characteristic sequences $\mathbf{a}_n$ of intervals $[N, N_1]$, $N \leq N_1 \leq 2N$ (we recall that $S(Q, Y, N, s)$ is defined in (8.4)).

By (8.7) we get

$$\begin{aligned}
S(Q, Y, N, 0) &\ll \sum_{Q < q \leq 16Q} \sum_{c=1}^{\infty} \frac{1}{qc} \left| \sum_{N \leq m, n \leq N_1} \varphi\left(\frac{4\pi\sqrt{mn}}{qc}\right) S(m, n; qc) \right| \\
&\quad + (Q + N^{1+\varepsilon}) N
\end{aligned}$$

where $\varphi(x)$ is the test function described in (8.6). Here c runs effectively over the interval $[\frac{1}{40}C, 16C]$ with $C = \pi NYQ^{-1}$. Next, by partial summation we get

rid of the weights $\varphi(4\pi\sqrt{mn}/qc)$, then we make $qc=k$ as one variable (the number of representations of k in this form does not exceed $\tau(k) \ll k^\varepsilon$) and finally we apply Theorem 14, that yields

$$\begin{aligned} S(Q, Y, N, 0) &\ll (QCN)^\varepsilon (QC + N^2) + (Q + N^{1+\varepsilon})N \\ &\ll (NY)^\varepsilon (Q + N + Y)N. \end{aligned}$$

This can be further rectified in Y -aspect by a standard device

$$\begin{aligned} (8.18) \quad S(Q, Y, N, 0) &\ll (1 + \sqrt{Y/Y_1}) S(Q, Y_1, N, 0) \\ &\ll (1 + \sqrt{Y/Y_1}) (Q + N + Y_1) N (NY_1)^\varepsilon \\ &\ll (QN)^\varepsilon (Q + N + \sqrt{NY} + \sqrt{QY}) N \end{aligned}$$

on taking $Y_1 = \sqrt{Q + N}$. Hence (8.17) is proved for $N > Q^{1-2\varepsilon}$, and it remains to get rid of the term $\sqrt{QY}$ in (8.18) in order to obtain (8.17) for all N . To this end we apply much similar induction arguments as that in the proof of Theorem 6. We first prove that

$$(8.19) \quad S(Q, Y, N, 0) \ll Q^{1+4\varepsilon} N$$

for $1 \leq N \leq Q$ and $Y = Q^{2-2\varepsilon} N^{-1}$. If $Q \leq Q_0(\varepsilon)$, the above follows by

$$S(Q, Y, N, 0) \leq Y^{\frac{1}{2}} S(Q, 1, N, 0) \ll Q(Q + N^{1+\varepsilon}) N \ll Q^{1+2\varepsilon} N$$

on applying the large sieve inequality (Theorem 2), and if $Q^{1-2\varepsilon} < N \leq Q$, (8.19) follows from (8.18) as we already pointed it out. Now let $Q > Q_0(\varepsilon)$ and $1 \leq N \leq Q^{1-2\varepsilon}$. By Lemma 8.1

$$S(Q, Y, N, 0) \ll \int_{-\infty}^{\infty} \frac{1}{t^4 + 1} S(Q_1, Y, N, it) dt + Q^{1+3\varepsilon} N$$

where $Q_1 = \pi NYQ^{-1} = \pi Q^{1-2\varepsilon} \leq Q - 1$, provided the constant $Q_0(\varepsilon)$ is chosen sufficiently large. Moreover we write $Y_1 = Q_1^{2-2\varepsilon} N^{-1}$, $Y = (\pi^{-1} Q^{2\varepsilon})^{2(1-\varepsilon)} Y_1$ and

$$\begin{aligned} S(Q_1, Y, N, it) &\leq \sqrt{Y/Y_1} S(Q_1, Y_1, N, it) \\ &\leq Q^{2\varepsilon(1-\varepsilon)} S(Q_1, Y_1, N, it) \end{aligned}$$

with the intention of making use of the induction hypothesis. However, it is not possible to do it straight because of varying coefficients $a_n = n^{it}$ in $S(Q_1, Y_1, N, it)$. But immediately after partial summation

$$\sum_{N < n \leq N_1} n^{it} \rho_{j\infty}(n) = N_1^{it} \sum_{N < n \leq N_1} \rho_{j\infty}(n) - it \int_N^{N_1} \sum_{N < n \leq \xi} \rho_{j\infty}(n) \xi^{it-1} d\xi$$

the induction hypothesis is applicable giving

$$S(Q_1, Y_1, N, it) \ll (t^2 + 1) Q_1^{1+4\varepsilon} N.$$

From all together we obtain

$$\begin{aligned} S(Q, Y, N, 0) &\ll Q^{2\varepsilon(1-\varepsilon)} Q_1^{1+4\varepsilon} N + Q^{1+3\varepsilon} N \\ &\ll Q^{1+4\varepsilon-8\varepsilon^2} N. \end{aligned}$$

An appropriate manipulation with the constants implied in $\ll$ completes the proof of (8.19).

In the general case, (8.17) results from (8.19) as follows

$$\begin{aligned} S(Q, Y, N, 0) &\leq (1 + \sqrt{Y/Y_1}) S(Q, Y_1, N, 0) \\ &\leq (1 + \sqrt{Y/Y_1}) Q^{1+4\varepsilon} N \ll Q^{5\varepsilon} (Q + \sqrt{NY}) N \end{aligned}$$

where $Y_1 = Q^{2-2\varepsilon} N^{-1}$. The proof of Theorem 7 is complete. ■

9. Upper Bounds for Sums of Kloosterman Sums. II

In this section we enhance the methods of section 7 by the weighted large sieve inequalities with the aim of gaining something if the exceptional eigenvalues do exist.

9.1 Quintilinear Forms in Kloosterman Sums

The objects we named in the title are sums in five variables

$$\begin{aligned} \mathcal{L}(C, M, N, R, S) &= \sum_{\substack{R < r \leq 2R \\ S < s \leq 2S \\ (r, s) = 1}} \sum_{\substack{M < m \leq 2M \\ N < n \leq 2N}} a_m b_{n, r, s} \\ &\cdot \sum_{(c, s) = 1} g(c, m, n, r, s) S(m\bar{r}, \pm n; sc). \end{aligned}$$

In this subsection we prove Theorems 10 and 11. In order to avoid tedious complications, mainly of a notational character, we restrict ourselves to give the proof for weight functions of the type

$$(9.1) \quad g(c, m, n, r, s) = \frac{CS\sqrt{R}}{cs\sqrt{r}} f\left(\frac{4\pi\sqrt{mn}}{cs\sqrt{r}}\right)$$

where $f(x)$ is a function of $\mathcal{C}^\infty$ class with support in $[X^{-1}, 2X^{-1}]$, $X = CS\sqrt{R}/4\pi\sqrt{MN}$ such that

$$|f^{(v)}(x)| \ll X^{-v}, \quad v = 0, 1, 2, \dots$$

the constant implied in $\ll$ depending on v alone.

Proofs of the general forms of these theorems can be quickly supplied by any reader who went through section 7 where the basic technique for the relevant transforms of smooth weight functions (separation of variables) is clearly demonstrated.

By the formula (1.6) we have

$$(9.2) \quad \mathcal{L}(C, M, N, R, S) = CS\sqrt{R} \sum_{\substack{R < r \leq 2R \\ S < s \leq 2S \\ (r, s) = 1}} \sum_{\substack{M < m \leq 2M \\ N < n \leq 2N}} a_m b_{n, r, s} e\left(-n \frac{\bar{s}}{r}\right) \\ \cdot \sum_{\gamma} \frac{1}{\gamma} f\left(\frac{4\pi\sqrt{mn}}{\gamma}\right) S_{\infty 1/s}(m, \pm n; \gamma)$$

where the innermost summation is taken over numbers $\gamma = cs\sqrt{r}$, c running over positive integers, which are the entries of the left-lower corner of matrices from $\Gamma_0(rs)\sigma_{1/s}$. Next for the inner sum we apply the Kuznetsov trace formulas (1.19) and (1.20). It is only in the treatment of the exceptional spectrum terms that we depart from the patterns of the previous proof of Theorem 9.

Note that for real r one has by Lemma 7.1

$$(9.3) \quad \hat{f}(r), \check{f}(r), \tilde{f}(r) \ll (r^2 + 1)^{-1},$$

thus the regular spectrum series, by the large sieve inequalities (Theorem 2), contribute to $\mathcal{L}(C, M, N, R, S)$ at most

$$(9.4) \quad \ll CS\sqrt{R} \sum_r \sum_s \frac{1 + |\log X|}{1 + X} \left(1 + X + \sqrt{\frac{M^{1+\varepsilon}}{rs}}\right) \\ \cdot \left(1 + X + \sqrt{\frac{N^{1+\varepsilon}}{rs}}\right) \|a_M\|_2 \|b_{N, R, S}\|_2 \\ \ll (CMNRS)^\varepsilon \sqrt{RS} \frac{(CS\sqrt{R} + \sqrt{MN} + C\sqrt{SM})(CS\sqrt{R} + \sqrt{MN} + C\sqrt{SN})}{CS\sqrt{R} + \sqrt{MN}} \\ \cdot \|a_M\|_2 \|b_{N, R, S}\|_2.$$

The exceptional spectrum series is equal to

$$CS\sqrt{R} \sum_{r, s} \sum_{\lambda_j \text{-except}}^{(rs)} \frac{\hat{f}(\kappa_j)}{\text{ch } \pi \kappa_j} \left(\sum_m a_m \overline{\rho_{j\infty}(m)}\right) \left(\sum_n e\left(-n \frac{\bar{s}}{r}\right) b_{n, r, s} \rho_{j1/s}(n)\right)$$

which by $\hat{f}(\kappa_j) \ll X(1+X)^{2i\kappa_j-1} \leq \frac{X}{1+X} \left(1 + \frac{X}{\sqrt{X_0}}\right)^{2i\kappa_j} (1+X_0)^{i\kappa_j}$ with $X_0 = \frac{RS}{N}$ yields

$$(9.5) \quad \ll CS\sqrt{R} X(1+X)^{-1} \left(\sum_{r, s} \sum_{\lambda_j \text{-except}}^{(rs)} \left(1 + \frac{X}{\sqrt{X_0}}\right)^{4i\kappa_j} \left|\sum_m \bar{a}_m \rho_j(m)^2\right)\right)^{\frac{1}{2}} \\ \cdot \left(\sum_{r, s} \sum_{\lambda_j \text{-except}}^{(rs)} (1+X_0)^{2i\kappa_j} \left|\sum_n c\left(-n \frac{\bar{s}}{r}\right) b_{n, r, s} \rho_{j1/s}(n)\right|^2\right)^{\frac{1}{2}}.$$

For the sums in the first line, we apply either Theorem 6 or Theorem 7 (in case it is permitted), obtaining the two bounds

$$(9.6) \quad (CNRS)^\gamma \left(RS + M + M \frac{X}{\sqrt{X_0}}\right) \|a_M\|_2^2$$

and

(9.7)
$$(CNRS)^{\varepsilon} \left(RS + M + \sqrt{\frac{M}{X_0}} X \right) \| \mathbf{a}_M \|_2^2$$

respectively. For the sum in the second line, we apply Theorem 5 getting the bound

(9.8)
$$(1 + N^{1+\varepsilon}/RS) \| \mathbf{b}_{N,R,S} \|_2^2.$$

Combining (9.2), (9.4), (9.5), (9.6) and (9.8) we complete the proof of Theorem 10 and, replacing (9.6) by (9.7), that of Theorem 11. ■

9.2 Incomplete Kloosterman Sums

There is a number of methods for transforming incomplete exponential sums over finite fields to complete ones. Here we apply a method which is based on the Poisson summation formula.

Lemma 9.1. *Let $f(x)$ be a function of $\mathcal{C}^2$ class with compact support. For $q \geq 1$, a an integer we have*

(9.9)
$$\sum_{m \equiv a \pmod{q}} f(m) = \frac{1}{q} \sum_m e\left(-\frac{am}{q}\right) \check{f}\left(\frac{m}{q}\right)$$

where $\check{f}(y)$ is the Fourier transform of $f(x)$, i.e.

$$\check{f}(y) = \int_{-\infty}^{\infty} f(x) e(x y) dx.$$

Our aim is to prove Theorem 12 or rather to reduce it to Theorem 11. By (9.9) we proceed as follows

$$\begin{aligned} &\sum_{(d,sc)=1} g(c,d,n,r,s) e\left(n \frac{\overline{rd}}{sc}\right) \\ &= \sum_{\substack{\delta \pmod{sc} \\ (\delta,sc)=1}} e\left(n \frac{\overline{r\delta}}{sc}\right) \sum_{d \equiv \delta \pmod{sc}} g(c,d,n,r,s) \\ &= \frac{1}{sc} \sum_{m=-\infty}^{\infty} S(n\overline{r}, -m; sc) \check{g}\left(c, \frac{m}{sc}, n, r, s\right) \end{aligned}$$

where

$$\check{g}\left(c, \frac{m}{sc}, n, r, s\right) = \int g(c, \xi, n, r, s) e\left(\frac{m}{sc} \xi\right) d\xi.$$

For $m=0$ we have $g(c,0,n,r,s) \ll D$ and the Kloosterman sum reduces to the Ramanujan one for which we have

$$|S(n\bar{r}, 0; sc)| \ll (n, sc).$$

Therefore the zero-term contributes to $\mathcal{H}(C, D, N, R, S)$ at most

$$(9.11) \quad D \sum_r \sum_s \sum_n |b_{n,r,s}| \sum_c \frac{(n, sc)}{sc} \ll N^\varepsilon D(NR/S)^{\frac{1}{2}} \|\mathbf{b}_{N,R,S}\|_2.$$

For $|m| \geq (CDS)^\varepsilon CSD^{-1} = M$, say, by iterated partial integration (the number of iterations depending on ε) we get

$$\check{g}(c, m/sc, n, r, s) \ll m^{-2}$$

whence the series on the *RHS* of (9.10) converges absolutely and its terms with $|m| \geq M$ contribute to $\mathcal{H}(C, D, N, R, S)$ at most

$$(9.12) \quad \sum_r \sum_s \sum_n |b_{n,r,s}| \sum_c \sum_{|m| \geq M} \frac{1}{m^2} \ll (CDS)^\varepsilon D(NR/S)^{\frac{1}{2}} \|\mathbf{b}_{N,R,S}\|_2.$$

It remains to evaluate the contribution of terms with $0 < |m| \leq M$. For such terms we divide the interval $(0, M]$ into $\ll \log(M+2)$ subintervals of the type $(M_1, 2M_1]$, $\frac{1}{2} \leq M_1 \leq M$, and restrict $|m|$ to run over $(M_1, 2M_1]$. For given ξ from the range of integration, i.e. such that

$$(9.13) \quad DM_1/SC \ll \xi \ll DM_1/SC$$

Theorem 11 is applicable with the weight function $m^{-1} M_1 g(c, \xi sc/m, n, r, s)$ giving

$$(9.14) \quad \sum_{\substack{R < r \leq 2R \\ S < s \leq 2S \\ (r,s)=1}} \sum_{\substack{N < n \leq 2N \\ M_1 < |m| \leq 2M_1}} b_{n,r,s} \sum_{(c,s)=1} \frac{1}{m} \\ \cdot g(c, \xi sc/m, n, r, s) S(n\bar{r}, -m; sc) \\ \ll (CMNR S)^\varepsilon L^*(C, M_1, N, R, S) M_1^{-\frac{1}{2}} \|\mathbf{b}_{N,R,S}\|_2.$$

where

$$L^*(C, M_1, N, R, S)$$

$$= \sqrt{RS} \frac{(CS\sqrt{R} + \sqrt{M_1 N} + C\sqrt{SM_1})(CS\sqrt{R} + \sqrt{M_1 N} + C\sqrt{SN})}{CS\sqrt{R} + \sqrt{M_1 N}} \\ + S(R(RS+N))^{\frac{1}{4}} C^{\frac{3}{4}}.$$

Integration over ξ from (9.13) enlarges this bound by the factor DM_1/SC . Then it is obvious that the maximum of

$$(DM_1/SC) L^*(C, M_1, N, R, S) M_1^{-\frac{1}{2}}$$

is attained at $M_1 = M$ and it is equal to

$$(CDS)^\varepsilon L^*(C, M, N, R, S) M^{-\frac{1}{2}}.$$

We further have

$$\begin{aligned}
& L^*(C, M, N, R, S) M^{-\frac{1}{2}} \\
& \leq \{ \sqrt{RS} (CS \sqrt{R} + \sqrt{MN} + C \sqrt{SN} + C \sqrt{SM} + C \sqrt{MN/R}) \\
& \quad + S(R(RS + N))^{\frac{1}{2}} C^{\frac{1}{2}} \} M^{-\frac{1}{2}} \\
& \ll (CDS)^{\varepsilon} \{ \sqrt{RS} (\sqrt{CDSR} + \sqrt{CDN} + C \sqrt{S} + C \sqrt{N/R}) \\
& \quad + C(DS)^{\frac{1}{2}} (R(RS + N))^{\frac{1}{2}} \} \\
& \leq (CDS)^{\varepsilon} K(C, D, N, R, S).
\end{aligned}$$

Gathering together (9.10), (9.11), (9.12), and (9.14) we get (1.57) and the proof of Theorem 12 is complete. ■

9.3 Multidimensional Exponential Sums

In this last section we shall estimate exponential sums of the type

$$\mathcal{F}_{r/s}(C, D, L, M) = \sum_c \sum_d \sum_l \sum_m f(c, d, l, m, r, s) e \left(l m \frac{dr}{cs} \right)$$

as well as

$$\mathcal{F}(C, D, L, M, R, S) = \sum_{R \leq r \leq 2R} \sum_{S \leq s \leq 2S} a_{r,s} \mathcal{F}_{r/s}(C, D, L, M).$$

Here $f(c, d, l, m, r, s)$ is a function of $\mathcal{C}^\infty$ class with support in $(0, \infty]^6$ and with derivatives satisfying

$$\begin{aligned}
(9.15) \quad & \left| \frac{\partial^{v_1 + v_2 + v_3 + v_4 + v_5 + v_6}}{\partial c^{v_1} \partial d^{v_2} \partial l^{v_3} \partial m^{v_4} \partial r^{v_5} \partial s^{v_6}} f(c, d, l, m, r, s) \right| \\
& \ll \frac{C^{2-v_1}}{c^2 + C^2} \frac{D^{2-v_2}}{d^2 + D^2} \frac{L^{2-v_3}}{l^2 + L^2} \frac{M^{2-v_4}}{m^2 + M^2} \frac{R^{2-v_5}}{r^2 + R^2} \frac{S^{2-v_6}}{s^2 + S^2}
\end{aligned}$$

for any $v_1, \dots, v_6 \geq 0$ and some parameters $C, D, L, M, R, S \geq 1$, the constant implied in $\ll$ depending on $v_1, \dots, v_6$ alone. The first result is conditional

Theorem 15. *If the smallest positive eigenvalue λ_1 of the Laplacian acting on $L^2(\Gamma \setminus H)$ for Hecke groups $\Gamma = \Gamma_0(q)$ is $\geq \frac{1}{4}$ then for $(r, s) = 1$ we have*

$$\begin{aligned}
(9.16) \quad & \mathcal{F}_{r/s}(C, D, L, M) \ll (rs CDLM)^{\varepsilon} \\
& \cdot \{ (s C(C+D)(C^2 s + LM r))^{\frac{1}{2}} + DLM s^{-1} + CD \}
\end{aligned}$$

the constant implied in $\ll$ depending on ε alone.

When an averaging over r and s is performed we have an unconditional result

Theorem 16. *For any complex sequence $\mathbf{a}_{r,s}$ we have*

$$(9.17) \quad \mathcal{F}(C, D, L, M, R, S) \ll (CDLMRS)^{\varepsilon} F(C, D, L, M, R, S) \|\mathbf{a}_{R,S}\|_2$$

with

$$F(C, D, L, M, R, S) = (C\sqrt{S} + \sqrt{LMR})(\sqrt{CS} + \sqrt{DRS} + \sqrt{D(LMR)^{\frac{1}{2}}})\sqrt{CS} \\ + DLM\sqrt{R/S} + CD\sqrt{RS},$$

the constant implied in $\ll$ depending on ε alone.

Theorems 15 and 16 can be derived from Theorems 11 and 12 respectively. A connection of $\mathcal{F}_{r/s}(C, D, L, M)$ with sums of incomplete Kloosterman sums is given through the following

Lemma 9.2. *Let $f(x, y)$ be a function of $\mathcal{C}^2$ class with compact support. Then for $q \geq 1$, $(a, q) = 1$ we have*

$$(9.18) \quad \sum_l \sum_m f(l, m) e\left(lm \frac{a}{q}\right) = \frac{1}{q} \sum_l \sum_m \check{f}\left(\frac{l}{q}, \frac{m}{q}\right) e\left(-lm \frac{\bar{a}}{q}\right)$$

where $\bar{a}$ is a solution of $\bar{a}a \equiv 1 \pmod{q}$ and $\check{f}(\alpha, \beta)$ is the Fourier transform

$$(9.19) \quad \check{f}(\alpha, \beta) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e(\alpha x + \beta y) dx dy.$$

Proof. This result constitutes a generalization of the two dimensional Poisson formula

$$(9.20) \quad \sum_l \sum_m f(l, m) = \sum_l \sum_m \check{f}(l, m).$$

We derive (9.18) from (9.20) as follows

$$\begin{aligned} \sum_l \sum_m f(l, m) e\left(lm \frac{a}{q}\right) &= \sum_{h \pmod{q}} \sum_{k \pmod{q}} e\left(\frac{ahk}{q}\right) \sum_{l \equiv h \pmod{q}} \sum_{m \equiv k \pmod{q}} f(l, m) \\ &= q^{-2} \sum_l \sum_m \check{f}\left(\frac{l}{q}, \frac{m}{q}\right) \sum_{h \pmod{q}} \sum_{k \pmod{q}} e\left(\frac{ahk - hl - km}{q}\right) \end{aligned}$$

which by the orthogonality of additive characters yields the RHS of (9.18). ■

In practice the non-oscillating terms, i.e. those with $lm=0$, furnish a main term. Having this in mind we add to Lemma 9.2 the following

Lemma 9.3. *Under the same notation and assumptions as in Lemma 9.2 we have*

$$(9.21) \quad - \sum_{lm=0} \check{f}\left(\frac{l}{q}, \frac{m}{q}\right) = \iint f(x, y) dx dy \\ + q \iint \left[\frac{x}{q}\right] f_x(x, y) dx dy \\ + q \iint \left[\frac{y}{q}\right] f_y(x, y) dx dy,$$

where $[\xi]$ stands for the integral part of ξ . In particular if $\text{Supp } f(x, y) \subset [0, q]^2$ the second line in (9.21) vanishes.

Proof. For $m=l=0$ we have

$$\ddot{f}(0, 0) = \iint f(x, y) \, dx \, dy.$$

If $m=0$ and $l \neq 0$ we integrate by parts with respect to x and appeal to the well-known expansion

$$\sum_{l \neq 0} \frac{1}{2\pi i l} e(l\xi) = [\xi] + \frac{1}{2} - \xi$$

giving

$$\begin{aligned} - \sum_{l \neq 0} \ddot{f}\left(\frac{l}{q}, 0\right) &= - \sum_{l \neq 0} \iint f(x, y) e\left(\frac{l}{q}x\right) \, dx \, dy \\ &= q \sum_{l \neq 0} \frac{1}{2\pi i l} \iint f_x(x, y) e\left(\frac{l}{q}x\right) \, dx \, dy \\ &= q \iint f_x(x, y) \left(\left[\frac{x}{q}\right] + \frac{1}{2} - \frac{x}{q} \right) \, dx \, dy \\ &= \iint f(x, y) \, dx \, dy + q \iint f_x(x, y) \left[\frac{x}{q}\right] \, dx \, dy. \end{aligned}$$

Analogously we obtain

$$- \sum_{m \neq 0} \ddot{f}\left(0, \frac{m}{q}\right) = \iint f(x, y) \, dx \, dy + q \iint f_y(x, y) \left[\frac{y}{q}\right] \, dx \, dy.$$

Gathering the three above relations we get (9.21). ■

Proof of Theorem 16. We first observe that it is sufficient to prove (9.17) for weight functions $f(c, d, l, m, r, s)$ with compact support in $[C, 2C] \times [D, 2D] \times [L, 2L] \times [M, 2M] \times [R, 2R] \times [S, 2S]$. Indeed, choose functions $\chi_j(\xi)$ of $\mathcal{C}^\infty$ class such that

$$\begin{aligned} \sum_{j=-\infty}^{\infty} \chi_j(t) &= 1 \quad \text{if } t > 0 \\ \text{Supp } \chi_j(t) &= [\tfrac{1}{2} T_j, T_j], \quad T_j = 2^{j/2} \\ |\chi_j^{(v)}(t)| &\ll t^{-v}, \quad v = 0, 1, \dots \end{aligned}$$

the constant implied in $\ll$ depending on v alone. Hence

$$\mathcal{F}_{r/s}(C, D, L, M) = \sum_{(j) \geq 0} \sum_c \sum_d \sum_l \sum_m f_{(j)}(c, d, l, m, r, s) e\left(lm \frac{dr}{cs}\right)$$

where for $(j) = (j_1, j_2, j_3, j_4)$ we denote

$$(9.22) \quad f_{(j)}(c, d, l, m, r, s) = \chi_{j_1}(c) \chi_{j_2}(d) \chi_{j_3}(l) \chi_{j_4}(m) f(c, d, l, m, r, s).$$

For (j) with $|j| \geq J = 8 \log CDLM$ we treat all terms trivially showing that they contribute to $\mathcal{F}_{r/s}(C, D, L, M)$

$$\begin{aligned} &\ll \sum_{cdlm \geq (CDLM)^3} (CDLM/c d l m)^2 \\ &\ll (CDLM)^2 \sum_{k \geq (CDLM)^3} k^{-2} \tau_4(k) \ll 1 \end{aligned}$$

which is admissible.

For each remaining (j) we apply the assumed case of Theorem 16 for the weight function $f_{(j)}(c, d, l, m, r, s)$ which by (9.15) and (9.22) satisfies

$$\begin{aligned} &\left| \frac{\partial^{v_1+v_2+v_3+v_4+v_5+v_6}}{\partial c^{v_1} \partial d^{v_2} \partial l^{v_3} \partial m^{v_4} \partial r^{v_5} \partial s^{v_6}} f_{(j)}(c, d, l, m, r, s) \right| \\ &\ll \Omega(j) c^{-v_1} d^{-v_2} l^{-v_3} m^{-v_4} r^{-v_5} s^{-v_6} \end{aligned}$$

with $\Omega(j) = (CDLM)^2 / (T_{j_1}^2 + C^2)(T_{j_2}^2 + D^2)(T_{j_3}^2 + L^2)(T_{j_4}^2 + M^2)$, the constant implied in $\ll$ depending on $v_1, \dots, v_6$ at most. We get

$$\begin{aligned} &\sum_{R < r \leq 2R} \sum_{S < s \leq 2S} a_{r,s} \sum_c \sum_d \sum_l \sum_m f_{(j)}(c, d, l, m, r, s) e \left(l m \frac{dr}{s c} \right) \\ &\ll (T_{j_1} T_{j_2} T_{j_3} T_{j_4})^\epsilon \Omega(j) F(T_{j_1}, T_{j_2}, T_{j_3}, T_{j_4}, R, S) \|\mathbf{a}_{R,S}\|_2. \end{aligned}$$

The maximum of the *RHS* with respect to (j) is taken for $T_{j_1}, T_{j_2}, T_{j_3}, T_{j_4}$ of order of magnitude C, D, L, M respectively. Multiplying the resulting bound by the number of (j)'s, i.e. by a number $\leq (J+1)^4 \ll (\log CDLM)^4$, we obtain (9.17) for $\mathcal{F}(C, D, L, M, R, S)$.

Let $\mathcal{F}^*(C, D, L, M, R, S)$ stand for the partial sum of $\mathcal{F}(C, D, L, M, R, S)$ with the variables in the summation restricted by the condition $(dr, cs) = 1$. Without loss of generality it suffices to show (9.17) for $\mathcal{F}^*(C, D, L, M, R, S)$. Indeed, introducing the greatest common divisors $\alpha = (c, d)$, $\beta = (r, s)$, $\gamma = (d/\alpha, s/\beta)$, $\delta = (c/\alpha, r/\beta)$ and denoting $C_1 = C/\alpha\delta$, $D_1 = D/\alpha\gamma$, $R_1 = R/\beta\delta$, $S_1 = S/\beta\gamma$ and $f_{\alpha\beta\gamma\delta}(c, d, l, m, r, s) = f(\alpha\delta c, \alpha\gamma d, l, m, r, s)$ we may write

$$\mathcal{F}(C, D, L, M, R, S) = \sum_{\alpha\beta\gamma\delta} \mathcal{F}^*(C_1, D_1, L, M, R_1, S_1)$$

where $\alpha\delta \leq 2C$, $\alpha\gamma \leq 2D$, $\beta\delta \leq 2R$, $\beta\gamma \leq 2S$ and $\mathcal{F}^*(C_1, D_1, L, M, R_1, S_1)$ is the sum related to the weight function $f_{\alpha\beta\gamma\delta}(c, d, l, m, r, s)$ which satisfies (9.15) with parameters C_1, D_1, L, M . Therefore, by (9.17) for $\mathcal{F}^*(C_1, D_1, L, M, R_1, S_1)$ we obtain

$$\begin{aligned} &\mathcal{F}(C, D, L, M, R, S) \\ &\ll (CDLMRS)^\epsilon \sum_{\alpha\beta\gamma\delta} F(C_1, D_1, L, M, R_1, S_1) \|\mathbf{a}_{R_1, S_1}\|_2 \\ &\ll (CDLMRS)^{2\epsilon} F(C, D, L, M, R, S) \|\mathbf{a}_{R,S}\|_2, \end{aligned}$$

i.e. (9.17) for $\mathcal{F}(C, D, L, M, R, S)$.

Now we proceed to prove (9.17) for $\mathcal{F}^*(C, D, L, M, R, S)$ with the weight function $f(c, d, l, m, r, s)$ with compact support in $[C, 2C] \times [D, 2D] \times [L, 2L] \times [M, 2M] \times [R, 2R] \times [S, 2S]$. By Lemma 9.2 we may write

$$(9.23) \quad \mathcal{F}^*(C, D, L, M, R, S) = \sum_{r,s} a_{r,s} \sum_{\substack{c,d \\ (cs, dr)=1}} \frac{1}{cs} \\ \cdot \sum_{lm} \check{f}\left(c, d, \frac{l}{cs}, \frac{m}{cs}, r, s\right) e\left(-lm \frac{\overline{dr}}{cs}\right)$$

where $\check{f}(c, d, \alpha, \beta, r, s)$ is the Fourier transform of $f(c, d, x, y, r, s)$ with respect to x and y . The constant terms, i.e. those with $lm=0$, contribute to $\mathcal{F}^*(C, D, L, M, R, S)$

$$\ll \sum_{r,s} |a_{r,s}| \sum_{c,d} \frac{1}{cs} LM \ll DLM \sqrt{R/S} \|\mathbf{a}_{R,S}\|_2$$

by Lemma 9.3. This yields one term in $F(C, D, L, M, R, S)$.

If $lm \neq 0$ we integrate by parts getting

$$\check{f}\left(c, d, \frac{l}{cs}, \frac{m}{cs}, r, s\right) = c^2 \left(\frac{-cs}{2\pi i l}\right)^{j_3} \left(\frac{-cs}{2\pi i m}\right)^{j_4} \\ \iint \partial^{(j)} f(c, d, cx, cy, r, s) e\left(\frac{lx+my}{s}\right) dx dy$$

where $(j)=(0, 0, j_3, j_4, 0, 0)$ and $\partial^{(j)} f$ is the partial derivative of f of order (j) , thus

$$|\partial^{(j)} f| \ll L^{-j_3} M^{-j_4}.$$

The effective range of the above integration is the rectangle

$$(9.24) \quad \frac{1}{2} \frac{L}{C} < x < 2 \frac{L}{C}, \quad \frac{1}{2} \frac{M}{C} < y < 2 \frac{M}{C}.$$

We intent to apply Theorem 12 for the weight function

$$g(c, d, n, r, s) = c^{j_3+j_4+1} \partial^{(j)} f(c, d, cx, cy, r, s)$$

and for the coefficients

$$b_{n,r,s} = s^{j_3+j_4-1} \sum_{lm=n} l^{-j_3} m^{-j_4} e((lx+my)s^{-1}) a_{r,s}.$$

In order to control the size of $n=lm$, we split up the range of variables l, m into subintervals of the type $(L_1, \sqrt{2}L_1]$ and $(M_1, \sqrt{2}M_1]$ respectively. Then $n=lm$ lays in $(N_1, 2N_1]$ with $N_1=L_1M_1$. By Theorem 12 we infer that

$$\begin{aligned}
 (9.24) \quad & \sum_{r,s} a_{r,s} \sum_{c,d} \frac{1}{c s} \sum_{\substack{L_1 < l \leq \sqrt{2} L \\ M_1 < m \leq \sqrt{2} M_1}} f\left(c, d, \frac{l}{s c}, \frac{m}{s c}, r, s\right) e\left(-l m \frac{\overline{d r}}{s c}\right) \\
 & \ll (C D N_1 R S)^{\varepsilon} \frac{L}{C} \frac{M}{C} \frac{C}{S} \left(\frac{C S}{L L_1}\right)^{j_3} \left(\frac{C S}{M M_1}\right)^{j_4} \\
 & \cdot K(C, D, N_1, R, S) N_1^{\frac{1}{2}} \|\mathbf{a}_{R,S}\|_2.
 \end{aligned}$$

To obtain a bound for $\mathcal{F}^*(C, D, L, M, R, S)$ it remains to choose j_3, j_4 appropriately and to sum the resulting bounds over L_1 and M_1 . We take $j_3=0$ if $L_1 \leq CS/L$ and $j_3=2$ otherwise, and analogously we take $j_4=0$ if $M_1 \leq CS/M$ and $j_4=2$ otherwise. Then it turns out that the maximum of the *RHS* of (9.24) with respect to L_1, M_1 is assumed for L_1 and M_1 of order of magnitude CS/L and CS/M respectively giving

$$\begin{aligned}
 & \ll N_1^{-\varepsilon} (C D L M R S)^{2\varepsilon} \sqrt{L M} K(C, D, C^2 S^2 / L M, R, S) \|\mathbf{a}_{R,S}\|_2 \\
 & \ll N_1^{-\varepsilon} (C D L M R S)^{2\varepsilon} F(C, D, L, M, R, S) \|\mathbf{a}_{R,S}\|_2.
 \end{aligned}$$

Summing over $N_1 = L_1 M_1, L_1 = 2^{\lambda/2}, M_1 = 2^{\mu/2}$, λ, μ positive integers this bound increases by a factor depending on ε alone. The proof of Theorem 16 is complete. ■

The proof of Theorem 15 is similar and the reader can easily construct it from the former arguments.

References

- [Bru] Bruggeman, R.W.: Fourier coefficients of cusp forms. *Invent. Math.* **45**, 1–18 (1978)
- [Bus] Buser, P.: On Cheeger's inequality $\lambda_1 \geq h^2/4$. *Proc. Symposia in Pure Math.*, XXXVI, pp. 29–77 AMS, Providence 1981
- [Che] Cheeger, J.: A lower bound for the smallest eigenvalue of the Laplacian. *Problems in Analysis*, A symposium in honour of Bochner, Princeton University Press, Princeton, N.J. 1970 pp. 195–199
- [Del] Deligne, P.: La conjecture de Weil I. *Publ. Math. I.H.E.S.*, **43**, 273–307 (1974)
- [DI1] Deshouillers, J.-M., Iwaniec, H.: On the greatest prime factor of n^2+1 . *Ann. Inst. Fourier*. in press (1982)
- [DI2] Deshouillers, J.-M., Iwaniec, H.: An additive divisor problem. *J. London Math. Soc.* **26**, (2), 1–14 (1982)
- [DI3] Deshouillers, J.-M., Iwaniec, H.: Power mean-values for the Riemann zeta-function. *Mathematika* in press (1982)
- [DI4] Deshouillers, J.-M., Iwaniec, H.: On the Brun-Titchmarsh theorem and the greatest prime factor of $p+a$. *Proc. Janos Bolyai Soc. Conf.* in press (1982)
- [EMOT] Erdélyi, A., Magnus, W., Oberhettinger, F., Tricomi, F.G.: *Higher Transcendental Functions II*. New York-Toronto-London: Mc Graw-Hill 1953
- [Fad] Faddeev, L.D.: Expansions in eigenfunctions of the Laplace operator in the fundamental domain of a discrete group on the Lobachevskii plane. *Trudy Moscov. Mat. Obsc.*, **17**, 323–350 (1967)
- [G-R] Gradsztejn, I.S., Rzyżyk, I.M.: *Tablice całek, sum, szeregów i iloczynów*. PWN-Warszawa, 1964
- [Gun] Gunning, R.C.: *Lectures on modular forms*. Princeton University Press, 1962

- [H.-B.] Heath-Brown, D.R.: The fourth power moment of the Riemann zeta-function. *Proceeding London Math. Soc.* **38**(3), 385–422 (1979)
- [Hej] Hejhal, D.A.: Some observations concerning eigenvalues of the Laplacian and Dirichlet L -series in Recent progress in analytic number theory. London: Academic Press 1981
- [Hoo] Hooley, C.: On the greatest prime factor of a quadratic polynomial. *Acta Math.* **117**, 281–299 (1967)
- [Iwa1] Iwaniec, H.: Fourier coefficients of cusp forms and the Riemann zeta-function. *Sém. Th. Nb. Bordeaux* (1979–1980), exposé n°18, 36 pages
- [Iwa2] Iwaniec, H.: Mean values for Fourier coefficients of cusp forms and sums of Kloosterman sums. *Proceedings from the Journées Arithmétiques at Exeter* in press (1982)
- [Iwa3] Iwaniec, H.: On mean values for Dirichlet's polynomials and the Riemann zeta-function. *J. London Math. Soc.* **22**, (2), 39–45 (1980)
- [Iwa4] Iwaniec, H.: On the Brun-Titchmarsh theorem. *J. Math. Soc. Japan* **34**, 95–123 (1982)
- [Kub] Kubota, T.: *Elementary Theory of Eisenstein Series*. New-York: John Wiley and Sons 1973
- [Kuz1] Kuznetsov, N.V.: Petersson hypothesis for forms of weight zero and Linnik hypothesis (in Russian), Preprint n°02, Khab. KHII, Khabarovsk 1977
- [Kuz2] Kuznetsov, N.V.: Petersson hypothesis for parabolic forms of weight zero and Linnik hypothesis. Sums of Kloosterman sums. *Math. Sbornik* **111**, (153), n°3, 334–383 (1980)
- [Lin] Linnik, Yu.V.: Additive problems and eigenvalues of the modular operators. *Proc. Internat. Congr. Math. Stockholm* 270–284 (1962)
- [Maa1] Maass, H.: Über eine neue Art von nichtanalytischen automorphen Funktionen. *Math. Ann.* **121**, (n°2) 141–183 (1949)
- [Maa2] Maass, H.: *On modular functions of one complex variable*. Tata Institute, Bombay, 1964
- [Mon] Montgomery, H.L.: *Topics in multiplicative number theory*. *Lect. Notes in Math.* vol. 227. Berlin-New-York: Springer 1971
- [Pet] Petersson, H.: Über die Entwicklungskoeffizienten der automorphen Formen. *Acta Math.* **58**, 169–215 (1932)
- [Pro1] Proskurin, N.V.: Summation formulas for generalized Kloosterman sums (in Russian). *Zap. Naučn. Sem. Leningrad. Otdel. Mat. Inst. Steklov.* **82**, 103–135 (1979)
- [Pro2] Proskurin, N.V.: On a hypothesis of Yu.V. Linnik (in Russian). *Zap. Naučn. Sem. Leningrad Otdel. Mat. Inst. Steklov.* **91**, 94–118 (1979)
- [Pro3] Proskurin, N.V.: The estimations of the eigenvalues of Hecke's operators in the space of parabolic forms of weight zero. *Studies in Number Theory*, vol.5, 136–143 (1979). *Pub. Math. Inst. Steklov*
- [Ran] Rankin, R.: *Modular forms and functions*. Cambridge-London-New-York: Cambridge University Press 1977
- [Roe1] Roelcke, W.: Über die Wellengleichung bei Grenzkreisgruppen erster Art. *Sitz. Ber. Heidl. Akad. der Wiss. (Math. natur. Kl.)* 1956, 4. Abh.
- [Roe2] Roelcke, W.: Das Eigenwertproblem der automorphen Formen in der hyperbolischen Ebene. I. II. *Math. Ann.* **167**, 292–337 (1966); **168**, 261–324 (1967)
- [Sel1] Selberg, A.: Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet's series. *J. Indian Mat. Soc.* **20**, 47–87 (1956)
- [Sel2] Selberg, A.: On the estimation of Fourier coefficients of modular forms. *Proc. Symposia in Pure Math.* VIII, A.M.S., Providence 1965, pp. 1–15
- [Shi] Shimura, G.: *Introduction to the arithmetic theory of automorphic functions*. Princeton: Princeton University Press 1971